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1 & 2 Semester

Physics Cycle

2010 Scheme

Engineering Mathematics - 1

[10MAT11]

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SEM: 1/2

University: VTU

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Engineering Mathematics – I

(10 MAT11)

LECTURE NOTES
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ENGINEERING MATHEMATICS – I

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DIFFERENTIAL CALCULUS – I

Introduction:

The mathematical study of change like motion, growth or decay is calculus. The Rate of change of given function is derivative or differential.

The concept of derivative is essential in day to day life. Also applicable in Engineering, Science, Economics, Medicine etc.

Successive Differentiation:

Let $y = f(x)$ be a real valued function.

The first order derivative of y denoted by $\frac{dy}{dx}$ or y' or y_1 or Δ^1

The Second order derivative of y denoted by $\frac{d^2y}{dx^2}$ or y'' or y_2 or Δ^2

Similarly differentiating the function (1) n -times, successively,
the n^{th} order derivative of y exists denoted by $\frac{d^n y}{dx^n}$ or y^n or y_n or Δ^n

The process of finding 2nd and higher order derivatives is known as Successive Differentiation.

n^{th} derivative of some standard functions:

1. $y = e^{ax}$

Sol: $y_1 = a e^{ax}$

$$y_2 = a^2 e^{ax}$$

Differentiating Successively

$$y_n = a^n e^{ax}$$

ie. $D^n[e^{ax}] = a^n e^{ax}$

For, $a = 1$ $D^n[e^x] = e^x$

2. $y = \log(ax + b)$

Solution : $y_1 = \frac{a}{ax+b}$

$$y_2 = \frac{(-1)a.a}{(ax+b)^2} = \frac{(-1)^1 a^2}{(ax+b)^2}$$

$$y_3 = \frac{(-1)(-2)a^2.a}{(ax+b)^3} = \frac{(-1)^2(1)(2)a^3}{(ax+b)^3} = \frac{(-1)^{3-1}(3-1)!a^3}{(ax+b)^3}$$

$$D^n [\log(ax + b)] = y_n = \frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$$

Similarly $D^n [\log x] = y_n = \frac{(-1)^{n-1}(n-1)!}{x^n}$

3. $y = (ax + b)^m$

Solution: $y_1 = m(ax + b)^{m-1} a$

$$y_2 = m(m-1)(ax + b)^{m-2} a^2$$

$$y_3 = m(m-1)(m-2)(ax + b)^{m-3} a^3$$

Similarly

$$y_n = m(m-1)(m-2)\dots\dots(m-n+1)(ax + b)^{m-n} a^n$$

...(*)

Case (i) :- If $m = n$ in (*)

$$\begin{aligned} D^n[(ax+b)^n] &= n(n-1)(n-2)\dots\dots 3 \cdot 2 \cdot 1 \cdot a^n \\ &= n! a^n \end{aligned}$$

$$D^n[x^n] = n!$$

Case (ii) :- If $m > n$ in (*)

$$D^n[(ax+b)^m] = \frac{m(m-1)\dots(m-n+1)(m-n)(m-n-1)\dots 3 \cdot 2 \cdot 1}{(m-n)(m-n-1)\dots\dots 3 \cdot 2 \cdot 1} (ax+b)^{m-n} a^n$$

$$D^n[(ax+b)^m] = \frac{m!}{(m-n)!} (ax+b)^{m-n} a^n$$

$$D^n[x^m] = \frac{m!}{(m-n)!} x^{m-n} a^n$$

Case iii :- If $m < n$ in (*)

$$D^n[(ax+b)^m] = 0$$

Case iv :- If $m = -1$ in (*)

$$D^n\left[\frac{1}{ax+b}\right] = (-1)(-2)\dots\dots(-n)(ax+b)^{-1-n} a^n = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$$

$$D^n\left[\frac{1}{(ax+b)^p}\right] = \frac{(-1)^n p(p+1)\dots\dots(p+n-1)a^n}{(ax+b)^{p+n}}$$

$$D^n\left[\frac{1}{(ax+b)^p}\right] = (-1)^n \frac{(p+n-1)!}{(p-1)!} \frac{a^n}{(ax+b)^{p+n}}$$

$$\text{If } a=1, \quad D^n\left[\frac{1}{x^p}\right] = (-1)^n \frac{(p+n-1)!}{(p-1)!} \frac{1}{x^{p+n}}$$

4. $y = \cos(ax + b)$

$$y_1 = -\sin(ax + b) \cdot a = a \cos(ax + b + \pi/2)$$

$$y_2 = -\sin(ax + b + \pi/2) \cdot a^2 = a^2 \cos(ax + b + 2\pi/2)$$

$$y_n = D^n [\cos(ax + b)] = a^n \cos(ax + b + n\pi/2)$$

If $a=1, b=0$

$$D^n [\cos x] = \cos(x + n\pi/2)$$

5. $y = \sin(ax + b)$

$$y_n = D^n [\sin(ax + b)] = a^n \sin(ax + b + n\pi/2)$$

If $a=1, b=0$

$$D^n [\sin x] = \sin(x + n\pi/2)$$

6. $y = e^{ax} \sin(bx + c)$

$$y_1 = a e^{ax} \sin(bx + c) + b e^{ax} \cos(bx + c)$$

$$= e^{ax} [a \sin(bx + c) + b \cos(bx + c)]$$

Put $a = r \cos \theta$ $b = r \sin \theta$ then $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1} \frac{b}{a}$

$$y_1 = e^{ax} [r \cos \theta \sin(bx + c) + r \sin \theta \cos(bx + c)]$$

$$y_1 = r e^{ax} [\sin(bx + c + \theta)]$$

$$y_2 = r [e^{ax} a \sin(bx + c + \theta) + e^{ax} b \cos(bx + c + \theta)]$$

Put $a = r \cos \theta$ $b = r \sin \theta$ then $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1} \frac{b}{a}$

$$y_2 = r e^{ax} [r \cos \theta \sin(bx + c + \theta) + r \sin \theta \cos(bx + c + \theta)]$$

$$y_2 = r^2 e^{ax} [\sin(bx + c + 2\theta)]$$

Similarly,

$$y_n = r^n e^{ax} [\sin(bx + c + n\theta)]$$

$$y_n = D^n [e^{ax} \sin(bx + c)] = (a^2 + b^2)^{n/2} e^{ax} [\sin(bx + c + n \tan^{-1} \frac{b}{a})]$$

For $a=b=1, c=0$

$$D^n [e^x \sin x] = (2)^{n/2} e^x [\sin(x + n\pi/4)]$$

$$7. y = e^{ax} [\cos(bx + c)]$$

$$y_n = D^n [e^{ax} \cos(bx + c)] = (a^2 + b^2)^{n/2} e^{ax} [\cos(bx + c + n \tan^{-1} \frac{b}{a})]$$

For $a=b=1, c=0$

$$D^n [e^x \cos x] = (2)^{n/2} e^x [\cos(x + n\pi/4)]$$

8.

$$y = a^{mx}$$

$$y_1 = a^{mx} (\log a^m) = a^{mx} (m \log a)$$

$$y_2 = a^{mx} (m \log a)^2$$

Differentiating Successively

$$y_n = a^{mx} (m \log a)^n$$

$$\text{For } m=1, D^n [a^x] = a^x (\log a)^n$$

Leibnitz's Theorem :

It provides a useful formula for computing the n^{th} derivative of a product of two functions.

Statement : If u and v are any two functions of x with u_n and v_n as their n^{th} derivative. Then the n^{th} derivative of uv is

$$(uv)_n = u_0 v_n + {}^n C_1 u_1 v_{n-1} + {}^n C_2 u_2 v_{n-2} + \dots + {}^n C_{n-1} u_{n-1} v_1 + u_n v_0$$

Note : We can interchange u & v $(uv)_n = (vu)_n$,

$${}^n C_1 = n, {}^n C_2 = n(n-1)/2!, {}^n C_3 = n(n-1)(n-2)/3! \dots$$

1. Find the n^{th} derivations of $e^{ax} \cos(bx + c)$

Solution: $y_1 = e^{ax} - b \sin(bx + c) + a e^{ax} \cos(bx + c)$, by product rule.

$$\text{i.e., } y_1 = e^{ax} [a \cos(bx + c) - b \sin(bx + c)]$$

Let us put $a = r \cos \theta$, and $b = r \sin \theta$.

$$\therefore a^2 + b^2 = r^2 \text{ and } \tan \theta = b/a$$

$$\text{i.e., } r = \sqrt{a^2 + b^2} \text{ and } \theta = \tan^{-1}(b/a)$$

$$\text{Now, } y_1 = e^{ax} [r \cos \theta \cos(bx + c) - r \sin \theta \sin(bx + c)]$$

$$\text{i.e., } y_1 = r e^{ax} \cos(\theta + bx + c)$$

where we have used the formula $\cos A \cos B - \sin A \sin B = \cos(A + B)$

Differentiating again and simplifying as before,

$$y_2 = r^2 e^{ax} \cos(2\theta + bx + c).$$

$$\text{Similarly } y_3 = r^3 e^{ax} \cos(3\theta + bx + c).$$

.....

$$\text{Thus } y_n = r^n e^{ax} \cos(n\theta + bx + c)$$

$$\text{Where } r = \sqrt{a^2 + b^2} \text{ and } \theta = \tan^{-1}(b/a).$$

$$\text{Thus } D^n [e^{ax} \cos(bx + c)]$$

$$= \left[(\sqrt{a^2 + b^2})^n e^{ax} \cos \left[n \tan^{-1}(b/a) + bx + c \right] \right]$$

2. Find the n^{th} derivative of $\log \sqrt{4x^2 + 8x + 3}$

Solution : Let $y = \log \sqrt{4x^2 + 8x + 3} = \log (4x^2 + 8x + 3)^{\frac{1}{2}}$

$$\text{ie., } y = \frac{1}{2} \log (4x^2 + 8x + 3) \because \log x^n = n \log x$$

$$y = \frac{1}{2} \log \{ (2x + 3) (2x + 1) \}, \text{ by factorization.}$$

$$\therefore y = \frac{1}{2} \{ \log (2x + 3) + \log (2x + 1) \}$$

$$\text{Now } y_n = \frac{1}{2} \left\{ \frac{(-1)^{n-1} (n-1)! 2^n}{(2x+3)^n} + \frac{(-1)^{n-1} (n-1)! 2^n}{(2x+1)^n} \right\}$$

$$\text{ie., } y_n = 2^{n-1} (-1)^{n-1} (n-1)! \left\{ \frac{1}{(2x+3)^n} + \frac{1}{(2x+1)^n} \right\}$$

3. Find the n^{th} derivative of $\log_{10} \{(1-2x)^3 (8x+1)^5\}$

Solution : Let $y = \log_{10} \{(1-2x)^3 (8x+1)^5\}$

It is important to note that we have to convert the logarithm to the base e by the property:

$$\log_{10} x = \frac{\log_e x}{\log_e 10}$$

$$\text{Thus } y = \frac{1}{\log_e 10} \log_e \{(1-2x)^3 (8x+1)^5\}$$

$$\text{ie., } y = \frac{1}{\log_e 10} \{ 3 \log(1-2x) + 5 \log(8x+1) \}$$

$$\therefore y_n = \frac{1}{\log_e 10} \left\{ 3 \cdot \frac{(-1)^{n-1} (n-1)! (-2)^n}{(1-2x)^n} + 5 \frac{(-1)^{n-1} (n-1)! 8^n}{(8x+1)^n} \right\}$$

$$\text{ie., } y_n = \frac{(-1)^{n-1} (n-1)! 2^n}{\log_e 10} \left\{ \frac{3(-1)^n}{(1-2x)^n} + \frac{5(4)^n}{(8x+1)^n} \right\}$$

4. Find the n^{th} derivative of $e^{2x} \cos^2 x \sin x$

Solution : $\gg$ let $y = e^{2x} \cos^2 x \sin x = e^{2x} \left[\frac{1 + \cos 2x}{2} \right] \sin x$

$$\text{ie., } y = \frac{e^{2x}}{2} (\sin x + \sin x \cos 2x)$$

$$= \frac{e^{2x}}{2} \left\{ \sin x + \frac{1}{2} [\sin 3x + \sin(-x)] \right\}$$

$$= \frac{e^{2x}}{4} (2 \sin x + \sin 3x - \sin x) \because \sin(-x) = -\sin x$$

$$\therefore y = \frac{e^{2x}}{4} (\sin x + \sin 3x)$$

Now $y_n = \frac{1}{4} \{ D^n (e^{2x} \sin x) + D^n (e^{2x} \sin 3x) \}$

Thus $y_n = \frac{1}{4} \left\{ (\sqrt{5})^n e^{2x} \sin [n \tan^{-1}(1/2) + x] + (\sqrt{13})^n e^{2x} \sin [n \tan^{-1}(3/2) + 3x] \right\}$

$$\therefore y_n = \frac{e^{2x}}{4} \left\{ (\sqrt{5})^n \sin [n \tan^{-1}(1/2) + x] + (\sqrt{13})^n \sin [n \tan^{-1}(3/2) + 3x] \right\}$$

5. Find the n^{th} derivative of $e^{2x} \cos^3 x$

Solution : Let $y = e^{2x} \cos^3 x = e^{2x} \cdot \frac{1}{4} (3 \cos x + \cos 3x)$

ie., $y = \frac{1}{4} (3 e^{2x} \cos x + e^{2x} \cos 3x)$

$$\therefore y_n = \frac{1}{4} \{ 3 D^n (e^{2x} \cos x) + D^n (e^{2x} \cos 3x) \}$$

$$y_n = \frac{1}{4} \left\{ (3\sqrt{5})^n e^{2x} \cos [n \tan^{-1}(1/2) + x] + (\sqrt{13})^n e^{2x} \cos [n \tan^{-1}(3/2) + 3x] \right\}$$

Thus $y_n = \frac{e^{2x}}{4} \left\{ 3(\sqrt{5})^n \cos [n \tan^{-1}(1/2) + x] + (\sqrt{13})^n \cos [n \tan^{-1}(3/2) + 3x] \right\}$

6. Find the n^{th} derivative of $\frac{x^2}{(2x+1)(2x+3)}$

Solution : $y = \frac{x^2}{(2x+1)(2x+3)}$ is an improper fraction because; the degree of the numerator being 2 is equal to the degree of the denominator. Hence we must divide and rewrite the fraction.

$$y = \frac{x^2}{4x^2 + 8x + 3} = \frac{1}{4} \cdot \frac{4x^2}{4x^2 + 8x + 3} \text{ for convenience.}$$

$$4x^2 + 8x + 3 \overline{) \begin{array}{r} 1 \\ 4x^2 \\ \hline 4x^2 + 8x + 3 \\ \hline -8x - 3 \end{array}}$$

$$\therefore y = \frac{1}{4} \left[1 + \frac{-8x - 3}{4x^2 + 8x + 3} \right]$$

$$\text{ie., } y = \frac{1}{4} - \frac{1}{4} \left[\frac{8x + 3}{4x^2 + 8x + 3} \right]$$

The algebraic fraction involved is a proper fraction.

$$\text{Now } y_n = 0 - \frac{1}{4} D^n \left[\frac{8x + 3}{4x^2 + 8x + 3} \right].$$

$$\text{Let } \frac{8x + 3}{(2x+1)(2x+3)} = \frac{A}{2x+1} + \frac{B}{2x+3}$$

Multiplying by $(2x + 1)(2x + 3)$ we have, $8x + 3 = A(2x + 3) + B(2x + 1)$

.....(1)

By setting $2x + 1 = 0$, $2x + 3 = 0$ we get $x = -1/2$, $x = -3/2$.

Put $x = -1/2$ in (1): $-1 - 1 + A(2) \Rightarrow A = -1/2$

Put $x = -3/2$ in (1): $-9 = B(-2) \Rightarrow B = 9/2$

$$\therefore y_n = -\frac{1}{4} \left\{ -\frac{1}{2} D^n \left[\frac{1}{2x+1} \right] + \frac{9}{2} D^n \left[\frac{1}{2x+3} \right] \right\}$$

$$= -\frac{1}{8} \left\{ (-1) \cdot \frac{(-1)^n n! 2^n}{(2x+1)^{n+1}} + 9 \cdot \frac{(-1)^n n! 2^n}{(2x+3)^{n+1}} \right\}$$

$$\text{ie., } y_n = \frac{(-1)^{n+1} n! 2^n}{8} \left\{ \frac{1}{(2x+1)^{n+1}} + \frac{9}{(2x+3)^{n+1}} \right\}$$

7. Find the n^{th} derivative of $\frac{x^4}{(x+1)(x+2)}$

Solution : $y = \frac{x^4}{(x+1)(x+2)}$ is an improper fraction.

(deg of nr. = 4 > deg. of dr. = 2)

On dividing x^4 by $x^2 + 3x + 2$, We get

$$y = (x^2 - 3x + 7) + \left[\frac{-15x - 14}{x^2 + 3x + 2} \right]$$

$$\therefore y_n = D^n (x^2 - 3x + 7) - D^n \left[\frac{15x - 14}{x^2 + 3x + 2} \right]$$

$$\text{But } D(x^2 - 3x + 7) = 2x - 3, D^2(x^2 - 3x + 7) = 2$$

$$D^3(x^2 - 3x + 7) = 0 \dots \dots \dots D^n(x^2 - 3x + 7) = 0 \text{ if } n > 2$$

$$\text{Hence } y_n = -D^n \left[\frac{15x + 14}{(x+1)(x+2)} \right]$$

$$\text{Now, let } D^n \frac{15x + 14}{x^2 + 3x + 2} = \frac{A}{(x+1)} + \frac{B}{(x+2)}$$

$$\Rightarrow 15x + 14 = A(x+2) + B(x+1)$$

$$\text{Put } x = -1 ; -1 = A(1) \text{ or } A = -1$$

$$\text{Put } x = -2 ; -16 = B(-1) \text{ or } B = 16$$

$$Y_n = \left\{ -D^n \left[\frac{1}{x+1} \right] + 16D^n \left[\frac{1}{x+2} \right] \right\}$$

$$= \frac{(-1)^n n! 1^n}{(x+1)^{n+1}} - 16 \frac{(-1)^n n! 1^n}{(x+2)^{n+1}}$$

$$y_n = (-1)^n n! \left\{ \frac{1}{(x+1)^{n+1}} - \frac{16}{(x+2)^{n+1}} \right\} n > 2$$

8. Show that

$$\frac{d^n}{dx^n} \left(\frac{\log x}{x} \right) = \frac{(-1)^n n!}{x^{n+1}} \left\{ \log x - 1 - \frac{1}{2} - \frac{1}{3} - \frac{1}{n} \right\}$$

Solution : Let $y = \frac{\log x}{x} = \log x \cdot \frac{1}{x}$ and let $u = \log x$, $v = \frac{1}{x}$

We have Leibnitz theorem,

$$(uv)_n = uv_n + n_{C_1} u_1 v_{n-1} + n_{C_2} u_2 v_{n-2} + \dots + u_n v \quad \dots (1)$$

$$\text{Now, } u = \log x \quad \therefore u_n = \frac{(-1)^{n-1} (n-1)!}{x^n}$$

$$v = \frac{1}{x} \quad \therefore v_n = \frac{(-1)^n n!}{x^{n+1}}$$

Using these in (1) by taking appropriate values for n we get,

$$D_n = \left(\frac{\log x}{x} \right) = \log x \cdot \frac{(-1)^n n!}{x^{n+1}} + n \frac{1}{x} \cdot \frac{(-1)^{n-1} (n-1)!}{x^n}$$

$$+ \frac{n(n-1)}{1 \cdot 2} \left(-\frac{1}{x^2} \right) \frac{(-1)^{n-2} (n-2)!}{x^{n-1}}$$

$$+ \dots + \frac{(-1)^{n-1} (n-1)!}{x^n} \cdot \frac{1}{x}$$

$$\text{ie..} = \log x \frac{(-1)^n n!}{x^{n+1}} + \frac{(-1)^{n-1} n!}{x^{n+1}}$$

$$- \frac{(-1)^{n-2} n!}{2x^{n+1}} + \dots + \frac{(-1)^{n-1} (n-1)!}{x^{n+1}}$$

$$- \frac{(-1)^{n-2} n!}{x^{n+1}} \left[\log x (-1)^{-1} - \frac{(-1)^{-2}}{2} + \dots + \frac{(-1)^{-1} (n-1)!}{n^1} \right]$$

$$\text{Note : } (-1)^{-1} = \frac{1}{-1} = -1; (-1)^{-2} = \frac{1}{(-1)^2} = 1$$

$$\text{Also } \frac{(n-1)!}{n!} = \frac{(n-1)!}{n(n-1)!} = \frac{1}{n}$$

$$\therefore \frac{d^n}{dx^n} \left[\frac{\log x}{x} \right] = \frac{(-1)^n n!}{x^{n+1}} \left[\log x - 1 - \frac{1}{2} - \frac{1}{3} \dots - \frac{1}{n} \right]$$

9. If $y_n = D^n (x^n \log x)$

Prove that $y_n = n y_{n-1} + (n-1)!$ and hence deduce that

$$y_n = n \left(\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right)$$

Solution : $y_n = D^n (x^n \log x) = D^{n-1} \{D (x^n \log x)\}$

$$= D^{n-1} \left\{ x^n \cdot \frac{1}{x} + nx^{n-1} \log x \right\}$$

$$= D^{n-1} (x^{n-1}) + n D^{n-1} (x^{n-1} \log x)$$

$\therefore y_n = (n-1)! + n y_{n-1}$. This proves the first part.

Now Putting the values for $n = 1, 2, 3 \dots$ we get

$$y_1 = 0! + 1 \cdot y_0 = 1 + \log x = 1! (\log x + 1)$$

$$y_2 = 1! + 2y_1 = 1 + 2(1 + \log x)$$

$$\text{ie., } y_2 = 2! \log x + 3 = 2(\log x + 3/2) = 2! \left(\log x + 1 + \frac{1}{2} \right)$$

$$y_3 = 2! + 3y_2 = 2 + 3(2 \log x + 3)$$

$$\text{ie., } y_3 = 6 \log x + 11 = 6(\log x + 11/6) = 3! \left(\log x + 1 + \frac{1}{2} + \frac{1}{3} \right)$$

.....

$$y_n = n! \left(\log x + 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right)$$

10. If $y = a \cos (\log x) + b \sin (\log x)$, show that

$x^2 y_2 + x y_1 + y = 0$. Then apply Leibnitz theorem to differentiate this result n times.

or

If $y = a \cos (\log x) + b \sin (\log x)$, show that

$$x^2 y_{n+2} + (2n+1) x y_{n+1} + (n^2+1) y_n = 0.$$

[July-03]

Solution : $y = a \cos (\log x) + b \sin (\log x)$

Differentiate w.r.t x

$$\therefore y_1 = -a \sin (\log x) \cdot \frac{1}{x} + b \cos (\log x) \cdot \frac{1}{x}$$

(we avoid quotient rule to find y_2) .

$$\Rightarrow xy_1 = -a \sin (\log x) + b \cos (\log x)$$

Differentiating again w.r.t x we have,

$$xy_2 + 1 \cdot y_1 = -a \cos (\log x) + b \sin (\log x) \cdot \frac{1}{x}$$

$$\text{or } x^2y_2 + xy_1 = -[a \cos (\log x) + b \sin (\log x)] = -y$$

$$\therefore x^2y_2 + xy_1 + y = 0$$

Now we have to differentiate this result n times.

$$\text{ie., } D^n (x^2y_2) + D^n (xy_1) + D^n (y) = 0$$

We have to employ Leibnitz theorem for the first two terms.

Hence we have,

$$\left\{ x^2 \cdot D^n (y_2) + n \cdot 2x \cdot D^{n-1} (y_2) + \frac{n(n-1)}{1 \cdot 2} \cdot 2 \cdot D^{n-2} (y^2) \right\}$$

$$\{x \cdot D^n (y_1) + n \cdot 1 \cdot D^{n-1} (y_1)\} + y_n = 0$$

$$\text{ie., } \{x^2y_{n+2} + 2n x y_{n+1} + n(n-1)y_n\} + \{xy_{n+1} + ny_n\} + y_n = 0$$

$$\text{ie., } x^2y_{n+2} + 2n x y_{n+1} + n^2y_n - ny_n + xy_{n+1} + ny_n + y_n = 0$$

$$\text{ie., } x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$$

11. If $\cos^{-1} (y/b) = \log (x/n)^n$, then show that

$$x^2y_{n+2} + (2n+1) xy_{n+1} + 2n^2y_n = 0$$

Solution : By data, $\cos^{-1} (y/b) = n \log (x/n) \therefore \log(a^m) = m \log a$

$$\Rightarrow \frac{y}{b} = \cos [n \log (x/n)]$$

$$\text{or } y = b \cdot \cos [n \log (x/n)]$$

Differentiating w.r.t x we get,

$$y_1 = -b \sin [n \log (x/n)] \cdot n \cdot \frac{1}{(x/n)} \cdot \frac{1}{n}$$

$$\text{or } xy_1 = -n b \sin [n \log (x/n)]$$

Differentiating w.r.t x again we get,

$$xy_2 + 1 \cdot y_1 = -n \cdot b \cos [n \log (x/n)] \cdot n \cdot \frac{1}{(x/n)} \cdot \frac{1}{n}$$

$$\text{or } x(xy_2 + y_1) = n^2 b \cos [n \log (x/n)] = -n^2 y, \text{ by using (1).}$$

$$\text{or } x^2 y_2 + xy_1 + n^2 y = 0$$

Differentiating each term n times we have,

$$D(x^2 y_2) + D^n(xy_1) + n^2 D^n(y) = 0$$

Applying Leibnitz theorem to the product terms we have,

$$\left\{ x^2 y_{n+2} + n \cdot 2x \cdot y_{n+1} + \frac{n(n-1)}{1 \cdot 2} \cdot 2 \cdot y_n \right\} \\ + \{ xy_{n+1} + n \cdot 1 \cdot y_n \} + n^2 y_n = 0$$

$$\text{ie } x^2 y_{n+2} + 2x y_{n+1} + n^2 y_n + xy_{n+1} + ny_n + n^2 y_n = 0$$

$$\text{or } x^2 y_{n+2} + (2n+1) xy_{n+1} + 2n^2 y_n = 0$$

12. If $y = \sin(\log(x^2 + 2x + 1))$,

or

[Feb-03]

If $\sin^{-1} y = 2 \log(x+1)$, show that

$$(x+1)^2 y_{n+2} + (2n+1)(x+1)y_{n+1} + (n^2 + 4)y_n = 0$$

Solution : By data $y = \sin \log(x^2 + 2x + 1)$

$$\therefore y_1 = \cos \log(x^2 + 2x + 1) \cdot \frac{1}{(x+1)^2} \cdot 2x + 2$$

$$\text{ie., } y_1 = \cos \log(x^2 + 2x + 1) \cdot \frac{1}{x^2 + 2x + 1} \cdot 2(x+1)$$

$$\text{ie., } y_1 = \frac{2 \cos \log(x^2 + 2x + 1)}{(x+1)}$$

$$\text{or } (x+1)y_1 = 2 \cos \log(x^2 + 2x + 1)$$

Differentiating w.r.t x again we get

$$(x+1)y_2 + 1 y_1 = -2 \sin \log (x^2 + 2x + 1) \frac{1}{(x+1)^2} \cdot 2(x+1)$$

$$\text{or } (x+1)^2 y_2 + (x+1) y_1 = -4y$$

$$\text{or } (x+1)^2 y_2 + (x+1) y_1 + 4y = 0 ,$$

Differentiating each term n times we have,

$$D^n [(x+1)^2 y_2] + D^n [(x+1) y_1] + D^n [y] = 0$$

Applying Leibnitz theorem to the product terms we have,

$$\left\{ (x+1)^2 y_{n+2} + n \cdot 2(x+1) \cdot y_{n+1} + \frac{n(n-1)}{1 \cdot 2} \cdot 2 \cdot y_n \right\}$$

$$+ \{(x+1) y_{n+1} + n \cdot 1 \cdot y_n\} + 4y_n = 0$$

$$\text{ie., } (x+1)^2 y_{n+2} + 2n(x+1) y_{n+1}$$

$$+ n^2 y_n - n y_n + (x+1) y_{n+1} + n y_n + 4y_n = 0$$

$$\text{ie., } (x+1)^2 y_{n+2} + (2n+1)(x+1) y_{n+1} + (n^2+4) y_n = 0$$

13. If $y = \log (x + \sqrt{1+x^2})$ prove that

$$(1+x^2) y_{n+2} + (2n+1) x y_{n+1} + n^2 y_n = 0$$

$$>> \text{By data, } y = \log (x + \sqrt{1+x^2})$$

$$\therefore y_1 = \frac{1}{(x + \sqrt{1+x^2})} \left\{ 1 + \frac{1}{2\sqrt{1+x^2}} \cdot 2x \right\}$$

$$\text{ie., } y_1 \frac{1}{(x + \sqrt{1+x^2})} \frac{\sqrt{1+x^2} + x}{\sqrt{1+x^2}} = \frac{1}{\sqrt{1+x^2}}$$

$$\text{or } \sqrt{1+x^2} y_1 = 1$$

Differentiating w.r.t.x again we get

$$\sqrt{1+x^2} y_2 + \frac{1}{2\sqrt{1+x^2}} \cdot 2x \cdot y_1 = 0$$

$$\text{or } (1+x^2) y_2 + x y_1 = 0$$

$$\text{Now } D^n [(1+x^2) y_2] + D^n [x y_1] = 0$$

Applying Leibnitz theorem to each term we get,

$$\left\{ (1+x^2)y_{n+2} + n \cdot 2x \cdot y_{n+1} + \frac{n(n-1)}{1 \cdot 2} \cdot 2 \cdot y_n \right\}$$

$$+ [x \cdot y_{n+1} + n \cdot 1 \cdot y_n] = 0$$

$$\text{i.e., } (1+x^2)y_{n+2} + 2nx y_{n+1} + n^2 y_n - n y_n + x y_{n+1} + n y_n = 0$$

$$\text{or } (1+x^2)y_{n+2} + (2n+1)xy_{n+1} + n^2 y_n = 0$$

14. If $x = \sin t$ and $y = \cos mt$, prove that

$$(1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0. \quad [\text{Feb-04}]$$

Solution : By data $x = \sin t$ and $y = \cos mt$

$$x = \sin t \Rightarrow t = \sin^{-1} x \text{ and } y = \cos mt \text{ becomes}$$

$$y = \cos [m \sin^{-1} x]$$

Differentiating w.r.t. x we get

$$y_1 = -\sin (m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\text{or } \sqrt{1-x^2} y_1 = -m \sin (m \sin^{-1} x)$$

Differentiating again w.r.f. x we get,

$$\sqrt{1-x^2} y_2 + \frac{1}{2\sqrt{1-x^2}} (-2x) y_1 = -m \cos (m \sin^{-1} x) \cdot \frac{m}{\sqrt{1-x^2}}$$

$$\text{or } (1-x^2)y_2 - xy_1 = -m^2 y$$

$$\text{or } (1-x^2)y_2 - xy_1 + m^2 y = 0$$

$$\text{Thus } (1-x^2)y_{n+2} - (2n+1)xy_{n+1} + (m^2-n^2)y_n = 0$$

15. If $x = \tan (\log y)$, find the value of

$$(1+x^2)y_{n+1} + (2nx-1)y_n + n(n-1)y_{n-1} \quad [\text{July-04}]$$

Solution : By data $x = \tan(\log y) \Rightarrow \tan^{-1} x = \log y$ or $y = e^{\tan^{-1} x}$ Since the desired relation involves y_{n+1} , y_n and y_{n-1} we can find y_1 and differentiate n times the result associated with y_1 and y .

$$\text{Consider } y = e^{\tan^{-1} x} \therefore y = e^{\tan^{-1} x} \cdot \frac{1}{1+x^2}$$

$$\text{or } (1+x^2)y_1 = y$$

Differentiating n times we have

$$D^n[(1+x^2)y_1] = D^n[y]$$

Applying Leibnitz theorem onto L.H.S, we have,

$$\{(1+x^2)D^n(y_1) + n \cdot 2x \cdot D^{n-1}(y_1) + \frac{n(n-1)}{1 \cdot 2} \cdot 2 \cdot D^{n-2}(y_1)\} = y_n$$

$$\text{ie., } (1+x^2)y_{n+1} + 2nx y_n + n(n-1)y_{n-1} = 0$$

$$\text{Or } (1+x^2)y_{n+1} + (2nx-1)y_n + n(n-1)y_{n-1} = 0$$

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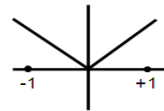
Continuity & Differentiability

Some Fundamental Definitions

A function $f(x)$ is defined in the interval I , then it is said to be continuous at a point $x = a$ if $\lim_{x \rightarrow a} f(x) = f(a)$

A function $f(x)$ is said to be differentiable at $x = a$ if $\lim_{h \rightarrow 0} \frac{f(x+h) - f(a)}{h} = f'(a)$ exists $a \in I$

Ex : Consider a function $f(x)$ is defined in the interval $[-1,1]$ by $f(x) = |x| = \begin{cases} -x & -1 \leq x \leq 0 \\ x & 0 \leq x \leq 1 \end{cases}$



It is continuous at $x = 0$

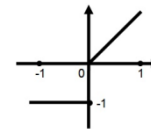
But not differentiable at $x = 0$

Note : If a function $f(x)$ is differentiable then it is continuous, but converse need not be true.
Geometrically :

(1) If $f(x)$ is Continuous at $x=a$ means, $f(x)$ has no breaks or jumps at the point $x = a$

Ex : $f(x) = \begin{cases} -1 & -1 \leq x \leq 0 \\ x & 0 < x \leq 1 \end{cases}$

Is discontinuous at $x=0$



(2) If $f(x)$ is differentiable at $x = a$ means, the graph of $f(x)$ has a unique tangent at the point or graph is smooth at $x = a$

1. Give the definitions of Continuity & Differentiability:

Solution: A function $f(x)$ is said to be continuous at $x = a$, if corresponding to an arbitrary positive number ϵ , however small, there exists another positive number δ such that.

$$|f(x) - f(a)| < \epsilon, \text{ where } |x - a| < \delta$$

It is clear from the above definition that a function $f(x)$ is continuous at a point 'a'.

If (i) it exists at $x = a$

(ii) $\lim_{x \rightarrow a} f(x) = f(a)$

i.e, limiting value of the function at $x = a$ is to the value of the function at $x = a$

Differentiability:

A function $f(x)$ is said to be differentiable in the interval (a, b) , if it is differentiable at every point in the interval.

In Case $[a, b]$ the function should possess derivatives at every point and at the end points a & b i.e., $Rf'(a)$ and $Lf'(a)$ exists.

2. State Rolle's Theorem with Geometric Interpretation.

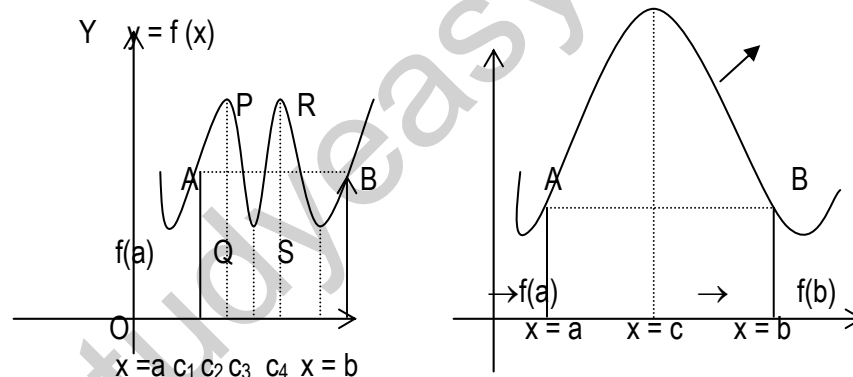
Statement: Let $f(x)$ be a function defined on $[a, b]$ & it satisfies the following Conditions.

- (i) $f(x)$ is continuous in $[a, b]$
- (ii) $f(x)$ is differentiable in (a, b)
- (iii) $f(a) = f(b)$

Then there exists at least a point $C \in (a, b)$, Here $a < b$ such that $f'(c) = 0$

Proof:

Geometrical Interpretation of Rolle's Theorem:



Let us consider the graph of the function $y = f(x)$ in xy -plane. $A(a, f(a))$ and $B(b, f(b))$ be the two points in the curve $f(x)$ and a, b are the corresponding end points of A & B respectively. Now, explained the conditions of Rolle's theorem as follows.

- (i) $f(x)$ is continuous function in $[a, b]$, Because from figure without breaks or jumps in between A & B on $y = f(x)$.
- (ii) $f(x)$ is differentiable in (a, b) , that means let us joining the points A & B , we get a line AB .

$\therefore$ Slope of the line $AB = 0$ then $\exists$ a point C at P and also the tangent at P (or Q or R or S) is Parallel to x -axis.

$\therefore$ Slope of the tangent at P (or Q or R or S) to be Zero even the curve $y = f(x)$ decreases or increases, i.e., $f(x)$ is Constant.

$$f'(x) = 0$$

$$\therefore f'(c) = 0$$

(iii) The Slope of the line AB is equal to Zero, i.e., the line AB is parallel to x – axis.

$$\therefore f(a) = f(b)$$

3. Verify Rolle's Theorem for the function $f(x) = x^2 - 4x + 8$ in the interval $[1, 3]$

Solution: We know that every Poly nominal is continuous and differentiable for all points and hence $f(x)$ is continuous and differentiable in the interval $[1, 3]$.

$$\text{Also } f(1) = 1 - 4 + 8 = 5, f(3) = 3^2 - 4 \cdot 3 + 8 = 5$$

$$\text{Hence } f(1) = f(3)$$

Thus $f(x)$ satisfies all the conditions of the Rolle's Theorem. Now $f'(x) = 2x - 4$ and $f'(x) = 0 \Rightarrow 2x - 4 = 0$ or $x = 2$. Clearly $1 < 2 < 3$. Hence there exists $2 \in (1, 3)$ such that $f'(2) = 0$. This shows that Rolle's Theorem holds good for the given function $f(x)$ in the given interval.

4. Verify Rolle's Theorem for the function $f(x) = x(x+3)e^{-x/2}$ in the interval $[-3, 0]$

Solution: Differentiating the given function W.r.t 'x' we get

$$\begin{aligned} f'(x) &= (x^2 + 3x) \left(-\frac{1}{2} \right) e^{-x/2} + (2x + 3) e^{-x/2} \\ &= -\frac{1}{2} (x^2 - x - 6) e^{-x/2} \end{aligned}$$

$\therefore f'(x)$ exists (i.e finite) for all x and hence continuous for all x .

Also $f(-3) = 0, f(0) = 0$ so that $f(-3) = f(0)$ so that $f(-3) = f(0)$. Thus $f(x)$ satisfies all the conditions of the Rolle's Theorem.

$$\text{Now, } f'(x) = 0$$

$$\Rightarrow -\frac{1}{2} (x^2 - x - 6) e^{-x/2} = 0$$

Solving this equation we get $x = 3$ or $x = -2$

Clearly $-3 < -2 < 0$. Hence there exists $-2 \in (-3, 0)$ such that $f'(-2) = 0$

This proves that Rolle's Theorem is true for the given function.

5. Verify the Rolle's Theorem for the function $\sin x$ in $[-\pi, \pi]$

Solution: Let $f(x) = \sin x$

Clearly $\sin x$ is continuous for all x .

Also $f'(x) = \cos x$ exists for all x in $(-\pi, \pi)$ and $f(-\pi) = \sin(-\pi) = 0$; $f(\pi) = \sin(\pi) = 0$ so that $f(-\pi) = f(\pi)$

Thus $f(x)$ satisfies all the conditions of the Rolle's Theorem.

Now $f'(x) = 0 \Rightarrow \cos x = 0$ so that

$$x = \pm \frac{\pi}{2}$$

Both these values lie in $(-\pi, \pi)$. These exist $c = \pm \frac{\pi}{2}$

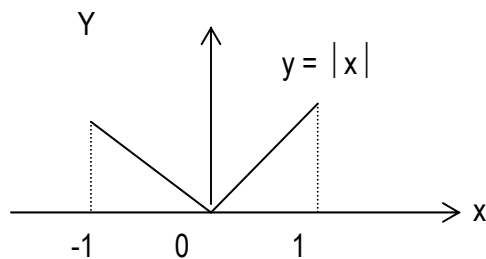
Such that $f'(c) = 0$

Hence Rolle's theorem is verified.

6. Discuss the applicability of Rolle's Theorem for the function $f(x) = |x|$ in $[-1, 1]$.

Solution: Now $f(x) = |x| = \begin{cases} x & \text{for } 0 \leq x \leq 1 \\ -x & \text{for } -1 \leq x \leq 0 \end{cases}$

$f(x)$ being a linear function is continuous for all x in $[-1, 1]$. $f(x)$ is differentiable for all x in $(-1, 1)$ except at $x = 0$. Therefore Rolle's Theorem does not hold good for the function $f(x)$ in $[-1, 1]$. Graph of this function is shown in figure. From which we observe that we cannot draw a tangent to the curve at any point in $(-1, 1)$ parallel to the x -axis.



Exercise:

7. Verify Rolle's Theorem for the following functions in the given intervals.

a) $x^2 - 6x + 8$ in $[2, 4]$

b) $(x - a)^3 (x - b)^3$ in $[a, b]$

c) $\log \left\{ \frac{x^2 + ab}{(a + b)x} \right\}$ in $[a, b]$

8. Find whether Rolle's Theorem is applicable to the following functions. Justify your answer.

a) $f(x) = |x - 1|$ in $[0, 2]$

b) $f(x) = \tan x$ in $[0, \pi]$.

9. State & prove Lagrange's (1st) Mean Value Theorem with Geometric meaning.

Statement: Let $f(x)$ be a function of x such that

(i) f is continuous in $[a, b]$

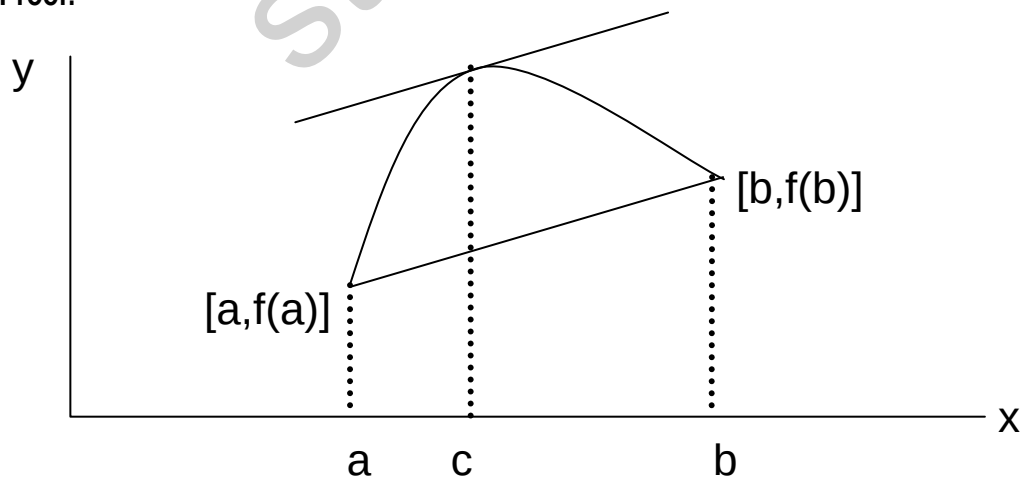
(ii) f is differentiable in (a, b)

Then there exists atleast a point (or value) $C \in (a, b)$ such that.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

i.e., $f(b) = f(a) + (b - a) f'(c)$

Proof:



Define a function $g(x)$ so that $g(x) = f(x) - Ax$ ----- (1)

Where A is a Constant to be determined.

So that $g(a) = g(b)$

Now, $g(a) = f(a) - Aa$

$g(b) = f(b) - Ab$

$\therefore g(a) = g(b) \Rightarrow f(a) - Aa = f(b) - Ab.$

$$\text{i.e., } A = \frac{f(b) - f(a)}{b - a} \text{ ----- (2)}$$

Now, $g(x)$ is continuous in $[a, b]$ as rhs of (1) is continuous in $[a, b]$

$g(x)$ is differentiable in (a, b) as r.h.s of (1) is differentiable in (a, b) .

Further $g(a) = g(b)$, because of the choice of A .

Thus $g(x)$ satisfies the conditions of the Rolle's Theorem.

$\therefore$ There exists a value $x = c$ so that $a < c < b$ at which $g'(c) = 0$

$\therefore$ Differentiate (1) W.r.t 'x' we get

$$g'(x) = f'(x) - A$$

$$\therefore g'(c) = f'(c) - A (\because x = c)$$

$$\Rightarrow f'(c) - A = 0 (\because g'(c) = 0)$$

$$\therefore f'(c) = A \text{ ----- (3)}$$

From (2) and (3) we get

$$f'(c) = \frac{f(b) - f(a)}{b - a} \text{ (or) } f(b) = f(a) + (b - a) f'(c) \text{ For } a < c < b$$

Corollary: Put $b - a = h$

i.e., $b = a + h$ and $c = a + \theta h$

Where $0 < \theta < 1$

Substituting in $f(b) = f(a) + (b - a) f'(c)$

$$\therefore f(a + h) = f(a) + h f'(a + \theta h), \text{ where } 0 < \theta < 1.$$

Geometrical Interpretation:-

Since $y = f(x)$ is continuous in $[a, b]$, it has a graph as shown in the figure below,

At $x = a$, $y = f(a)$

At $x = b$, $y = f(b)$

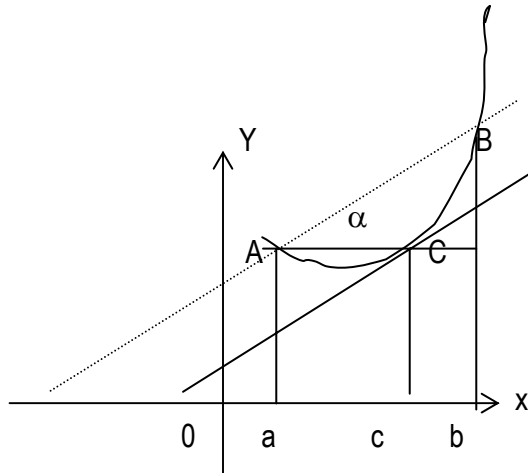


Figure (i)

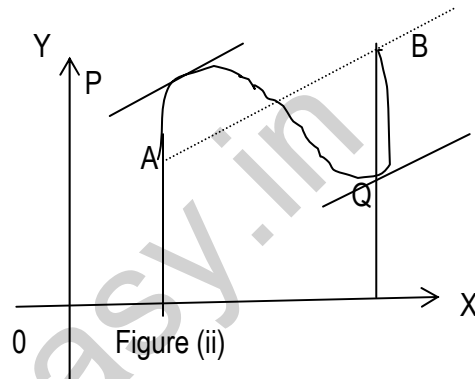


Figure (ii)

Slope of the line joining the points A $(a, f(a))$ and B $(b, f(b))$

$$\text{Is } \frac{f(b) - f(a)}{b - a} \quad (\because \text{Slope} = m = \tan \theta)$$

$$= \tan \alpha$$

Where α is the angle made by the line AB with x – axis

$$= \text{Slope of the tangent at } x = c$$

$$= f'(c), \text{ where } a < c < b$$

Geometrically, it means that there exists at least one value of $x = c$, where $a < c < b$ at which the tangent will be parallel to the line joining the end points at $x = a$ & $x = b$.

Note: There can be more than one value at which the tangents are parallel to the line joining points A & B (from Fig (ii)).

10. Verify Lagrange's Mean value theorem for $f(x) = (x - 1)(x - 2)(x - 3)$ in $[0,4]$.

Solution: Clearly given function is continuous in $[0,4]$ and differentiable in $(0,4)$, because $f(x)$ is in polynomial.

$$f(x) = (x - 1)(x - 2)(x - 3)$$

$$f(x) = x^3 - 6x^2 + 11x - 6$$

$$\text{and } f(0) = 0^3 - 6(0)^2 + 11(0) - 6 = -6$$

$$f(4) = 4^3 - 6(4)^2 + 11(4) - 6 = 6$$

Differentiate $f(x)$ W.r.t x , we get

$$f'(x) = 3x^2 - 6x + 11$$

$$\text{Let } x = c, f'(c) = 3c^2 - 6c + 11$$

By Lagrange's Mean value theorem, we have

$$\begin{aligned} f'(c) &= \frac{f(b) - f(a)}{b - a} = \frac{f(4) - f(0)}{(4 - 0)} \\ &= \frac{6 - (-6)}{4} = 3 \end{aligned}$$

$$\therefore 3c^2 - 6c + 11 = 3$$

$$\Rightarrow 3c^2 - 6c + 8 = 0$$

Solving this equation, we get

$$c = 2 \pm \frac{2}{\sqrt{3}} \in (0,4)$$

Hence the function is verified.

11. Verify the Lagrange's Mean value theorem for $f(x) = \log x$ in $[1,e]$.

Solution: Now $\log x$ is continuous for all $x > 0$ and hence $[1,e]$.

Also $f'(x) = \frac{1}{x}$ which exists for all x in $(1,e)$

Hence $f(x)$ is differentiable in $(1,e)$

$\therefore$ by Lagrange's Mean Value theorem, we get

$$\frac{\text{Log} e - \text{Log} 1}{e-1} = \frac{1}{c} \Rightarrow \frac{1}{e-1} = \frac{1}{c}$$

$$\Rightarrow C = e - 1$$

$$\Rightarrow 1 < e - 1 < 2 < e$$

Since $e \in (2, 3)$

$\therefore$ So that $c = e - 1$ lies between 1 & e

Hence the Theorem.

12. Find θ for $f(x) = Lx^2 + mx + n$ by Lagrange's Mean Value theorem.

Solution: $f(x) = Lx^2 + mx + n$

$$\therefore f'(x) = 2Lx + m$$

$$\text{We have } f(a+h) = f(a) + hf'(a+\theta h)$$

$$\text{Or } f(a+h) - f(a) = hf'(a+\theta h)$$

$$\text{i.e., } \{L(a+h)^2 + m(a+h) + n\} - \{La^2 + ma + n\} = h\{2L(a+\theta h) + m\}$$

Comparing the Co-efficient of h^2 , we get

$$1 = 2\theta \quad \therefore \theta = \frac{1}{2} \in (0, 1)$$

Exercise:

13. Verify the Lagrange's Mean Value theorem for $f(x) = \sin^2 x$ in $\left[0, \frac{\pi}{2}\right]$

14. Prove that, $\frac{b-a}{1+b^2} < \tan^{-1} b - \tan^{-1} a < \frac{b-a}{1+a^2}$ if $0 < a < b$ and reduce that

$$\frac{\pi}{4} + \frac{3}{25} < \tan^{-1} \frac{4}{3} < \frac{\pi}{4} + \frac{1}{6}$$

15. Show that $\frac{2}{\pi} < \frac{\sin x}{x} < 1$ in $\left(0, \frac{\pi}{2}\right)$

16. Prove that $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$ Where $a < b$. Hence reduce

$$\frac{\pi}{6} - \frac{1}{2\sqrt{3}} < \sin^{-1} \frac{1}{4} < \frac{\pi}{6} - \frac{1}{\sqrt{15}}$$

17. State & prove Cauchy's Mean Value Theorem with Geometric meaning.

Proof: The ratio of the increments of two functions called Cauchy's Theorem.

Statement: Let $f(x)$ and $g(x)$ be two functions of x such that,

- (i) Both $f(x)$ and $g(x)$ are continuous in $[a, b]$
- (ii) Both $f(x)$ and $g(x)$ are differentiable in (a, b)
- (iii) $g'(x) \neq 0$ for any $x \in (a, b)$

These three conditions exist at least at one value $x = c \in (a, b)$ at which

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

Proof: Define a function,

$$\phi(x) = f(x) - A \cdot g(x) \quad \text{----- (1)}$$

So that $\phi(a) = \phi(b)$ and A is a Constant to be determined.

$$\text{Now, } \phi(a) = f(a) - A \cdot g(a)$$

$$\phi(b) = f(b) - A \cdot g(b)$$

$$\therefore f(a) - A \cdot g(a) = f(b) - A \cdot g(b)$$

$$\Rightarrow A = \frac{f(b) - f(a)}{g(b) - g(a)} \quad \text{----- (2)}$$

Now, ϕ is continuous in $[a, b]$ as r.h.s of (1) is continuous in $[a, b]$ and $\phi(x)$ is differentiable in (a, b) as r.h.s of (1) is differentiable in $[a, b]$.

$$\text{Also } \phi(a) = \phi(b)$$

Hence all the conditions of Rolle's Theorem are satisfied then there exists a value $x = c \in (a, b)$ such that $\phi'(c) = 0$.

Now, Differentiating (1) W.r.t x , we get

$$\phi'(x) = f'(x) - A \cdot g'(x)$$

$$\text{at } x = c \in (a, b)$$

$$\therefore \phi^1(c) = f^1(c) - A g^1(c)$$

$$0 = f^1(c) - A g^1(c) \quad (\because g^1(x) \neq 0)$$

$$\Rightarrow A = \frac{f^1(c)}{g^1(c)} \quad \text{----- (3)}$$

Substituting (3) in (2), we get

$$\frac{f^1(c)}{g^1(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}, \text{ where } a < c < b$$

Hence the proof.

18. Verify Cauchy's Mean Value Theorem for the function $f(x) = x^2 + 3$, $g(x) = x^3 + 1$ in $[1, 3]$

Solution: Here $f(x) = x^2 + 3$, $g(x) = x^3 + 1$

Both $f(x)$ and $g(x)$ are Polynomial in x . Hence they are continuous and differentiable for all x and in particular in $[1, 3]$

$$\text{Now, } f^1(x) = 2x, g^1(x) = 3x^2$$

$$\text{Also } g^1(x) \neq 0 \text{ for all } x \in (1, 3)$$

Hence $f(x)$ and $g(x)$ satisfy all the conditions of the Cauchy's mean value theorem. Therefore

$$\frac{f(3) - f(1)}{g(3) - g(1)} = \frac{f^1(c)}{g^1(c)}, \text{ for some } c : 1 < c < 3$$

$$\text{i.e., } \frac{12 - 4}{28 - 2} = \frac{26}{3c^2}$$

$$\text{i.e., } \frac{2}{13} = \frac{1}{3c} \Rightarrow C = \frac{13}{6} = 2\frac{1}{6}$$

Clearly $C = 2\frac{1}{6}$ lies between 1 and 3.

Hence Cauchy's theorem holds good for the given function.

19. Verify Cauchy's Mean Value Theorem for the functions $f(x) = \sin x$, $g(x) = \cos x$ in $\left[0, \frac{\pi}{2}\right]$

Solution: Here $f(x) = \sin x$, $g(x) = \cos x$ so that

$$f'(x) = \cos x, g'(x) = -\sin x$$

Clearly both $f(x)$ and $g(x)$ are continuous in $\left[0, \frac{\pi}{2}\right]$, and differentiable in $\left(0, \frac{\pi}{2}\right)$

Also $g'(x) = -\sin x \neq 0$ for all $x \in \left(0, \frac{\pi}{2}\right)$

$\therefore$ From Cauchy's mean value theorem we obtain

$$\frac{f\left(\frac{\pi}{2}\right) - f(0)}{g\left(\frac{\pi}{2}\right) - g(0)} = \frac{f'(c)}{g'(c)} \text{ for some } C : 0 < C < \frac{\pi}{2}$$

$$\text{i.e., } \frac{1-0}{0-1} = \frac{\cos c}{-\sin c} \quad \text{i.e., } -1 = -\cot c \text{ (or) } \cot c = 1$$

$$\therefore C = \frac{\pi}{4}, \text{ clearly } C = \frac{\pi}{4} \text{ lies between } 0 \text{ and } \frac{\pi}{2}$$

Thus Cauchy's Theorem is verified.

Exercises:

20. Find C by Cauchy's Mean Value Theorem for

a) $f(x) = e^x, g(x) = e^{-x}$ in $[0,1]$

b) $f(x) = x^2, g(x) = x$ in $[2,3]$

21. Verify Cauchy's Mean Value theorem for

a) $f(x) = \tan^{-1} x, g(x) = x$ in $\left[\frac{1}{\sqrt{3}}, 1\right]$

b) $f(x) = \log x, g(x) = \frac{1}{x}$ in $[1,e]$

Generalized Mean Value Theorem:

22. State Taylor's Theorem and hence obtain Maclaurin's expansion (series)

Statement: If $f(x)$ and its first $(n - 1)$ derivatives are continuous in $[a, b]$ and its n^{th} derivative exists in (a, b) then

$$f(b) = f(a) + (b - a) f'(a) + \frac{(b - a)^2}{2!} f''(a) + \dots + \frac{(b - a)^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{(b - a)^n}{n!} f^{(n)}(c)$$

Where $a < c < b$

Remainder in Taylor's Theorem:

We have

$$f(x) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots + \frac{(x - a)^{n-1}}{(n-1)!} f^{(n-1)}(a) + \frac{(x - a)^n}{n!} f^{(n)}[a + (x - a)\theta]$$

$$f(x) = S_n(x) + R_n(x)$$

Where $R_n(x) = \frac{(x - a)^n}{n!} f^{(n)}[a + (x - a)\theta]$ is called the Lagrange's form of the Remainder.

$$\text{Where } a = 0, R_n(x) = \frac{x^n}{n!} f^{(n)}(\theta x), 0 < \theta < 1$$

Taylor's and Maclaurin's Series:

$$\text{We have } f(x) = S_n(x) + R_n(x)$$

$$\therefore \lim_{n \rightarrow \infty} [f(x) - S_n(x)] = \lim_{n \rightarrow \infty} R_n(x)$$

$$\text{If } \lim_{n \rightarrow \infty} R_n(x) = 0 \text{ then } f(x) = \lim_{n \rightarrow \infty} S_n(x)$$

Thus $\lim_{n \rightarrow \infty} S_n(x)$ converges and its sum is $f(x)$.

This implies that $f(x)$ can be expressed as an infinite series.

$$\text{i.e., } f(x) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots \text{ to } \infty$$

This is called Taylor's Series.

Putting $a = 0$, in the above series, we get

$$F(x) = f(0) + x f'(0) + \frac{(x)^2}{2!} f''(0) + \dots \text{to } \infty$$

This is called Maclaurin's Series. This can also denoted as

$$Y = y(0) + x y_1(0) + \frac{(x)^2}{2!} y_2(0) + \dots + \frac{(x)^n}{n!} y_n(0) \dots \text{to } \infty$$

Where $y = f(x)$, $y_1 = f_1(x)$, $\dots$ $y_n = f_n(x)$

23. By using Taylor's Theorem expand the function e^x in ascending powers of $(x - 1)$

Solution: The Taylor's Theorem for the function $f(x)$ is ascending powers of $(x - a)$ is

$$f(x) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \dots \quad (1)$$

Here $f(x) = e^x$ and $a = 1$

$$f'(x) = e^x \Rightarrow f'(a) = e$$

$$f''(x) = e^x \Rightarrow f''(a) = e$$

$\therefore$ (1) becomes

$$e^x = e + (x - 1)e + \frac{(x - 1)^2}{2} e + \dots$$

$$= e \left\{ 1 + (x - 1) + \frac{(x - 1)^2}{2} + \dots \right\}$$

24. By using Taylor's Theorem expand $\log \sin x$ in ascending powers of $(x - 3)$

Solution: $f(x) = \log \sin x$, $a = 3$ and $f(3) = \log \sin 3$

$$\text{Now } f'(x) = \frac{\cos x}{\sin x} = \cot x, f'(3) = \cot 3$$

$$f''(x) = -\operatorname{cosec}^2 x, f''(3) = -\operatorname{cosec}^2 3$$

$$f'''(x) = -2\operatorname{cosec} x (-\operatorname{cosec} x \cot x) = 2\operatorname{cosec}^3 x \cot x$$

$$\therefore f'''(3) = 2\operatorname{cosec}^3 3 \cot 3$$

$$\therefore f(x) = f(a) + (x - a) f'(a) + \frac{(x - a)^2}{2!} f''(a) + \frac{(x - a)^3}{3!} f'''(a) + \dots$$

$$\therefore \log \sin x = f(3) + (x - 3) f'(3) + \frac{(x - 3)^2}{2!} f''(3) + \frac{(x - 3)^3}{3!} f'''(3) + \dots$$

$$= \log \sin 3 + (x - 3) \cot 3 + \frac{(x - 3)^2}{2!} (-\operatorname{cosec}^2 3) + \frac{(x - 3)^3}{3!} 2 \operatorname{cosec}^3 3 \cot 3 + \dots$$

Exercise:

25. Expand $\sin x$ in ascending powers of $\left(x - \frac{\pi}{2}\right)$
26. Express $\tan^{-1} x$ in powers of $(x - 1)$ up to the term containing $(x - 1)^3$
27. Apply Taylor's Theorem to prove

$$e^{x+h} = e^x \left[1 + h + \frac{h^2}{2!} + \frac{h^3}{3!} + \dots \right]$$

Problems on Maclaurin's Expansion:

28. Expand the $\log(1+x)$ as a power series by using Maclaurin's theorem.

Solution: Here $f(x) = \log(1+x)$, Hence $f(0) = \log 1 = 0$

We know that

$$f^n(x) = \frac{d^n}{dx^n} \{\log(1+x)\} = \frac{d^{n-1}}{dx^{n-1}} \left\{ \frac{1}{1+x} \right\}$$

$$= \frac{(-1)^{n-1} (n-1)!}{(1+x)^n}, \quad n = 1, 2, \dots$$

Hence $f^n(0) = (-1)^{n-1} (n-1)!$

$$f^1(0) = 1, f^{11}(0) = -1, f^{111}(0) = 3!, f^{1111}(0) = -3!$$

Substituting these values in

$$f(x) = f(0) + x f^1(0) + \frac{x^2}{2!} f^{11}(0) + \dots + \frac{x^n}{n!} f^n(0) + \dots$$

$$\therefore \log(1+x) = 0 + x \cdot 1 + \frac{x^2}{2!} (-1) + \frac{x^3}{3!} 2! + \frac{x^4}{4!} - 3! + \dots$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

This series is called Logarithmic Series.

29. Expand $\tan^{-1} x$ by using Maclaurin's Theorem up to the term containing x^5

Solution: let $y = \tan^{-1} x$, Hence $y(0) = 0$

We find that $y_1 = \frac{1}{1+x^2}$ which gives $y_1(0) = 1$

Further $y_1(1+x^2) = 1$, Differentiating we get

$$y_1 \cdot 2x + (1+x^2) y_2 = 0 \text{ (or) } (1+x^2) y_2 + 2xy_1 = 0$$

$$\text{Hence } y_2(0) = 0$$

Taking n^{th} derivative on both sides by using Leibniz's Theorem, we get

$$(1+x^2) y_{n+2} + n \cdot 2xy_{n+1} + \frac{n(n-1)}{1.2} \cdot 2 \cdot y_n + 2xy_{n-1} + n \cdot 2 \cdot y_n = 0$$

$$\text{i.e., } (1+x^2) y_{n+2} + 2(n+1)xy_{n+1} + n(n+1)y_n = 0$$

Substituting $x = 0$, we get, $y_{n+2}(0) = -n(n+1)y_n(0)$

$$\text{For } n = 1, \text{ we get } y_3(0) = -2y_1(0) = -2$$

$$\text{For } n = 2, \text{ we get } y_4(0) = -2 \cdot 3 \cdot y_2(0) = 0$$

$$\text{For } n = 3, \text{ we get } y_5(0) = -3 \cdot 4 \cdot y_3(0) = 24$$

Using the formula

$$Y = y(0) + xy_1(0) + \frac{x^2}{2!} y_2(0) + \frac{x^3}{3!} y_3(0) + \dots$$

$$\text{We get } \tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$$

Exercise:

30. Using Maclaurin's Theorem prove the following:

a) $\sec x = 1 + \frac{x^2}{2!} + \frac{5x^4}{4!} + \dots$

b) $\sin^{-1} x = x + \frac{x^3}{6} + \frac{3x^5}{40} + \dots$

c) $e^x \cos x = 1 + x - \frac{x^3}{3} + \dots$

d) Expand $e^{ax} \cos bx$ by Maclaurin's Theorem as far as the term containing x^3

Exercise : Verify Rolle's Theorem for

(i) $f(x) = e^x (\sin x - \cos x)$ in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$,

(ii) $f(x) = x(x-2)e^{x/2}$ in $[0, 2]$

(iii) $f(x) = \frac{\sin 2x}{e^{2x}}$ in $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$.

Exercise : Verify the Lagrange's Mean Value Theorem for

(i) $f(x) = x(x-1)(x-2)$ in $\left[0, \frac{1}{2}\right]$

(ii) $f(x) = \tan^{-1} x$ in $[0, 1]$

Exercise : Verify the Cauchy's Mean Value Theorem for

(i) $f(x) = \sqrt{x}$ and $g(x) = \frac{1}{\sqrt{x}}$ in $\left[\frac{1}{4}, 1\right]$

(ii) $f(x) = \frac{1}{x^2}$ and $g(x) = \frac{1}{x}$ in $[a, b]$

(iii) $f(x) = \sin x$ and $g(x) = \cos x$ in $[a, b]$

UNIT – II

DIFFERENTIAL CALCULUS-II

Give different types of Indeterminate Forms.

If $f(x)$ and $g(x)$ be two functions such that $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exists, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$$

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ then

$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{0}{0}$ Which do not have any definite value, such an expression is called

indeterminate form. The other indeterminate forms are $\frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 0^0 , ∞^0 and 1^∞

1. State & prove L' Hospital's Theorem (rule) for Indeterminate Forms.

L'Hospital rule is applicable when the given expression is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

Statement: Let $f(x)$ and $g(x)$ be two functions such that

$$(1) \lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0$$

$$(2) f'(a) \text{ and } g'(a) \text{ exist and } g'(a) \neq 0$$

$$\text{Then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)}$$

Proof: Now $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \left[\frac{1/g(x)}{1/f(x)} \right]$, which takes the indeterminate form $\frac{0}{0}$. Hence applying the

L' Hospitals theorem, we get

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{-g^1(x)/[g(x)]^2}{-f^1(x)/[f(x)]^2} = \lim_{x \rightarrow a} \frac{g^1(x)}{f^1(x)} \left[\frac{f(x)}{g(x)} \right]^2 \\ &= \left[\lim_{x \rightarrow a} \frac{g^1(x)}{f^1(x)} \right] \left[\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \right]^2 \end{aligned}$$

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \neq 0$ and $\neq \infty$ then

$$1 = \left[\lim_{x \rightarrow a} \frac{g^1(x)}{f^1(x)} \right] \left[\lim_{x \rightarrow a} \frac{f(x)}{g(x)} \right]$$

i.e. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f^1(x)}{g^1(x)}$

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = 0$ or ∞ the above theorem still holds good.

2. Evaluate $\lim_{x \rightarrow a} \frac{\sin x}{x} = \frac{0}{0}$ form

Solution: Apply L'Hospital rule, we get

$$\lim_{x \rightarrow a} \frac{\cos x}{1} = \frac{\cos \theta}{1} = \frac{1}{1} = 1$$

$$\therefore \lim_{x \rightarrow a} \frac{\sin x}{x} = 1$$

3. Evaluate $\lim_{x \rightarrow a} \frac{\log \sin x}{\cot x}$

Solution: $\lim_{x \rightarrow a} \frac{\log \sin x}{\cot x} = \frac{\log \sin 0}{\cot 0} = \frac{\log 0}{\infty} = \frac{-\infty}{\infty}$ form

Apply L' Hospital rule

$$= \lim_{x \rightarrow a} \frac{-\operatorname{Cosec}^2 x}{-2 \operatorname{Cosec} x \operatorname{Cosec} x \cot x}$$

$$= \lim_{x \rightarrow a} \frac{1}{2 \cot x} = 0$$

$$\therefore \lim_{x \rightarrow a} \frac{\log \sin x}{\cot x} = 0$$

Exercise: 1

Evaluate

a) $\lim_{x \rightarrow 0} \frac{\tan x}{x}$

b) $\lim_{x \rightarrow 0} (1+x)^{1/x}$

c) $\lim_{x \rightarrow \infty} \frac{a^x - 1}{x}$

d) $\lim_{x \rightarrow 0} \frac{x^n - a^n}{x - a}$

4. Explain $\infty - \infty$ and $0 \times \infty$ Forms:

Solution: Suppose $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ in this case

$$\lim_{x \rightarrow a} f(x) - g(x) = 0 \times \infty, \text{ reduce this to } \frac{0}{0} \text{ or } \frac{\infty}{\infty} \text{ form}$$

$$\text{Let } \lim_{x \rightarrow a} [f(x).g(x)] = \lim_{x \rightarrow a} \left\{ \frac{f(x)}{1/g(x)} \right\} = \frac{0}{0} \text{ form}$$

$$\text{Or } \lim_{x \rightarrow a} [f(x).g(x)] = \lim_{x \rightarrow a} \left\{ \frac{g(x)}{1/f(x)} \right\} = \frac{\infty}{\infty} \text{ form}$$

L' Hospitals rule can be applied in either case to get the limit.

Suppose $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$ in this case $\lim_{x \rightarrow a} [f(x).g(x)] = \infty - \infty$ form, reduce

this $\frac{0}{0}$ or $\frac{\infty}{\infty}$ form and then apply L'Hospitals rule to get the limit

5. Evaluate $\lim_{x \rightarrow 0} \left\{ \frac{1}{x} - \frac{\log(1+x)}{x^2} \right\}$

Solution: Given $\lim_{x \rightarrow 0} \left\{ \frac{1}{x} - \frac{\log(1+x)}{x^2} \right\} = \infty - \infty$ form

$\therefore$ Required limit = $\lim_{x \rightarrow 0} \left\{ \frac{x - \log(1+x)}{x^2} \right\} = \frac{0}{0}$ form

Apply L'Hospital rule.

$$= \lim_{x \rightarrow 0} \frac{1 - \frac{1}{1+x}}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{x}{1+x}}{2x} = \lim_{x \rightarrow 0} \frac{1}{2(1+x)} = \frac{1}{2}$$

6. Evaluate $\lim_{x \rightarrow 0} \left\{ \frac{1}{x} - \cot x \right\}$

Solution: Given limit is $\infty - \infty$ form at $x = 0$. Hence we have

Required limit = $\lim_{x \rightarrow 0} \left\{ \frac{1}{x} - \frac{\cos x}{\sin x} \right\}$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin x - x \cos x}{x \sin x} \right) = \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospital's rule

$$= \lim_{x \rightarrow 0} \frac{\cos x - \cos x + x \sin x}{x \cos x + \sin x}$$

$$= \lim_{x \rightarrow 0} \frac{x \sin x}{x \cos x + \sin x} = \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospitals rule

$$= \lim_{x \rightarrow 0} \frac{x \cos x + \sin x}{\cos x - x \sin x + \cos x}$$

$$= \frac{0}{2-0} = \frac{0}{2} = 0$$

7. Evaluate $\lim_{x \rightarrow 0} \tan x \log x$

Solution: Given limit is $(0 \times -\infty)$ form at $x = 0$

$$\therefore \text{Required limit} = \lim_{x \rightarrow 0} \frac{\log x}{\cot x} = \left[\frac{-\infty}{\infty} \right] \text{ form}$$

Apply L' Hospitals rule

$$= \lim_{x \rightarrow 0} \frac{1/x}{-\operatorname{Cosec}^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{-\sin^2 x}{x} \left(\frac{0}{0} \right) \text{ form}$$

Apply L Hospitals rule

$$= \lim_{x \rightarrow 0} \frac{-2 \sin x \cos x}{1} = 0$$

8. Evaluate $\lim_{x \rightarrow 1} \sec\left(\frac{\pi x}{2}\right) \cdot \log x$

Solution: Given limit is $(\infty \times 0)$ form at $x = 1$

$$\therefore \text{Required limit} = \lim_{x \rightarrow 1} \frac{\log x}{\cos \frac{\pi x}{2}} \frac{0}{0} \text{ form}$$

Apply L' Hospitals rule

$$= \lim_{x \rightarrow 1} \frac{1/x}{-\sin \frac{\pi x}{2} \cdot \frac{\pi}{2}} = -\frac{2}{\pi}$$

Exercise: 2

Evaluate

a) $\lim_{x \rightarrow 1} \left(\frac{x}{x-1} - \frac{1}{\log x} \right)$

b) $\lim_{x \rightarrow 0} \left(\frac{a}{x} - \cot \left(\frac{x}{a} \right) \right)$

c) $\lim_{x \rightarrow \frac{\pi}{2}} \left(\sec x - \frac{1}{1 - \sin x} \right)$

d) $\lim_{x \rightarrow \infty} \left(a^{1/x} - 1 \right) x$

9. Explain Indeterminate Forms 0^0 , 1^∞ , ∞^0 , 0^∞

Solution: At $x = a$, $[f(x)]^{g(x)}$ takes the indeterminate form

(i) 0^0 if $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$

(ii) 1^∞ if $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \infty$

(iii) ∞^0 if $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = 0$ and

(iv) 0^∞ if $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$

In all these cases the following method is adopted to evaluate $\lim_{x \rightarrow a} [f(x)]^{g(x)}$

Let $L = \lim_{x \rightarrow a} [f(x)]^{g(x)}$ so that

$$\log L = \lim_{x \rightarrow a} [g(x) \log f(x)] = 0 \times \infty$$

Reducing this to $\frac{0}{0}$ or $\frac{\infty}{\infty}$ and applying L' Hospitals rule, we get $\log L = a$ Or

$$L = e^a$$

10. Evaluate $\lim_{x \rightarrow 0} x^{\sin x}$

Solution: let $L = \lim_{x \rightarrow 0} x^{\sin x} \Rightarrow 0^0$ form at $x = 0$

Hence $\log L = \lim_{x \rightarrow 0} \sin x \log x \Rightarrow 0 \times \infty$ form

$$\therefore \log L = \lim_{x \rightarrow 0} \frac{\log x}{\frac{1}{\sin x}} = \lim_{x \rightarrow 0} \frac{\log x}{\cos x} \left(\frac{\infty}{\infty} \right) \text{ form}$$

Apply L' Hospital rule,

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\cos x \cot x} = \lim_{x \rightarrow 0} \frac{-\sin x \cdot \tan x}{x} \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospitals rule we get

$$= \lim_{x \rightarrow 0} \frac{\sin x \sec^2 x - \cos x \tan x}{1} = -\infty$$

$$\therefore \log L = -\infty \Rightarrow L = e^{-\infty} = \frac{1}{e^{\infty}} = \frac{1}{\infty} = 0$$

$$\therefore L = 0$$

11. Evaluate $\lim_{x \rightarrow 1} (x)^{\frac{1}{1-x}}$

Solution: let $L = \lim_{x \rightarrow 1} (x)^{\frac{1}{1-x}}$ is 1^∞ form

$$\therefore \log L = \lim_{x \rightarrow 1} \left(\frac{1}{1-x} \log x \right) = \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospitals rule

$$= \lim_{x \rightarrow 1} \frac{\frac{1}{x}}{-1} = \lim_{x \rightarrow 1} \frac{1}{-x} = -1$$

$$\therefore \text{Log } L = -1$$

$$\Rightarrow L = e^{-1} = \frac{1}{e}$$

12. Evaluate $\lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2}$

Solution: let $L = \lim_{x \rightarrow 0} \left(\frac{\tan x}{x} \right)^{1/x^2} \equiv 1^\infty$ form

$$\therefore \text{Log}L = \lim_{x \rightarrow 0} \left\{ \frac{1}{x^2} \log \left(\frac{\tan x}{x} \right) \right\} \equiv (\infty \times 0) \text{ form}$$

$$\therefore \text{Log}L = \lim_{x \rightarrow 0} \left\{ \frac{\log(\tan x/x)}{x^2} \right\} \equiv \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospitals rule

$$= \lim_{x \rightarrow 0} \left\{ \frac{\frac{1}{\tan x}}{\frac{x \sec^2 x - \tan x}{x^2}} \right\} \quad \text{--- } 2x$$

$$LogL = \frac{1}{2} \lim_{x \rightarrow 0} \left[\frac{x \sec^2 x - \tan x}{x^3} \right] = \left(\frac{0}{0} \right) \text{ form}$$

Apply L' Hospital rule, we get

$$= \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sec^2 x + 2x \sec^2 x \tan x - \sec^2 x}{3x^2}$$

$$= \frac{1}{3} \lim_{x \rightarrow 0} (\sec^2 x) \left(\frac{\tan x}{x} \right)$$

$$\text{Log } L = \frac{1}{3}$$

$$\therefore L = e^{1/3}$$

Exercise: 3

Evaluate the following limits.

a) $\lim_{x \rightarrow 0} (\sec x)^{\cot x}$

b) $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right)^{\tan\left(\frac{\pi x}{2a}\right)}$

c) $\lim_{x \rightarrow 0} \frac{(1+x)^{1/x} - e}{x}$

d) $\lim_{x \rightarrow 0} (\cos ax)^{\infty/x^2}$

(i) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2 \log(1+x)}{x \sin x}$

(ii) $\lim_{x \rightarrow 0} \frac{\log(1+x^3)}{\sin^3 x}$

(iii) $\lim_{x \rightarrow 0} \frac{1 + \sin x - \cos x + \log(1-x)}{x \tan^2 x}$

(iv) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\log \sin x}{\left(x - \frac{\pi}{2}\right)^2}$

(v) $\lim_{x \rightarrow 0} \frac{\cosh x + \log(1-x) - 1 + x}{x^2}$

(vi) $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x \sin^{-1} x}{x^2}$

(vii) $\lim_{x \rightarrow 0} \frac{e^{2x} - (1+x)^2}{x \log(1+x)}$

Evaluate the following limits.

(i) $\lim_{x \rightarrow a} \left(2 - \frac{x}{a} \right) \cot(x-a)$

(ii) $\lim_{x \rightarrow 0} (\operatorname{cosec} x - \cot x)$

(iii) $\lim_{x \rightarrow \frac{\pi}{2}} \left[x \tan x - \frac{\pi}{2} \sec x \right]$

(iv) $\lim_{x \rightarrow 0} \left(\cot^2 x - \frac{1}{x^2} \right)$

(v) $\lim_{x \rightarrow 0} \left[\frac{1}{x^2} - \frac{1}{x \tan x} \right]$

(vi) $\lim_{x \rightarrow \frac{\pi}{2}} [2x \tan x - \pi \sec x]$

(i) $\lim_{x \rightarrow 0} (\cos ax)^{b/x^2}$

(ii) $\lim_{x \rightarrow 0} \left(\frac{1 + \cos x}{2} \right)^{1/x^2}$

$$(iii) \lim_{x \rightarrow 1} (1 - x^2)^{\frac{1}{\log(1-x)}}$$

$$(iv) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

$$(v) \lim_{x \rightarrow 0} (\sin x)^{\tan x}$$

$$(iv) \lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$$

$$(vii) \lim_{x \rightarrow 0} (\cos x)^{\sec^2 x}$$

$$(viii) \lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\tan 2x}$$

$$(ix) \lim_{x \rightarrow \infty} \left(\frac{ax+1}{ax-1} \right)^x$$

$$(x) \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{1}{x}}$$

Evaluate the following limits.

$$(i) \lim_{x \rightarrow 0} (\cos ax)^{\frac{b}{x^2}}$$

$$(ii) \lim_{x \rightarrow 0} \left(\frac{1 + \cos x}{2} \right)^{\frac{1}{x^2}}$$

$$(iii) \lim_{x \rightarrow 1} (1 - x^2)^{\frac{1}{\log(1-x)}}$$

$$(iv) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$$

$$(v) \lim_{x \rightarrow 0} (\sin x)^{\tan x}$$

$$(iv) \lim_{x \rightarrow 0} (1 + \sin x)^{\cot x}$$

$$(vii) \lim_{x \rightarrow 0} (\cos x)^{\sec^2 x}$$

$$(viii) \lim_{x \rightarrow \frac{\pi}{4}} (\tan x)^{\tan 2x}$$

$$(ix) \lim_{x \rightarrow \infty} \left(\frac{ax+1}{ax-1} \right)^x$$

$$(x) \lim_{x \rightarrow 0} \left(\frac{a^x + b^x + c^x}{3} \right)^{\frac{1}{x}}$$

Polar Curves

If we traverse in a hill section where the road is not straight, we often see caution boards hairpin bend ahead, sharp bend ahead etc. This gives an indication of the difference in the amount of bending of a road at various points which is the curvature at various points. In this chapter we discuss about the curvature, radius of curvature etc.

Consider a point P in the xy-Plane.

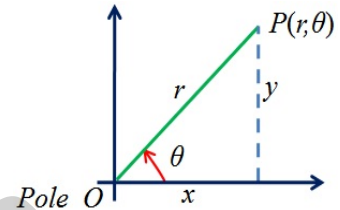
r = length of OP = radial distance

θ = Polar angle

$(r, \theta) \rightarrow$ Polar co-ordinates

Let $r = f(\theta)$ be the polar curve

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right) \quad \text{-----(1)}$$

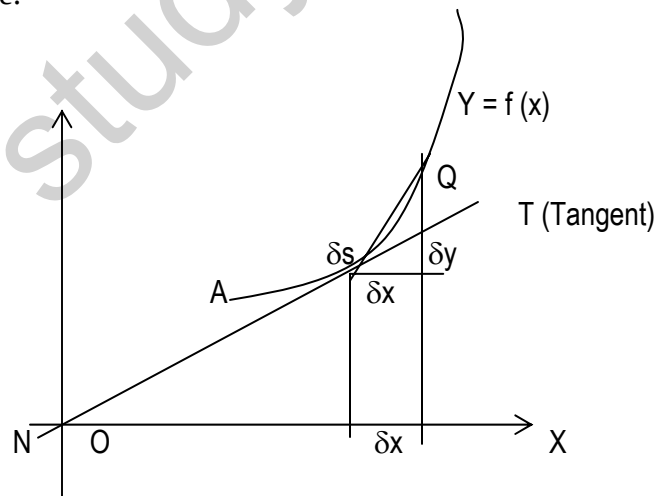


$$x = r \cos \theta \quad y = r \sin \theta$$

Relation (1) enables us to find the polar co-ordinates (r, θ) when the Cartesian co-ordinates (x, y) are known.

Expression for arc length in Cartesian form.

Proof: Let P (x, y) and Q $(x + \delta x, Y + \delta y)$ be two neighboring points on the graph of the function $y = f(x)$. So that they are at length S and $S + \delta s$ measured from a fixed point A on the curve.



From figure, $\widehat{PQ} = \delta S,$

$$\widehat{AP} = S$$

$$\angle TPR = \psi \quad \text{and} \quad PR = \delta x, RQ = \delta y$$

$$\therefore \text{Arc PQ} = \delta S$$

From $\triangle^{\text{le}} \text{PQR}$, we have

$$[\text{Chord PQ}]^2 = \text{PR}^2 + \text{QR}^2$$

$$[\text{Chord PQ}]^2 = (\delta x)^2 + (\delta y)^2 \quad (\because \text{from figure})$$

When Q is very close to point P, the length of arc PQ is equal to the length of Chord PQ.

$$\text{i.e arc PQ} = \text{Chord PQ} = \delta s$$

$$\therefore (\delta s)^2 = (\delta x)^2 + (\delta y)^2 \text{ ----- (1)}$$

$\div (\delta x)^2$, we get

$$\left(\frac{\delta s}{\delta x} \right)^2 = 1 + \left(\frac{\delta y}{\delta x} \right)^2$$

When $Q \rightarrow P$ along the curve, $\delta x \rightarrow 0$, $\delta s \rightarrow 0$

$$\therefore \lim_{\delta x \rightarrow 0} \left(\frac{\delta s}{\delta x} \right)^2 = 1 + \lim_{\delta x \rightarrow 0} \left(\frac{\delta y}{\delta x} \right)^2$$

$$\text{i.e., } \left(\frac{ds}{dx} \right)^2 = 1 + \left(\frac{dy}{dx} \right)^2$$

$$\therefore \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \text{ -----(2)}$$

Similarly, dividing (1) by δy and taking the limit as $\delta y \rightarrow 0$, we get

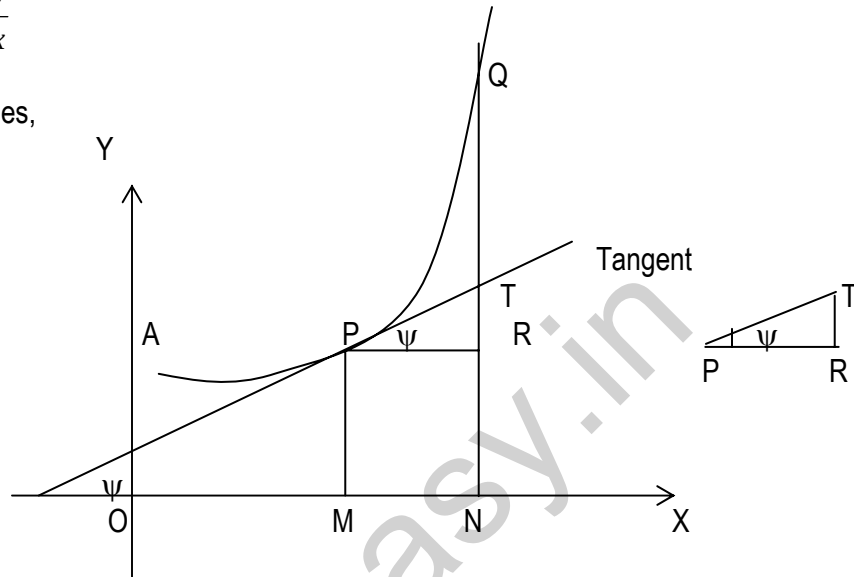
$$\frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy} \right)^2} \text{ -----(3)}$$

Expressions for $\frac{ds}{dx}$ & $\frac{ds}{dy}$

Trace a tangent to the curve at the point P, it makes an angle ψ with the x – axis. From $\triangle^{\text{le}}$ PRT, we have

$$\tan \psi = \frac{dy}{dx}$$

Equation (2) becomes,



$$\frac{ds}{dx} = \sqrt{1 + \tan^2 \psi}$$

$$= \sqrt{\sec^2 \psi} = \sec \psi$$

$$\therefore \frac{ds}{dx} = \sec \psi \text{ (Or) } \frac{dx}{ds} = \cos \psi$$

and equation (3) becomes

$$\frac{ds}{dy} = \sqrt{1 + \cot^2 \psi} = \sqrt{\operatorname{cosec}^2 \psi} = \operatorname{cosec} \psi$$

$$\therefore \frac{ds}{dy} = \operatorname{cosec} \psi \text{ or } \frac{dy}{ds} = \sin \psi$$

Derive an expression for arc length in parametric form.

Solution: Let the equation of the curve in Parametric form be $x = f(t)$ and $y = g(t)$.

We have,

$$(\delta s)^2 = (\delta x)^2 + (\delta y)^2$$

÷ by $(\delta t)^2$, we get

$$\left(\frac{\delta s}{\delta t}\right)^2 = \left(\frac{\delta x}{\delta t}\right)^2 + \left(\frac{\delta y}{\delta t}\right)^2$$

Taking the limit as $\delta t \rightarrow 0$ on both sides, we get

$$\lim_{\delta t \rightarrow 0} \left(\frac{\delta s}{\delta t}\right)^2 = \lim_{\delta t \rightarrow 0} \left(\frac{\delta x}{\delta t}\right)^2 + \lim_{\delta t \rightarrow 0} \left(\frac{\delta y}{\delta t}\right)^2$$

$$\therefore \left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2$$

$$(Or) \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \text{ ----- (4)}$$

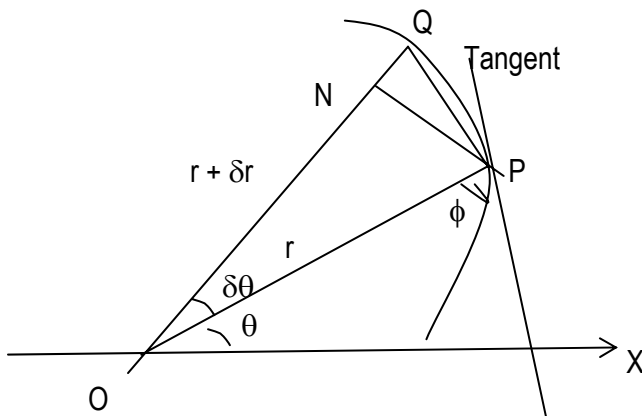
Derive an expression for arc length in Cartesian form.

Solution: Let P (r, θ) and Q $(r + \delta r, \theta + \delta \theta)$ be two neighboring points on the graph of the function $r = f(\theta)$. So that they are at lengths S and $s + \delta s$ from a fixed point A on the curve.

$$\therefore PQ = (S + \delta s) - s = \delta s$$

Draw $PN \perp OQ$

From $\triangle OPN$,



$$\frac{PN}{OP} = \sin \delta\theta$$

$$\text{i.e., } \frac{PN}{r} = \sin \delta\theta \text{ (or) } PN = r \sin \delta\theta$$

$$\text{and } \frac{ON}{OP} = \cos \delta\theta$$

$$\text{i.e. } \frac{ON}{r} = \cos \delta\theta \text{ (or) } ON = r \cos \delta\theta$$

PQ. When Q is very close to P, the length of arc PQ is equal to δs , where δs is the length of chord

In $\triangle PQN$,

$$(PQ)^2 = (PN)^2 + (QN)^2$$

$$\text{but } PN = r \sin \delta\theta \text{ (} \because \sin \delta\theta \approx \delta\theta \text{)}$$

$$\text{And } QN = OQ - ON$$

$$= (r + \delta r) - r \cos \delta\theta$$

$$= r + \delta r - r \text{ (} \because \cos \delta\theta \approx 1 \text{)}$$

$$\therefore QN = \delta r$$

$$\text{And } (PQ)^2 = (PN)^2 + (QN)^2$$

$$(\delta s)^2 + (r \delta\theta)^2 + (\delta r)^2 \text{ ----- (5)}$$

$\div$ by $(\delta\theta)^2$ we get

$$\left(\frac{\delta s}{\delta\theta} \right)^2 = r^2 + \left(\frac{\delta r}{\delta\theta} \right)^2$$

When $Q \rightarrow P$ along the curve $\delta\theta \rightarrow 0$ as $\delta s \rightarrow 0$ and $\delta r \rightarrow 0$

$$\lim_{\delta\theta \rightarrow 0} \left(\frac{\delta S}{\delta\theta} \right)^2 = r^2 + \lim_{\delta\theta \rightarrow 0} \left(\frac{\delta r}{\delta\theta} \right)^2$$

$$\text{i.e.} \left(\frac{ds}{d\theta} \right)^2 = r^2 + \left(\frac{dr}{d\theta} \right)^2$$

$$\text{i.e., } \frac{ds}{d\theta} = \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} \quad \text{----- (6)}$$

Similarly equation (5) $\div$ by $(\delta r)^2$ and taking the limits as $\delta r \rightarrow 0$ we get

$$\lim_{\delta r \rightarrow 0} \left(\frac{\delta S}{\delta r} \right)^2 = \lim_{\delta r \rightarrow 0} r^2 \left(\frac{\delta\theta}{\delta r} \right)^2 + 1$$

$$\text{i.e.} \left(\frac{ds}{dr} \right)^2 = r^2 \left(\frac{d\theta}{dr} \right)^2 + 1$$

$$\therefore \frac{ds}{dr} = \sqrt{r^2 \left(\frac{d\theta}{dr} \right)^2 + 1} \quad \text{----- (7)}$$

Note: Angle between Tangent and Radius Vector:-

We have,

$$\tan \phi = r \frac{d\theta}{dr}$$

$$\text{i.e., } \frac{\sin \phi}{\cos \phi} = r \frac{d\theta}{dr}$$

$$= r \frac{d\theta}{ds} \cdot \frac{ds}{dr}$$

$$\frac{\sin \phi}{\cos \phi} = \frac{r \frac{d\theta}{ds}}{\frac{dr}{ds}}$$

$$\therefore \sin \phi = r \frac{d\theta}{ds} \text{ and } \cos \phi = \frac{dr}{ds}$$

Find $\frac{ds}{dx}$ and $\frac{ds}{dy}$ for the following curves:-

1) $y = C \cosh\left(\frac{x}{c}\right)$

Solution: $y = C \cosh\left(\frac{x}{c}\right)$

Differentiating y w.r.t x. we get

$$\frac{dy}{dx} = \sinh\left(\frac{x}{c}\right)$$

$$\therefore \frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\equiv \sqrt{1 + \sinh^2\left(\frac{x}{c}\right)} = \sqrt{\cosh^2\left(\frac{x}{c}\right)}$$

$$\frac{ds}{dx} = \cosh\left(\frac{x}{c}\right)$$

Again Differentiating y w.r.t y we get

$$1 = C \sinh\left(\frac{x}{c}\right) \cdot \frac{1}{C} \cdot \frac{dx}{dy}$$

$$\text{i.e. } \frac{dx}{dy} = \operatorname{cosech}\left(\frac{x}{c}\right)$$

$$\therefore \frac{ds}{dy} = \sqrt{1 + \operatorname{cosec}^2\left(\frac{x}{c}\right)}$$

2) $x^3 = ay^2$

Solution : $x^3 = ay^2$

Differentiating w.r.t y and x separately we get

$$3x^2 \frac{dx}{dy} = 2ay \text{ and } 3x^2 = 2ay \frac{dy}{dx}$$

$$\text{i.e., } \frac{dx}{dy} = \frac{2ay}{3x^2} \text{ and } \frac{dy}{dx} = \frac{3x^2}{2ay}$$

We know that

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$= \sqrt{1 + \left(\frac{3x^2}{2ay}\right)^2}$$

$$= \sqrt{1 + \frac{9x^3x}{4a^2y^2}} = \sqrt{1 + \frac{9ay^2x}{4a^2y^2}}$$

$$\frac{ds}{dx} = \sqrt{1 + \frac{9x}{4a}} \text{ and}$$

$$\frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + \left(\frac{2ay}{3x^2}\right)^2}$$

$$\frac{ds}{dy} = \left(1 + \frac{4a^2y^2}{9x^4}\right)^{\frac{1}{2}} = \left(1 + \frac{4a}{9x}\right)^{\frac{1}{2}} \quad (\because x^3 = ay^2)$$

3. $y = \log \cos x$

Solution . $y = \log \cos x$

Differentiating w.r.t x and y separately, we get

$$\frac{dy}{dx} = \frac{1}{\cos x} (-\sin x) = -\tan x$$

$$\text{i.e., } \frac{dy}{dx} = -\tan x$$

$$\text{and } 1 = \frac{1}{\cos x} (-\sin x) \frac{dx}{dy}$$

$$\text{i.e., } 1 = -\tan x \frac{dx}{dy} \text{ (Or) } \frac{dx}{dy} = -\cot x$$

We have

$$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \quad \text{and} \quad \frac{ds}{dy} = \sqrt{1 + \left(\frac{dx}{dy}\right)^2}$$

$$\therefore \frac{ds}{dx} = \sqrt{1 + \tan^2 x} \quad \& \quad \frac{ds}{dy} = \sqrt{1 + \cot^2 x}$$

$$= \sqrt{\sec^2 x}$$

$$\therefore \frac{ds}{dx} = \sec x \text{ and } \frac{ds}{dy} = (1 + \cot^2 x)^{\frac{1}{2}}$$

Find $\frac{ds}{dt}$ for the following Curves:-

1. $x = a (\cos t + t \sin t), y = a (\sin t - t \cos t)$

2. $x = a \sec t, y = b \tan t$

3. $x = a \left(\cos t + \log \tan \frac{t}{2} \right), y = a \sin t$

Solution of 1

Given $x = a (\cos t + t \sin t), y = a (\sin t - t \cos t)$

Differentiating x & y W.r.t ' t ', we get

$$\frac{dx}{dt} = a (-\sin t + \sin t + t \cos t)$$

$$\frac{dx}{dt} = a t \cos t$$

$$\text{and } \frac{dy}{dt} = a (\cos t - \cos t + t \sin t)$$

$$\frac{dy}{dt} = a t \sin t$$

$$\therefore \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$= \sqrt{a^2 t^2 \cos^2 t + a^2 t^2 \sin^2 t}$$

$$\frac{ds}{dt} = at$$

Solution of 2

$$x = a \sec t, y = b \tan t$$

$$\therefore \frac{dx}{dt} = a \sec t \tan t, \frac{dy}{dt} = b \sec^2 t$$

We have

$$\therefore \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$= (a^2 \sec^2 t \tan^2 t + b^2 \sec^4 t)^{\frac{1}{2}}$$

$$= [a^2 \sec^2 t (\sec^2 t - 1) + b^2 \sec^4 t]^{\frac{1}{2}}$$

$$= [a^2 \sec^4 t - a^2 \sec^2 t + b^2 \sec^4 t]^{\frac{1}{2}}$$

$$\frac{ds}{dt} = [(a^2 + b^2) \sec^4 t - a^2 \sec^2 t]^{\frac{1}{2}}$$

Solution of 3

$$x = a (\cos t + \log \tan \frac{t}{2}), y = a \sin t$$

Differentiating x and y w.r.t 't' we get

$$\frac{dx}{dt} = a \left(-\sin t + \frac{1}{\tan \frac{t}{2}} \sec^2 \frac{t}{2} \cdot \frac{1}{2} \right), \frac{dy}{dt} = a \cos t$$

$$\therefore \frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$= \left[a^2 \left[-\sin t + \frac{\sec^2 \frac{t}{2}}{2 \tan \frac{t}{2}} \right]^2 + a^2 \cos^2 t \right]^{\frac{1}{2}}, = \left[a^2 - \frac{a^2 \sin t \sec^2 \frac{t}{2}}{\tan \frac{t}{2}} + \frac{a^2 \sec^4 \frac{t}{2}}{4 \tan^2 \frac{t}{2}} \right]$$

$$= a \cot t$$

Find $\frac{ds}{d\theta}$ and $\frac{dr}{d\theta}$ for the following curves

1. $r = a(1 - \cos \theta)$

2. $r^2 = a^2 \cos 2\theta$

3. $r = a e^{\theta \cot \alpha}$

Solution of 1

$$r = a(1 - \cos \theta)$$

Differentiating r w.r.t θ we get

$$\frac{dr}{d\theta} = a \sin \theta$$

Hence

$$\begin{aligned} \frac{ds}{d\theta} &= \left\{ r^2 + \left(\frac{dr}{d\theta} \right)^2 \right\}^{\frac{1}{2}} \\ &= \{ a^2 (1 - \cos \theta)^2 + a^2 \sin^2 \theta \}^{\frac{1}{2}} \\ &= \{ a^2 (1 - 2 \cos \theta + \cos^2 \theta) + a^2 \sin^2 \theta \}^{\frac{1}{2}} \\ &= \{ a^2 - 2a^2 \cos \theta + a^2 \}^{\frac{1}{2}} \\ &= \{ 2a^2 - 2a^2 \cos \theta \}^{\frac{1}{2}} \\ &= a \{ 2(1 - \cos \theta) \}^{\frac{1}{2}} \\ &= a \{ 2 (2 \sin^2 \frac{\theta}{2}) \}^{\frac{1}{2}} \end{aligned}$$

$$\frac{ds}{d\theta} = 2a \sin \frac{\theta}{2}$$

$$\begin{aligned} \text{And } \frac{ds}{dr} &= \left\{ 1 + r^2 \left(\frac{d\theta}{dr} \right)^2 \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + \frac{a^2 (1 - \cos \theta)^2}{a^2 \sin^2 \theta} \right\}^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned}
&= \left\{ \frac{\sin^2 \theta + (1 - \cos \theta)^2}{\sin^2 \theta} \right\}^{\frac{1}{2}} \\
&= \frac{\{2(1 - \cos \theta)\}^{\frac{1}{2}}}{\sin \theta} = \frac{2 \sin\left(\frac{\theta}{2}\right)}{\sin \theta} = \frac{2 \sin \frac{\theta}{2}}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \\
\frac{ds}{dr} &= \frac{1}{\cos \frac{\theta}{2}}
\end{aligned}$$

Solution of 2

$$r^2 = a^2 \cos 2\theta$$

Differentiating W.r.t 'θ' we get

$$2r \frac{dr}{d\theta} = -a^2 \sin 2\theta \cdot 2$$

$$r \frac{dr}{d\theta} = -a^2 \sin 2\theta$$

$$\frac{dr}{d\theta} = \frac{-a^2 \sin 2\theta}{r}$$

Hence

$$\begin{aligned}
\frac{ds}{d\theta} &= \left\{ r^2 + \left(\frac{dr}{d\theta} \right)^2 \right\}^{\frac{1}{2}} \\
&= \left\{ r^2 + \left(\frac{-a^2 \sin 2\theta}{r} \right)^2 \right\}^{\frac{1}{2}} \\
&= \left\{ r^2 + \frac{a^4 \sin^2 2\theta}{r^2} \right\}^{\frac{1}{2}} \\
&= \left\{ a^2 \cos 2\theta + \frac{a^4 \sin^2 \theta}{a^2 \cos 2\theta} \right\}^{\frac{1}{2}} \\
&= \left\{ \frac{a^4 \cos^2 2\theta + a^4 \sin^2 2\theta}{a^2 \cos 2\theta} \right\}^{\frac{1}{2}}
\end{aligned}$$

$$= \left\{ \frac{a^4}{a^2 \cos 2\theta} \right\}^{\frac{1}{2}} = \frac{a}{\sqrt{\cos 2\theta}}$$

$$\text{Or } \frac{ds}{d\theta} = \left\{ \frac{a^4}{r^2} \right\}^{\frac{1}{2}} = \frac{a^2}{r} \quad (\because r^2 = a^2 \cos 2\theta)$$

$$\begin{aligned} \text{And } \frac{ds}{dr} &= \left\{ 1 + r^2 \left(\frac{d\theta}{dr} \right)^2 \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + r^2 \left(\frac{r}{-a^2 \sin 2\theta} \right)^2 \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + \frac{r^4}{a^4 \sin^2 2\theta} \right\}^{\frac{1}{2}} \\ &= \left\{ \frac{a^4 \sin^2 2\theta + a^4 \cos^2 2\theta}{a^4 \sin^2 2\theta} \right\}^{\frac{1}{2}} \\ &= \left\{ \frac{a^4}{a^4 \sin^2 2\theta} \right\}^{\frac{1}{2}} = \frac{1}{\sin 2\theta} = \operatorname{Cosec} 2\theta \end{aligned}$$

Solution 3

$$r = a e^{\theta \cot \alpha}, \text{ here } \alpha \text{ is constant}$$

Differentiating w.r.t 'θ' we get

$$\frac{dr}{d\theta} = a e^{\theta \cot \alpha} \cdot \cot \alpha$$

Hence

$$\begin{aligned} \frac{ds}{d\theta} &= \left\{ r^2 + \left(\frac{dr}{d\theta} \right)^2 \right\}^{\frac{1}{2}} \\ &= \{ a^2 e^{2\theta \cot \alpha} + a^2 e^{2\theta \cot \alpha} \cot^2 \alpha \}^{\frac{1}{2}} \\ &= a e^{\theta \cot \alpha} \{ 1 + \cot^2 \alpha \}^{\frac{1}{2}} \\ &= a e^{\theta \cot \alpha} \{ \operatorname{Cosec}^2 \alpha \}^{\frac{1}{2}} \end{aligned}$$

$$\frac{ds}{d\theta} = a e^{\theta \cot \alpha} \cos \alpha$$

$$\begin{aligned} \text{and } \frac{ds}{dr} &= \left\{ 1 + r^2 \left(\frac{d\theta}{dr} \right)^2 \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + a^2 e^{2\theta \cot \alpha} \frac{1}{a^2 e^{2\theta \cot \alpha \cot^2 \alpha}} \right\}^{\frac{1}{2}} \\ &= \{1 + \tan^2 \alpha\}^{\frac{1}{2}} \\ &= \{\sec^2 \alpha\}^{\frac{1}{2}} = \sec \alpha \end{aligned}$$

Exercises:

Find $\frac{ds}{dr}$ and $\frac{ds}{d\theta}$ to the following curves.

1. $r^n = a^n \cos n\theta$

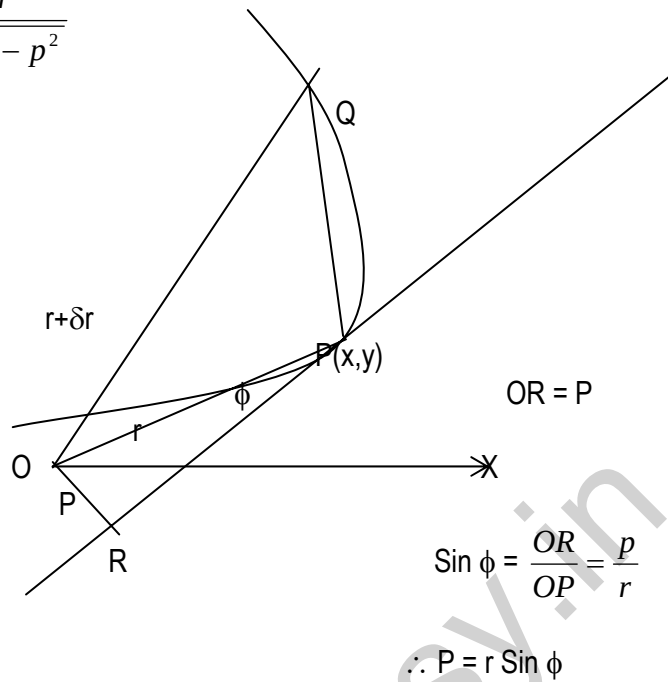
2. $r(1 + \cos \theta) = a$

3. $r\theta = a$

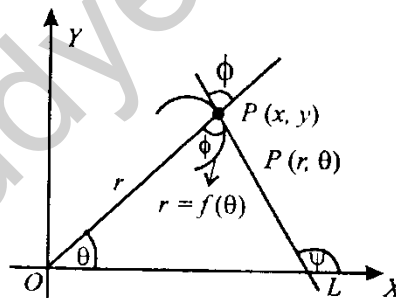
Note:

$$\begin{aligned} \text{We have } \sin \phi &= r \frac{d\theta}{ds} \text{ and} \\ \cos \phi &= \frac{dr}{ds} \\ \therefore \frac{dr}{ds} &= \cos \phi = (1 - \sin^2 \phi)^{\frac{1}{2}} \\ &= \left[1 - \frac{p^2}{r^2} \right]^{\frac{1}{2}} \text{ Since } p = r \sin \phi. \\ \frac{dr}{ds} &= \frac{\sqrt{r^2 - p^2}}{r} \end{aligned}$$

$$\therefore \frac{ds}{dr} = \frac{r}{\sqrt{r^2 - p^2}}$$



Prove that with usual notations $\tan \phi = r \left(\frac{d\theta}{dr} \right)$



Let P (r, θ) be any point on the curve $r = f(\theta)$

$\therefore \angle XOP = \theta$ and $OP = r$

Let PL be the tangent to the curve at P subtending an angle ψ with the positive direction of the initial line (x – axis) and ϕ be the angle between the radius vector OP and the tangent PL.

That is $\angle OPL = \phi$

From the figure we have

$$\psi = \phi + \theta$$

(Recall from geometry that an exterior angle is equal to the sum of the interior opposite angles)

$$\Rightarrow \tan \psi = \tan (\phi + \theta)$$

$$\text{or } \tan \psi = \frac{\tan \phi + \tan \theta}{1 - \tan \phi \tan \theta} \quad \dots(1)$$

Let (x, y) be the Cartesian coordinates of P so that we have,

$$x = r \cos \theta, \quad y = r \sin \theta$$

Since r is a function of θ , we can as well regard these as parametric equations in terms of θ .

We also know from the geometrical meaning of the derivative that

$$\tan \psi = \frac{dy}{dx} = \text{slope of the tangent } PL$$

$$\text{ie., } \tan \psi = \frac{dy}{d\theta} \bigg/ \frac{dx}{d\theta} \quad \text{since } x \text{ and } y \text{ are function of } \theta$$

$$\text{ie., } \tan \psi = \frac{\frac{d}{d\theta}(r \sin \theta)}{\frac{d}{d\theta}(r \cos \theta)} = \frac{r \cos \theta + r' \sin \theta}{-r \sin \theta + r' \cos \theta} \quad \text{where } r' = \frac{dr}{d\theta}$$

Dividing both the numerator and denominator by $r' \cos \theta$ we have,

$$\tan \psi = \frac{\frac{r \cos \theta}{r' \cos \theta} + \frac{r' \sin \theta}{r' \cos \theta}}{\frac{-r \sin \theta}{r' \cos \theta} + \frac{r' \cos \theta}{r' \cos \theta}}$$

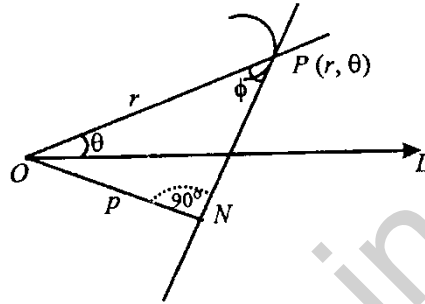
$$\text{Or } \tan \psi = \frac{\frac{r}{r'} + \tan \theta}{1 - \frac{r}{r'} \cdot \tan \theta} \quad \dots(2)$$

Comparing equations (1) and (2) we get

$$\tan \phi = \frac{r}{r'} = \frac{r}{\left(\frac{dr}{d\theta}\right)} \quad \text{or } \tan \phi = r \left(\frac{d\theta}{dr}\right)$$

Prove with usual notations $\frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$ **or**

$$\frac{1}{p^2} = u^2 + \left(\frac{du}{d\theta} \right)^2 \text{ where } u = \frac{1}{r}$$



Proof :

Let O be the pole and OL be the initial line. Let P (r, θ) be any point on the curve and hence we have OP = r and $\angle LOP = \theta$

Draw ON = p (say) perpendicular from the pole on the tangent at P and let φ be the angle made by the radius vector with the tangent.

From the figure $\angle ONP = 90^\circ$ $\angle LOP = \theta$

Now from the right angled triangle ONP

$$\sin \phi = \frac{ON}{OP}$$

$$\text{ie., } \sin \phi = \frac{p}{r} \quad \text{or} \quad p = r \sin \phi$$

$$\text{we have } p = r \sin \phi \quad \dots(1)$$

$$\text{and } \cot \phi = \frac{1}{r} \frac{dr}{d\theta} \quad \dots(2)$$

Squaring equation (1) and taking the reciprocal we get,

$$\frac{1}{p^2} = \frac{1}{r^2} \cdot \frac{1}{\sin^2 \phi} \quad \text{ie.,} \quad \frac{1}{p^2} = \frac{1}{r^2} \operatorname{cosec}^2 \phi$$

$$\text{Or} \quad \frac{1}{p^2} = \frac{1}{r^2} (1 + \cot^2 \phi)$$

Now using (2) we get,

$$\frac{1}{p^2} = \frac{1}{r^2} \left[1 + \frac{1}{r^2} \left(\frac{dr}{d\theta} \right)^2 \right]$$

$$\text{Or} \quad \frac{1}{p^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2 \quad \dots(3)$$

Further, let $\frac{1}{r} = u$

Differentiating w.r.t. θ we get,

$$-\frac{1}{r^2} \left(\frac{dr}{d\theta} \right) = \frac{du}{d\theta} \Rightarrow \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2 = \left(\frac{du}{d\theta} \right)^2, \text{ by squaring}$$

Thus (3) now becomes

$$\frac{1}{p^2} = u^2 + \left(\frac{du}{d\theta} \right)^2 \quad \dots(4)$$

1. Find the angle of intersection of the curves:

$$r = a(1 + \cos \theta) \text{ \& } r = b(1 - \cos \theta)$$

$$\text{Solution : } r = a(1 + \cos \theta) \quad : \quad r = b(1 - \cos \theta)$$

$$\Rightarrow \log r = \log a + \log(1 + \cos \theta) \quad : \quad \log r = \log b + \log(1 - \cos \theta)$$

Differentiating these w.r.t. θ we get

$$\frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{-\sin \theta}{1 + \cos \theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = 0 + \frac{\sin \theta}{1 - \cos \theta}$$

$$\cot \phi_1 = \frac{-2 \sin(\theta/2) \cos(\theta/2)}{2 \cos^2(\theta/2)} \quad : \quad \cot \phi_2 = \frac{2 \sin(\theta/2) \cos(\theta/2)}{2 \sin^2(\theta/2)}$$

$$\text{ie., } \cot \phi_1 = -\tan(\theta/2) = \cot(\pi/2 + \theta/2) \quad : \quad \cot \phi_2 = \cot(\theta/2)$$

$$\Rightarrow \phi_1 = \pi/2 + \theta/2 \quad : \quad \phi_2 = \theta/2$$

$$\therefore \text{angle of intersection} = |\phi_1 - \phi_2| = |\pi/2 + \theta/2 - \theta/2| = \pi/2$$

Hence the curves intersect orthogonally.

2. S.T. the curves $r = a(1 + \sin \theta)$ & $r = a(1 - \sin \theta)$ **cut each other orthogonally**

Solution : $\log r = \log a + \log (1 + \sin \theta)$: $\log r = \log a + \log (1 - \sin \theta)$

Differentiating these w.r.t θ we get

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta}{1 + \sin \theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = \frac{-\cos \theta}{1 - \sin \theta}$$

$$\text{ie.,} \quad \cot \phi_1 = \frac{\cos \theta}{1 + \sin \theta} \quad : \quad \cot \phi_2 = \frac{-\cos \theta}{1 - \sin \theta}$$

$$\text{We have } \tan \phi_1 = \frac{1 + \sin \theta}{\cos \theta} \text{ and } \tan \phi_2 = \frac{1 - \sin \theta}{-\cos \theta}$$

$$\therefore \tan \phi_1 \cdot \tan \phi_2 = \frac{1 - \sin^2 \theta}{-\cos^2 \theta} = \frac{\cos^2 \theta}{-\cos^2 \theta} = -1$$

Hence the curves intersect orthogonally.

3. Find the angle of intersection of the curves:

$$r = \sin \theta + \cos \theta, r = 2 \sin \theta$$

Solution : $\log r = \log (\sin \theta + \cos \theta)$: $r = 2 \sin \theta$

$$\Rightarrow \log r = \log (\sin \theta + \cos \theta) : \log r = \log 2 + \log (\sin \theta)$$

Differentiating these w.r.t θ we get

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta - \sin \theta}{\sin \theta + \cos \theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = \frac{\cos \theta}{\sin \theta}$$

$$\text{ie.,} \quad \cot \phi_1 = \frac{\cos \theta (1 - \tan \theta)}{\cos \theta (1 + \tan \theta)} \quad : \quad \cot \phi_2 = \cot \theta \Rightarrow \phi_2 = \theta$$

$$\text{ie.,} \quad \cot \phi_1 = \cot (\pi/4 + \theta) \Rightarrow \phi_1 = \pi/4 + \theta$$

$$\therefore |\phi_1 - \phi_2| = |\pi/4 + \theta - \theta| = \pi/4$$

The angle of intersection is $\pi/4$

4. Find the angle of the curves: $r = a \log \theta$ and $r = a / \log \theta$

Solution : $r = a \log \theta$: $r = a / \log \theta$

$$\Rightarrow \log r = \log a + \log (\log \theta) \quad : \quad \log r = \log a - \log (\log \theta)$$

Differentiating these w.r.t θ , we get,

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{1}{\log \theta \cdot \theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = -\frac{1}{\log \theta \cdot \theta}$$

$$\text{ie.,} \quad \cot \phi_1 = \frac{1}{\theta \log \theta} \quad : \quad \cot \phi_2 = -\frac{1}{\theta \log \theta}$$

Note : We can not find ϕ_1 and ϕ_2 explicitly.

$$\therefore \tan \phi_1 = \theta \log \theta \quad : \quad \tan \phi_2 = -\theta \log \theta$$

$$\text{Now consider, } \tan \phi_1 = \theta \log \theta \quad : \quad \tan \phi_2 = -\theta \log \theta$$

$$\text{Now consider, } \tan (\phi_1 - \phi_2) = \frac{2 \theta \log \theta}{1 + \tan \phi_1 \tan \phi_2} = \frac{2 \theta \log \theta}{1 - (\theta \log \theta)^2} \quad \dots\dots\dots(1)$$

We have to find θ by solving the given pair of equations :

$$r = a \log \theta \quad \text{and} \quad r = a / \log \theta$$

$$\text{Equating the R.H.S we have } a \log \theta = \frac{a}{\log \theta}$$

$$\text{ie., } (\log \theta = 1)^2 = 1 \text{ or } \log \theta \Rightarrow \theta = e$$

Substituting $\theta = e$ in (1) we get

$$\tan (\phi_1 - \phi_2) = \frac{2e}{1 - e^2} \quad (\because \log e = 1)$$

$$\therefore \text{ angle of intersection } \phi_1 - \phi_2 = \tan^{-1} \left(\frac{2e}{1 - e^2} \right) = 2 \tan^{-1} e$$

5. Find the angle of intersection of the curves:

$$r = a (1 - \cos \theta) \quad \text{and} \quad r = 2a \cos \theta$$

Solution : $r = a (1 - \cos \theta)$: $r = 2a \cos \theta$

Taking logarithms we have,

$$\log r = \log a + \log (1 - \cos \theta) \quad : \quad \log r = \log 2a + \log (\cos \theta)$$

Differentiating these w.r.t θ , we get,

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{\sin \theta}{1 - \cos \theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin \theta}{\cos \theta}$$

$$\text{ie., } \cot \phi_1 = \frac{2 \sin(\theta/2) \cos(\theta/2)}{2 \sin^2(\theta/2)} : \cot \phi_2 = -\tan \theta$$

$$\text{ie., } \cot \phi_1 = \cot(\theta/2) \quad : \quad \cot \phi_2 = \cot(\pi/2 + \theta)$$

$$\Rightarrow \phi_1 = \theta/2 \quad : \quad \phi_2 = \pi/2 + \theta$$

$$\therefore |\phi_1 - \phi_2| = |\theta/2 - \pi/2 - \theta| = |\pi/2 + \theta| \quad \dots(1)$$

$$\text{Now consider } r = a(1 - \cos \theta) = 2a \cos \theta$$

$$\text{Or } 3 \cos \theta = 1 \text{ or } \theta = \cos^{-1}(1/3)$$

Substituting this value in (1) we get,

$$\text{The angle of intersection } \pi/2 + 1/2 \cdot \cos^{-1}(1/3)$$

6. Find the angle of intersection of the curves :

$$r = a\theta \quad \text{and} \quad r = a/\theta$$

$$\text{Solution : } r = a\theta \quad : \quad r = a/\theta$$

$$\Rightarrow \log r = \log a + \log \theta \quad : \quad \log r = \log a - \log \theta$$

Differentiating these w.r.t θ , we get,

$$\frac{1}{r} \frac{dr}{d\theta} = \frac{1}{\theta} \quad : \quad \frac{1}{r} \frac{dr}{d\theta} = -\frac{1}{\theta}$$

$$\text{ie., } \cot \phi_1 = \frac{1}{\theta} \quad : \quad \cot \phi_2 = -\frac{1}{\theta}$$

$$\text{or } \tan \phi_1 = \theta \quad : \quad \tan \phi_2 = -\theta$$

Also by equating the R.H.S of the given equations we have

$$a\theta = a/\theta \text{ or } \theta^2 = 1 \Rightarrow \theta = \pm 1$$

$$\text{When } \theta = 1, \tan \phi_1 = 1, \tan \phi_2 = -1 \text{ and}$$

$$\text{When } \theta = -1, \tan \phi_1 = -1, \tan \phi_2 = 1.$$

$$\therefore \tan \phi_1 \cdot \tan \phi_2 = -1 \Rightarrow \phi_1 - \phi_2 = \pi/2$$

The curves intersect at right angles.

7. Find the pedal equation of the curve: $r(1 - \cos \theta) = 2a$

Solution : $r(1 - \cos \theta) = 2a$

$$\Rightarrow \log r + \log (1 - \cos \theta) = \log 2a$$

Differentiating w.r.t θ , we get

$$\frac{1}{r} \frac{dr}{d\theta} + \frac{\sin \theta}{1 - \cos \theta} = 0 \quad \text{or} \quad \frac{1}{r} \frac{dr}{d\theta} = \frac{-\sin \theta}{1 - \cos \theta}$$

$$\therefore \cot \phi = \frac{-2 \sin (\theta/2) \cos (\theta/2)}{2 \sin^2 (\theta/2)} = -\cot (\theta/2)$$

$$\text{ie., } \therefore \cot \phi = \cot (-\theta/2) \Rightarrow \phi = -(\theta/2)$$

Consider $p = r \sin \phi$

$$\therefore p = r \sin (-\theta/2) \quad \text{or} \quad p = -r \sin (\theta/2)$$

$$\text{Now we have, } r(1 - \cos \theta) = 2a \quad \dots(1)$$

$$p = -r \sin (\theta/2) \quad \dots(2)$$

We have to eliminate θ from (1) and (2)

$$(1) \text{ can be put in form } r \cdot 2 \sin^2 (\theta/2) = 2a$$

$$\text{ie., } r \sin^2 (\theta/2) = a$$

But $p/r = \sin (\theta/2)$, from (2)

$$\therefore r \left(\frac{p^2}{r^2} \right) = a \quad \text{or} \quad p^2 = ar$$

Thus $p^2 = ar$ is the required pedal equation.

8. Find the pedal equation of the curve: $r^2 = a^2 \sec 2\theta$

Solution : $r^2 = a^2 \sec 2\theta$

$$\Rightarrow 2 \log r = 2 \log a + \log (\sec 2\theta)$$

Differentiating w.r.t θ , we get,

$$\frac{2}{r} \frac{dr}{d\theta} = \frac{2 \sec 2\theta \tan 2\theta}{\sec 2\theta} \quad \text{ie., } \frac{1}{r} \frac{dr}{d\theta} = \tan 2\theta$$

$$\text{ie., } \cot \phi = \cot (\pi/2 - 2\theta) \Rightarrow \phi = \pi/2 - 2\theta$$

Consider $p = r \sin \phi \therefore p = r \sin (\pi/2 - 2\theta)$ i.e., $p = r \cos 2\theta$

Now we have, $r^2 = a^2 \sec 2\theta$... (1)

$$p = r \cos 2\theta \quad \dots (2)$$

From (2) $p/r = \cos 2\theta$ or $r/p = \sec 2\theta$

Substituting in (1) we get, $r^2 = a^2 (r/p)$ or $pr = a^2$

Thus $pr = a^2$ is the required pedal equation.

9. Find the pedal equation of the curve: $r^n = a^n \cos n\theta$

Solution : $r^n = a^n \cos n\theta$

$$\Rightarrow n \log r = n \log a + \log (\cos n\theta)$$

Differentiating w.r.t θ we get

$$\frac{n}{r} \frac{dr}{d\theta} = \frac{-n \sin n\theta}{\cos n\theta} \text{ i.e., } \frac{1}{r} \frac{dr}{d\theta} = -\tan n\theta$$

$$\therefore \cot \phi = \cot (\pi/2 + n\theta) \Rightarrow \phi = \pi/2 + n\theta$$

Consider $p = r \sin \phi$

$$\therefore p = r \sin (\pi/2 + n\theta) \text{ i.e., } p = r \cos n\theta$$

Now we have, $r^n = a^n \cos n\theta$... (1)

$$p = r \cos n\theta \quad \dots (2)$$

$\therefore$ (1) as a consequence of (2) is $r^n = a^n (p/r)$

Thus $r^{n+1} = pa^n$ is the required pedal equation.

10. Find the pedal equation of the curve: $r^m = a^m (\cos m\theta + \sin m\theta)$

Solution : $r^m = a^m (\cos m\theta + \sin m\theta)$

Differentiating w.r.t θ , we get,

$$\frac{m}{r} \frac{dr}{d\theta} = \frac{-m \sin m\theta + m \cos m\theta}{\cos m\theta + \sin m\theta}$$

$$\text{i.e., } \frac{1}{r} \frac{dr}{d\theta} = \frac{\cos m\theta - \sin m\theta}{\cos m\theta + \sin m\theta} = \frac{\cos m\theta (1 - \tan m\theta)}{\cos m\theta (1 + \tan m\theta)}$$

$$\therefore \cot \phi = \cot (\pi/4 + m\theta) \Rightarrow \phi = \pi/4 + m\theta$$

Consider $p = r \sin \phi$

$$\therefore p = r \sin (\pi/4 + m\theta)$$

$$\text{ie., } p = r [\sin (\pi/4) \cos m\theta + \cos (\pi/4) \sin m\theta]$$

$$\text{ie., } p = \frac{r}{\sqrt{2}} (\cos m\theta + \sin m\theta)$$

(we have used the formula of $\sin (A + B)$ and also the values $\sin (\pi/4) = \cos (\pi/4) = 1/\sqrt{2}$)

$$\text{Now we have, } r^m = a^m (\cos m\theta + \sin m\theta) \quad \dots(1)$$

$$p = \frac{r}{\sqrt{2}} (\cos m\theta + \sin m\theta) \quad \dots(2)$$

Using (2) in (1) we get,

$$r^m = a^m \cdot \frac{p\sqrt{2}}{r} \text{ or } r^{m+1} = \sqrt{2} a^m p$$

Thus $r^{m+1} = \sqrt{2} a^m p$ is the required pedal equation.

11. Establish the pedal equation of the curve:

$$r^n = a^n \sin n\theta + b^n \cos n\theta \text{ in the form } p^2 (a^{2n} + b^{2n}) = r^{2n+2}$$

Solution : We have $r^n = a^n \sin n\theta + b^n \cos n\theta$

$$\Rightarrow n \log r = \log (a^n \sin n\theta + b^n \cos n\theta)$$

Differentiating w.r.t θ we get

$$\frac{n}{r} \frac{dr}{d\theta} = \frac{na^n \cos n\theta - nb^n \sin n\theta}{a^n \sin n\theta + b^n \cos n\theta}$$

$$\text{Dividing by } n, \cot \phi = \frac{a^n \cos n\theta - b^n \sin n\theta}{a^n \sin n\theta + b^n \cos n\theta}$$

Consider $p = r \sin \phi$

Since ϕ cannot be found, squaring and taking the reciprocal we get,

$$\frac{1}{p^2} = \frac{1}{r^2} \operatorname{cosec}^2 \phi \text{ or } \frac{1}{p^2} = \frac{1}{r^2} (1 + \cot^2 \phi)$$

$$\therefore \frac{1}{p^2} = \frac{1}{r^2} \left\{ 1 + \frac{(a^n \cos n\theta - b^n \sin n\theta)^2}{(a^n \sin n\theta + b^n \cos n\theta)^2} \right\}$$

$$\text{ie., } \frac{1}{p^2} = \frac{1}{r^2} \left\{ \frac{(a^n \sin n\theta + b^n \cos n\theta)^2 + (a^n \cos n\theta - b^n \sin n\theta)^2}{(a^n \sin n\theta + b^n \cos n\theta)^2} \right\}$$

$$\text{ie., } \frac{1}{p^2} = \frac{1}{r^2} \left\{ \frac{a^{2n}(\sin^2 n\theta + \cos^2 n\theta) + b^{2n}(\cos^2 n\theta + \sin^2 n\theta)}{(a^n \sin n\theta + b^n \cos n\theta)^2} \right\}$$

(product terms cancels out in the numerator)

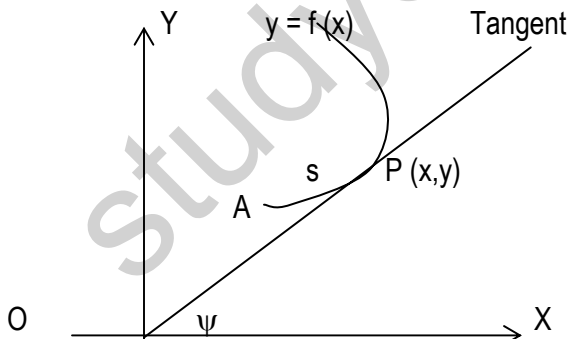
$$\text{ie., } \frac{1}{p^2} = \frac{1}{r^2} \cdot \frac{a^{2n} + b^{2n}}{(a^n \sin n\theta + b^n \cos n\theta)^2}$$

$$\text{or } \frac{1}{p^2} = \frac{1}{r^2} \cdot \frac{a^{2n} + b^{2n}}{(r^n)^2}, \text{ by using the given equation.}$$

$$\therefore p^2(a^{2n} + b^{2n}) = r^{2n+2} \text{ is the required pedal equation.}$$

12. Define Curvature and Radius of curvature

Solution: A Curve Cuts at every point on it. Which is determined by the tangent drawn.



Let P be a point on the curve $y = f(x)$ at the length 's' from a fixed point A on it. Let the tangent at 'P' makes an angle ψ with positive direction of x – axis. As the point 'P' moves along curve, both s and ψ vary.

The rate of change ψ w.r.t s, i.e., $\frac{d\psi}{ds}$ as called the Curvature of the curve at 'P'.

The reciprocal of the Curvature at P is called the radius of curvature at P and is denoted by ρ .

$$\therefore \rho = \frac{1}{\frac{d\psi}{ds}} = \frac{ds}{d\psi}$$

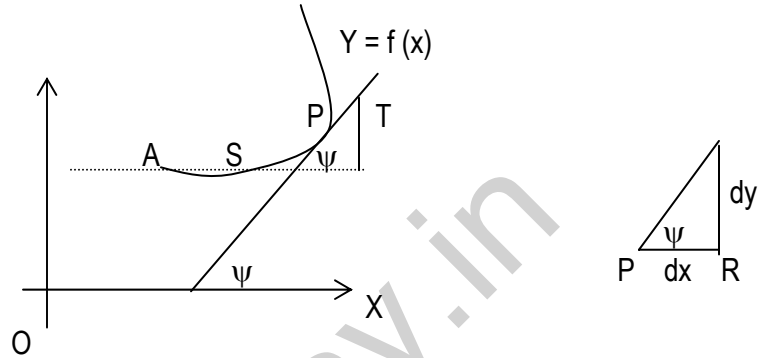
$$\therefore \rho = \frac{ds}{d\psi} \quad (\text{or}) \quad \frac{1}{\rho} = \frac{d\psi}{ds}$$

$$\text{Also denoted } \rho = \frac{1}{K} \quad \therefore K = \frac{d\psi}{ds}$$

K is read it as Kappa.

13. Derive an expression for radius of curvature in Cartesian form.

Solution : (1) Cartesian Form:



Let $y = f(x)$ be the curve in Cartesian form.

We know that, $\tan \psi = \frac{dy}{dx}$ (From Figure) ----- (1)

Where ψ is the angle made by the tangent at P with x – axis. Differentiating (1) W.r.t x, we get

$$\begin{aligned} \sec^2 \psi \cdot \frac{d\psi}{dx} &= \frac{d^2 y}{dx^2} \\ \text{i.e., } \frac{d^2 y}{dx^2} &= \sec^2 \psi \frac{d\psi}{dx} \\ &= (1 + \tan^2 \psi) \frac{d\psi}{dx} \\ &= (1 + \tan^2 \psi) \frac{d\psi}{dx} \cdot \frac{ds}{dx} \\ &= (1 + \tan^2 \psi) \frac{1}{\rho} \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{\frac{1}{2}} \\ &= \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\} \cdot \frac{1}{\rho} \cdot \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{\frac{1}{2}} \quad \text{from eqn (1)} \end{aligned}$$

$$= \frac{1}{\rho} \left\{ 1 + \left(\frac{dy}{dx} \right)^2 \right\}^{\frac{3}{2}}$$

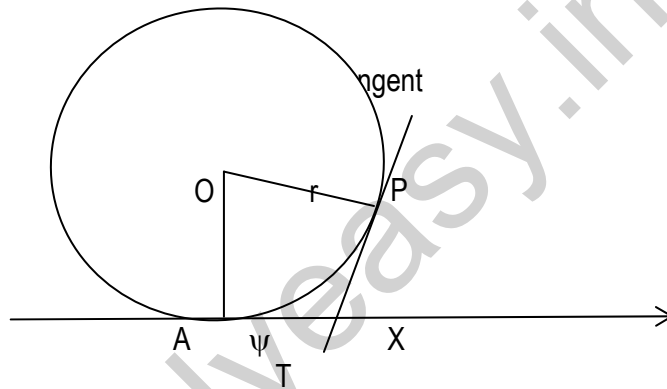
$$\therefore \rho = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}}}{d^2y/dx^2}$$

$$\therefore \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \dots \dots \dots (1)$$

Where $y_1 = \frac{dy}{dx}$, $y_2 = \frac{d^2y}{dx^2}$

This is the formula for Radius of Curvature in Cartesian Form.

14. Show that the Curvature of a Circle at any point on it, is a Constant



Solution: Consider a Circle of radius r . Let A be a fixed point and 'P' be a given point on the circle such that arc $AP = S$.

Let the angle between the tangent to the Circle at A and P be ψ . Then clearly $\angle AOP = \psi$.

$$\therefore AP = r\psi$$

$$\text{i.e., } S = r\psi$$

This is the intrinsic equation of the circle.

Differentiating w.r.t S we get

$$1 = r \frac{d\psi}{ds} \text{ Or } K = \frac{d\psi}{ds} = \frac{1}{r}$$

$$\therefore K = \frac{1}{r} \text{ which is Constant}$$

Hence the Curvature of the Circle at any point on it is constant.

15. Derive an expression for radius of curvature in parametric form.

Solution: We have $\rho = (1 + y_1^2)^{3/2}$

Let $x = f(t)$, $y = g(t)$ be the curve in Parametric Form.

$$\text{Then } y_1 = \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\dot{Y}}{\dot{X}}$$

$$Y_2 = \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dx} \left(\frac{\dot{Y}}{\dot{X}} \right) = \frac{d}{dt} \left(\frac{\dot{Y}}{\dot{X}} \right) \frac{dt}{dx}$$

$$= \frac{d}{dt} \left(\frac{\dot{Y}}{\dot{X}} \right) \cdot \left(\frac{1}{\frac{dx}{dt}} \right) = \frac{d}{dt} \left(\frac{\dot{Y}}{\dot{X}} \right) \cdot \frac{1}{\dot{X}}$$

$$= \frac{\ddot{X}\dot{Y} - \dot{X}\ddot{Y}}{(\dot{X})^2} \cdot \frac{1}{\dot{X}}$$

$$\therefore Y_2 = \frac{\ddot{X}\dot{Y} - \dot{X}\ddot{Y}}{(\dot{X})^3}$$

Substituting Y_1 and Y_2 in equation 1.

$$\rho = \frac{(1 + Y_1^2)^{3/2}}{Y_2}, \text{ we get}$$

$$\rho = \frac{\left[\left(1 + \left(\frac{\dot{Y}}{\dot{X}} \right)^2 \right)^{3/2} \right]}{\frac{\ddot{X}\dot{Y} - \dot{X}\ddot{Y}}{(\dot{X})^3}}$$

$$\rho = \frac{[(\dot{X})^2 + (\dot{Y})^2]^{3/2}}{\ddot{X}\dot{Y} - \dot{X}\ddot{Y}} \text{ ----- (2)}$$

Equation (2) is called the Radius of Curvature in Parametric Form.

16. Derive an expression for radius of curvature in polar form.

Solution: Let $r = f(\theta)$ be the curve in the Polar Form. We know that, Angle between the tangent and radius vector,

$$\begin{aligned}\tan \phi &= r \frac{d\theta}{dr} \\ &= r \cdot \frac{1}{dr/d\theta}\end{aligned}$$

$$\text{i.e., } \tan \phi = \frac{r}{(dr/d\theta)}$$

Differentiate w.r.t ' θ ' we get

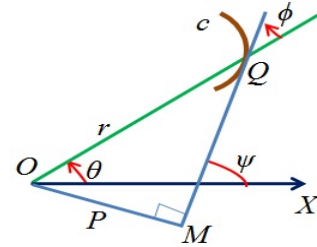
$$\sec^2 \phi \cdot \frac{d\phi}{d\theta} = \frac{\frac{dr}{d\theta} \cdot \frac{dr}{d\theta} - r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2}$$

$$\therefore \frac{d\phi}{d\theta} = \frac{1}{\sec^2 \phi} \left\{ \frac{\left(\frac{dr}{d\theta}\right)^2 r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2} \right\}$$

$$= \frac{1}{1 + \tan^2 \phi} \left\{ \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2} \right\}$$

$$= \frac{1}{1 + \frac{r^2}{\left(\frac{dr}{d\theta}\right)^2}} \left\{ \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2} \right\}$$

$$\frac{d\phi}{d\theta} = \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2 r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2 + r^2}$$



From figure $\psi = \theta + \phi$

Differentiating w.r.t θ , we get

$$\frac{d\psi}{d\theta} = 1 + \frac{d\phi}{d\theta}$$

$$\therefore \frac{d\psi}{d\theta} = 1 + \frac{\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2 + r^2}$$

$$\frac{d\psi}{d\theta} = \frac{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2}}{\left(\frac{dr}{d\theta}\right)^2 + r^2}$$

Also we know that $\frac{ds}{d\theta} = \left\{ r^2 + \left(\frac{dr}{d\theta}\right)^2 \right\}^{\frac{1}{2}}$

$$\begin{aligned} \text{Now, } \rho &= \frac{ds}{d\psi} = \frac{ds}{d\theta} \cdot \frac{d\theta}{d\psi} \\ &= \left\{ r^2 + \left(\frac{dr}{d\theta}\right)^2 \right\}^{\frac{1}{2}} \cdot \frac{\left(\frac{dr}{d\theta}\right)^2 + r^2}{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r \frac{d^2r}{d\theta^2}} \end{aligned}$$

$$\rho = \frac{(r^2 + r_1^2)^{\frac{3}{2}}}{r^2 + 2r_1^2 - rr_2} \quad (3)$$

Where $r_1 = \frac{dr}{d\theta}$, $r_2 = \frac{d^2r}{d\theta^2}$

Equation (3) is called the radius of curvature in Polar form.

17. Derive an expression for radius of curvature in pedal form.

Solution: Let $p = r \sin \phi$ be the curve in Polar Form.

We have $p = r \sin \phi$

Differentiating p w.r.t r , we get

$$\frac{dp}{dr} = \sin \phi + r \cos \phi \frac{d\phi}{dr}$$

$$\text{But } \sin \phi = r \frac{d\theta}{ds}, \cos \phi = \frac{dr}{ds}$$

$$\therefore \frac{dp}{dr} = r \frac{d\theta}{ds} + r \frac{dr}{ds} \frac{d\phi}{dr}$$

$$= r \frac{d\theta}{ds} + \frac{d\phi}{dr} \cdot \frac{dr}{ds}$$

$$= r \frac{d\theta}{ds} + r \frac{d\phi}{ds}$$

$$= r \frac{d}{ds} (\theta + \phi)$$

$$= r \frac{d\psi}{ds}$$

$$\therefore \frac{dp}{dr} = r \frac{d\psi}{ds}$$

$$\frac{dp}{dr} = r \cdot \frac{1}{\rho} \quad \therefore \frac{d\psi}{ds} = \frac{1}{\rho}$$

$$\therefore \rho = r \frac{dr}{dp} \quad \text{----- (4)}$$

Equation (4) is called Radius of Curvature of the Curve in Polar Form.

18. Find the radius of curvature at (x, y) for the curve $ay^2 = x^3$.

Solution: Given $ay^2 = x^3$ ----- (1) is in Cartesian form.

We have, Radius of curvature in Cartesian form.

$$\rho = \frac{(1 + y_1^2)^{3/2}}{y_2} \quad \text{----- (2)}$$

Differentiating (1) w.r.t x we get,

$$a 2y \frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = y_1 = \frac{3x^2}{2ay} = \frac{3x^2}{2a\left(\frac{x^3}{a}\right)^{\frac{1}{2}}} = \frac{3\sqrt{x}}{2\sqrt{a}}$$

$$\therefore y_1 = \frac{3\sqrt{x}}{2\sqrt{a}}$$

Differentiating y_1 w.r.t 'x' we get

$$y_2 = \frac{d^2y}{dx^2} = \frac{3}{2\sqrt{a}} \cdot \frac{1}{2\sqrt{x}} = \frac{3}{4\sqrt{a}\sqrt{x}}$$

Substitute Y_1 and Y_2 in (2), we get.

$$\rho = \frac{\sqrt{x}(4a+9x)^{\frac{3}{2}}}{6a}$$

19. Find the radius of curvature at (x,y) for the curve $y = c \log \sec \left(\frac{x}{c} \right)$

Solution: Given $y = c \log \sec \left(\frac{x}{c} \right)$ ----- (1)

Differentiating (1) w.r.t x, we get

$$y_1 = c \cdot \frac{1}{\sec\left(\frac{x}{c}\right)} \cdot \sec\left(\frac{x}{c}\right) \tan\left(\frac{x}{c}\right) = c \tan\left(\frac{x}{c}\right) \cdot \frac{1}{c} = \tan\left(\frac{x}{c}\right)$$

Differentiating y_1 W.r.t x we get

$$y_2 = \sec^2\left(\frac{x}{c}\right) \cdot \frac{1}{c} = \frac{1}{c} \sec^2\left(\frac{x}{c}\right)$$

Substitute y_1 and y_2 in $\delta = \frac{(1+y_1^2)^{\frac{3}{2}}}{y_2}$

$$\therefore \delta = \frac{\left[1 + \tan^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \sec^2\left(\frac{x}{c}\right)} = \frac{\left[\sec^2\left(\frac{x}{c}\right)\right]^{\frac{3}{2}}}{\frac{1}{c} \sec^2\left(\frac{x}{c}\right)}$$

$$\rho = c \sec\left(\frac{x}{c}\right)$$

20. Find the radius of curvature at the point 't' on the curve $x = a(t + \sin t)$, $y = a(1 - \cos t)$.

Solution: Given Curves are in Parametric Form

$$\therefore \text{Radius of Curvature, } \rho = \frac{\left((\dot{x})^2 + (\dot{y})^2\right)^{3/2}}{\dot{x}\ddot{y} - \dot{y}\ddot{x}} \text{ ----- (1)}$$

Differentiating the given Curves W.r.t t, we get

$$\dot{x} = \frac{dx}{dt} = a(1 + \cos t) \quad \dot{y} = \frac{dy}{dt} = a \sin t$$

Differentiating W.r.t 't' we get

$$\ddot{x} = -a \sin t, \quad \ddot{y} = a \cos t$$

Substitute $\dot{x}$, $\ddot{y}$, $\dot{y}$ and $\ddot{x}$ in (1), we get

$$\begin{aligned} \rho &= \frac{\left(a^2(1 + \cos t)^2 + a^2 \sin^2 t\right)^{3/2}}{a(1 + \cos t)a \cos t - a \sin t(-a \sin t)} \\ &= \frac{a^3 \{1 + 2\cos t + \cos^2 t + \sin^2 t\}^{3/2}}{a^2 \{ \cos t + \cos^2 t + \sin^2 t \}} \\ &= \frac{a \{2(1 + \cos t)\}^{3/2}}{(1 + \cos t)} = \frac{a \{2 \cdot 2 \cos^2 t/2\}^{3/2}}{2 \cos^2 t/2} = \frac{8a \cos^3 t/2}{2 \cos^2 t/2} \\ \rho &= 4a \cos t/2 \end{aligned}$$

21. Find the Radius of Curvature to $x = a\left\{\cos t + \log \tan\left(\frac{t}{2}\right)\right\}$, $y = a \sin t$ at t.

Solution: Here $x = a\left\{\cos t + \log \tan\left(\frac{t}{2}\right)\right\}$, $y = a \sin t$

$$\begin{aligned} \frac{dx}{dt} &= a \left\{ -\sin t + \frac{1}{\tan t/2} \cdot \sec^2 t/2 \cdot \frac{1}{2} \right\} \\ &= a \left\{ -\sin t + \frac{1}{2 \sin t/2 \cos t/2} \right\} \\ &= a \left\{ -\sin t + \frac{1}{2 \sin t/2} \right\} \\ &= a \left\{ (1 - \sin^2 t) / \sin t \right\} \end{aligned}$$

$$\frac{dx}{dt} = a \cos^2 t / \sin t$$

$$\text{and } \frac{dy}{dx} = a \cos t$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a \cos t}{a \cos t \cos t / \sin t} = \tan t$$

$$\therefore \frac{dy}{dx} = \tan t$$

Differentiating W.r.t 'x' we get

$$\frac{d^2 y}{dx^2} = \sec^2 t \frac{dt}{dx} = \sec^2 t \frac{1}{\frac{dx}{dt}} = \sec^2 t \cdot \frac{1}{a \cos^2 t / \sin t}$$

$$\frac{d^2 y}{dx^2} = \frac{1}{a} = \sec^4 t \sin t$$

Substitute $\frac{dy}{dx}$ & $\frac{d^2 y}{dx^2}$ in $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$, we get

$$\text{i.e., } \rho = \frac{(1 + \tan^2 t)^{3/2}}{\frac{1}{a} \sec^4 t \sin t} = \frac{a \sec^3 t}{\sec^4 t \sin t} = a \frac{\cos t}{\sin t} = a \cot t$$

$$\therefore \rho = a \cot t.$$

22. Find the Radius of Curvature to $\sqrt{x} + \sqrt{y} = \sqrt{a}$ at $\left(\frac{a}{4}, \frac{a}{4}\right)$

Solution: Given $\sqrt{x} + \sqrt{y} = \sqrt{a}$ ----- (1)

Differentiating (1) w.r.t to 'x' we get

$$\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0$$

$$\text{i.e., } y_1 = \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}} = -\frac{(\sqrt{a} - \sqrt{x})}{\sqrt{x}} \text{ (From (1))}$$

$$y_1 = 1 - \frac{\sqrt{a}}{\sqrt{x}} \text{ ----- (2)}$$

$$\text{Also } y_2 = \frac{d^2 y}{dx^2} = -\sqrt{a} \left(-\frac{1}{2} \right) x^{-3/2} = \frac{\sqrt{a}}{2x^{3/2}} \quad \text{----- (3)}$$

At the given point $\left(\frac{a}{4}, \frac{a}{4} \right)$

$$\text{Then } y_1 = 1 - \frac{\sqrt{a}}{\sqrt{a/2}} = -1 \text{ \& } y_2 = 1 - \frac{\sqrt{a}}{2(a/4)^{3/2}} = \frac{4}{a}$$

$$\therefore \text{ Substitute } y_1 \text{ and } y_2 \text{ in } \rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$$

$$\rho = \frac{(1 + (-1)^2)^{3/2}}{4/a} = \frac{2^{3/2} a}{4} = \frac{2\sqrt{2}a}{4} = \frac{a}{\sqrt{2}}$$

$$\rho = \frac{a}{\sqrt{2}}$$

23. Show that for the Cardioids $r = a(1 + \cos\theta)$, ρ^2 / r^2 is a constant

Solution: $r = a(1 + \cos\theta)$

$$\frac{dr}{d\theta} = -a \sin\theta$$

We have,

$$\frac{1}{P^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2, \text{ is Pedal Equation.}$$

$$= \frac{1}{r^2} + \frac{1}{r^4} a^2 \sin^2\theta$$

$$= \frac{r^2 + a^2 \sin^2\theta}{r^4} = \frac{a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta}{r^4}$$

$$= \frac{2a^2(1 + \cos\theta)}{r^4}$$

$$\therefore \frac{1}{P^2} = \frac{2a^2}{r^4} \left(\frac{r}{a} \right) \quad (\because r = a(1 + \cos\theta))$$

$$\therefore \frac{r}{a} = 1 + \cos\theta$$

$$\frac{1}{P^2} = \frac{2a}{r^3}$$

$$\therefore P^2 = \frac{r^3}{2a}$$

Differentiating w.r.t 'P' we get

$$2P = \frac{1}{2a} 3r^2 \frac{dr}{dp}$$

$$\frac{dr}{dp} = \frac{4ap}{3r^2}$$

Now, $\rho = r \frac{dr}{dp} = \frac{4ap}{3r}$

And $\frac{\rho^2}{r^2} = \frac{1}{r} \left[\frac{16a^2 p^2}{9r^2} \right] = \frac{16a^2}{9r^3} \cdot \frac{r^3}{2a} = \frac{8a}{9}$

24. Find the Radius of Curvature of the Curve $\frac{1}{P^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$

Solution: Given $\frac{1}{P^2} = \frac{1}{a^2} + \frac{1}{b^2} - \frac{r^2}{a^2 b^2}$

Differentiating w.r.t to P, we get

$$\frac{-2}{P^3} = \frac{-1}{a^2 b^2} 2r \frac{dr}{dp}$$

$$\therefore \frac{dr}{dp} = \frac{a^2 b^2}{p^3 r}$$

$$\therefore \rho = r \cdot \frac{dr}{dp} = r \cdot \frac{a^2 b^2}{p^3 r} = \frac{a^2 b^2}{p^3}$$

25. Find the Radius of Curvature at (r, θ) on $r = \frac{a}{\theta}$

Solution: Given $r = \frac{a}{\theta}$

Differentiating w.r.t to ' θ ' we get

$$\frac{dr}{d\theta} = \frac{-a}{\theta^2} = \frac{-a}{\theta} \cdot \frac{1}{\theta} = \frac{-r}{\theta}$$

$$\frac{dr}{d\theta} = \frac{-r}{\theta}$$

We have $\frac{1}{P^2} = \frac{1}{r^2} + \frac{1}{r^4} \left(\frac{dr}{d\theta} \right)^2$

$$= \frac{1}{r^2} + \frac{1}{r^4} \frac{r^2}{\theta^2} = \frac{1}{r^2} + \frac{1}{r^2 \theta^2}$$

$$\frac{1}{P^2} = \frac{1}{r^2} \left(1 + \frac{1}{\theta^2} \right) = \frac{1}{r^2} \left(\frac{\theta^2 + 1}{\theta^2} \right)$$

$$\therefore P = \frac{r\theta}{\sqrt{\theta^2 + 1}}$$

$$P = \frac{r\theta}{\sqrt{\theta^2 + 1}} = \frac{r \cdot \frac{a}{r}}{\sqrt{\frac{a^2}{r^2} + 1}} = \frac{a.r}{\sqrt{a^2 + r^2}}$$

Differentiating above result w.r.t to 'P' we get

$$1 = \frac{\sqrt{a^2 + r^2} \cdot a \frac{dr}{dp} - r \cdot a \frac{1}{2\sqrt{a^2 + r^2}} 2r \frac{dr}{dp}}{(a^2 + r^2)}$$

$$a^2 + r^2 = \left(\sqrt{a^2 + r^2} \cdot a - \frac{r^2 a}{\sqrt{a^2 + r^2}} \right) \frac{dr}{dp}$$

$$a^2 + r^2 = \frac{(a^2 + r^2) \cdot a - r^2 a}{\sqrt{a^2 + r^2}} \frac{dr}{dp}$$

$$(a^2 + r^2)\sqrt{a^2 + r^2} = a^3 \frac{dr}{dp}$$

$$\frac{(a^2 + r^2)^{3/2}}{a^3} = \frac{dr}{dp}$$

$$\text{Thus, } \rho = r \cdot \frac{dr}{dp} = \frac{r \cdot (a^2 + r^2)^{3/2}}{a^3}$$

$$\therefore \rho = \frac{r}{a^3} (a^2 + r^2)^{3/2}$$

Exercises:

- (1) Find the Radius of the Curvature at the point (s, ψ) on $S = a \log \tan \left(\frac{\pi}{4} + \frac{\psi}{2} \right)$
- (2) Find the Radius of the Curvature of $xy = C^2$ at (x,y)
- (3) Find the Radius of the Curvature of $xy^3 = a^4$ at (a,a)
- (4) Find the Radius of Curvature at the point θ on $x = C \sin 2\theta (1 + \cos 2\theta)$,
 $y = C \cos 2\theta (1 - \cos 2\theta)$
- (5) If ρ_1 and ρ_2 are the radii of curvature at the extremities of any chord of the cardioid
 $r = a (1 + \cos \theta)$ which passes through the Pole prove that $\rho_1^2 + \rho_2^2 = \frac{16a^2}{9}$

DIFFERENTIAL CALCULUS – III

PARTIAL DIFFERENTIATION

Introduction :-

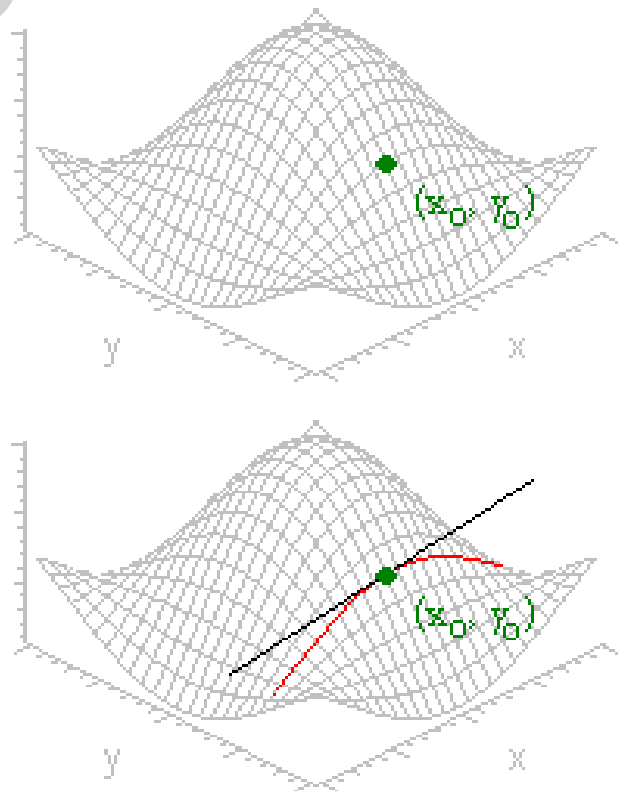
Partial differential equations abound in all branches of science and engineering and many areas of business. The number of applications is endless.

Partial derivatives have many important uses in math and science. We shall see that a partial derivative is not much more or less than a particular sort of directional derivative. The only trick is to have a reliable way of specifying directions ... so most of this note is concerned with formalizing the idea of direction.

So far, we had been dealing with functions of a single independent variable. We will now consider functions which depend on more than one independent variable; Such functions are called functions of several variables.

Geometrical Meaning

Suppose the graph of $z = f(x, y)$ is the surface shown. Consider the partial derivative of f with respect to x at a point (x_0, y_0) . Holding y constant and varying x , we trace out a curve that is the intersection of the surface with the vertical plane $y = y_0$. The partial derivative $f_x(x_0, y_0)$ measures the change in z per unit increase in x along this curve. That is, $f_x(x_0, y_0)$ is just the slope of the curve at (x_0, y_0) . The geometrical interpretation of $f_y(x_0, y_0)$ is analogous.



Real-World Applications:

Rates of Change:

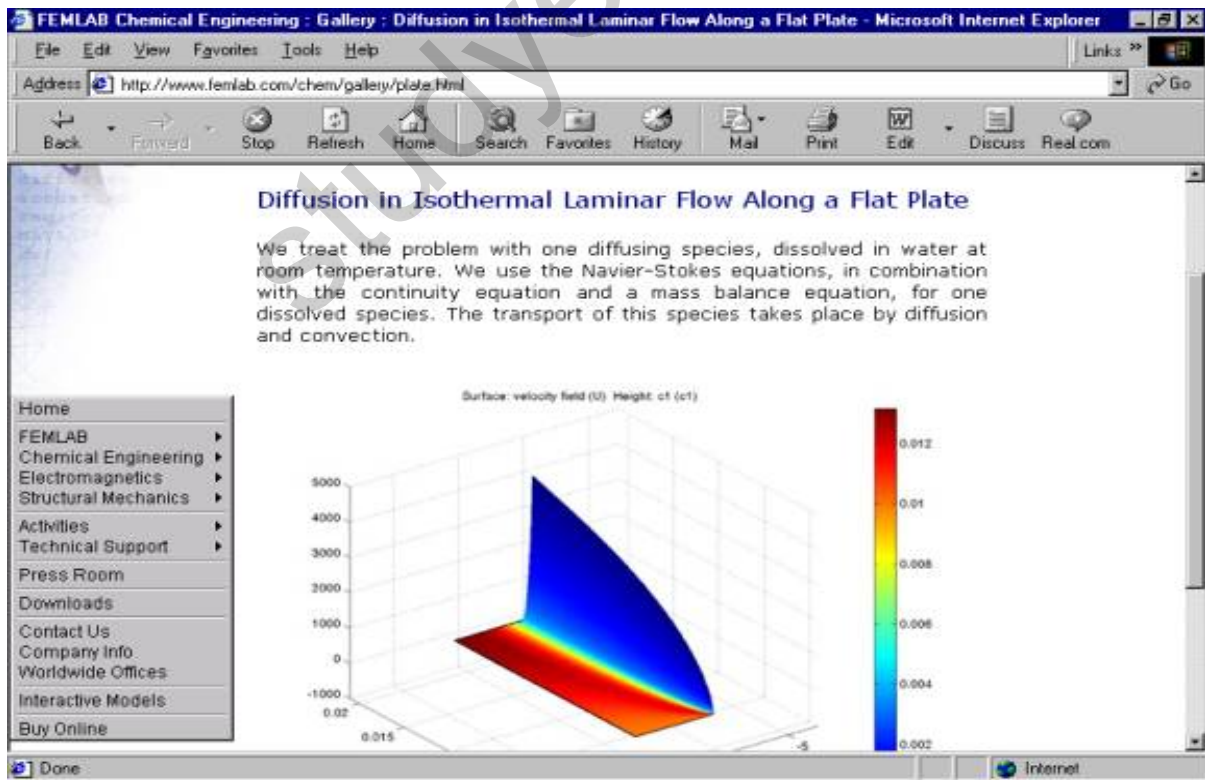
In the Java applet we saw how the concept of partial derivative could be applied geometrically to find the slope of the surface in the x and y directions. In the following two examples we present partial derivatives as rates of change. Specifically we explore an application to a temperature function (this example does have a geometric aspect in terms of the physical model itself) and a second application to electrical circuits, where no geometry is involved.

I. Temperature on a Metal Plate

The screen capture below shows a current website illustrating thermal flow for chemical engineering. Our first application will deal with a similar flat plate where temperature varies with position.

* The example following the picture below is taken from the current text in SM221,223:

[Multivariable Calculus](#) by James Stewart.



Suppose we have a flat metal plate where the temperature at a point (x,y) varies according to position. In particular, let the temperature at a point (x,y) be given by,

$$T(x, y) = 60 / (1 + x^2 + y^2)$$

where T is measured in $^{\circ}\text{C}$ and x and y in meters.

Question: what is the rate of change of temperature with respect to distance at the point $(2,1)$ in (a) the x -direction? and (b) in the y -direction ?

Let's take (a) first.

What is the rate of change of temperature with respect to distance at the point $(2,1)$ in (a) the x -direction?

What observations and translations can we make here?

Rate of change of temperature indicates that we will be computing a type of derivative.

Since the temperature function is defined on two variables we will be computing a partial derivative. Since the question asks for the rate of change in the x -direction, we will be holding y constant. Thus, our question now becomes:

What is $\frac{\partial T}{\partial x}$ at the point $(2,1)$?

$$T(x, y) = 60 / (1 + x^2 + y^2) = 60(1 + x^2 + y^2)^{-1}$$

$$\frac{\partial T}{\partial x} = -60(2x)(1 + x^2 + y^2)^{-2}$$

$$\frac{\partial T}{\partial x}(2,1) = -60(4)(1 + 4 + 1)^{-2} = -\frac{20}{3}$$

Conclusion :

The rate of change of temperature in the x -direction at $(2,1)$ is $-\frac{20}{3}$ degrees per meter;

note this means that the temperature is decreasing !

Part (b):

The rate of change of temperature in the y-direction at (2,1) is computed in a similar manner.

$$T(x, y) = 60 / (1 + x^2 + y^2) = 60(1 + x^2 + y^2)^{-1}$$

$$\frac{\partial T}{\partial x} = -60(2y)(1 + x^2 + y^2)^{-2}$$

$$\frac{\partial T}{\partial x}(2,1) = -60(2)(1 + 4 + 1)^{-2} = -10/3$$

Conclusion :

The rate of change of temperature in the y-direction at (2,1) is $-10/3$ degrees per meter;

note this means that the temperature is decreasing !

II. Electrical Circuits: Changes in Current

The following is adapted from an example in a former text for SM221,223 [Multivariable Calculus](#) by Bradley and Smith.

* In an electrical circuit with electromotive force (EMF) of E volts and resistance R ohms, the current, I, is

$I = E/R$ amperes.

Question: (a) At the instant when $E=120$ and $R=15$, what is the rate of change of current with respect to voltage.

(b) What is the rate of change of current with respect to resistance?

(a) Even though no geometry is involved in this example, the rate of change questions can be answered with **partial derivatives**.

we first note that I is a function of E and R, namely,

$$I(E,R) = ER^{-1}$$

The rate of change of current with respect to voltage =

the partial derivative of I with respect to voltage, holding resistance constant is

$$\frac{\partial I}{\partial E} = R^{-1}$$

when **E=120** and **R=15**, we have $\frac{\partial I}{\partial E} = 15^{-1} \approx 0.0667$

verbal conclusion : If the resistance is fixed at 15 ohms, the current is increasing with respect to voltage at the rate of 0.0667 amperes per volt when the EMF is 120 volts.

Part (b):

What is the rate of change of current with respect to resistance?

Using similar observations to part (a) we conclude:

The partial derivative of I with respect to resistance, holding voltage constant = $\frac{\partial I}{\partial R} = ER^{-1}$

when **E=120** and **R=15**, we have $\frac{\partial I}{\partial R}(120, 15) = -120(15)^{-1} \approx -0.5333$

Conclusion : If the EMF is fixed at 120 volts, the current is decreasing with respect to resistance at the rate of 0.5333 amperes per ohm when the resistance is 15 ohms.

Key Words :-

Then the partial derivative of z w.r.t x is given by

$$z_x = \frac{\partial z}{\partial x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x, y) - f(x, y)}{\delta x}$$

The partial derivative of z w.r.t y is given by

$$z_y = \frac{\partial z}{\partial y} = \lim_{\delta y \rightarrow 0} \frac{f(x, y + \delta y) - f(x, y)}{\delta y}$$

1. If $u = e^{ax-by} \sin(ax+by)$ show that $b \frac{\partial u}{\partial x} - a \frac{\partial u}{\partial y} = 2abu$

Solution : $u = e^{ax-by} \sin(ax+by)$

$$\therefore \frac{\partial u}{\partial x} = e^{ax-by} \cos(ax+by) \cdot a + a \cdot e^{ax-by} \sin(ax+by)$$

$$\text{ie., } \frac{\partial u}{\partial x} = a e^{ax-by} \cos(ax+by) + au \quad \dots(1)$$

$$\text{Also } \frac{\partial u}{\partial y} = e^{ax-by} \cos(ax+by) \cdot b + (-b) e^{ax-by} \sin(ax+by)$$

$$\text{ie., } \frac{\partial u}{\partial y} = b e^{ax-by} \cos(ax+by) - bu \quad \dots(2)$$

$$\text{Now } b \frac{\partial u}{\partial x} = a \frac{\partial u}{\partial y} \text{ by using (1) and (2) becomes}$$

$$\begin{aligned} &= a b e^{ax-by} \cos(ax+by) + abu - a b e^{ax-by} \cos(ax+by) + abu \\ &= 2abu \end{aligned}$$

$$\text{Thus } b \frac{\partial u}{\partial x} - a \frac{\partial u}{\partial y} = 2abu$$

2. If $u = e^{ax+by} f(ax-by)$, prove that

$$b \frac{\partial u}{\partial x} = a \frac{\partial u}{\partial y} = 2abu$$

Solution : $u = e^{ax+by} f(ax-by)$, by data

$$\frac{\partial u}{\partial x} = e^{ax+by} \cdot f'(ax-by) \cdot a + a e^{ax+by} f(ax-by)$$

$$\text{Or } \frac{\partial u}{\partial x} = a e^{ax+by} \cdot f'(ax-by) + a u \quad \dots(1)$$

$$\text{Next, } \frac{\partial u}{\partial y} = e^{ax+by} f'(ax-by) \cdot (-b) + b e^{ax+by} f(ax-by)$$

$$\text{Or } \frac{\partial u}{\partial y} = -b e^{ax+by} f'(ax-by) + bu \quad \dots(2)$$

$$\text{Now consider L.H.S} = b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y}$$

$$= b \left\{ a e^{ax+by} f'(ax-by) + au \right\} + a \left\{ b e^{ax+by} f'(ax-by) + bu \right\}$$

$$= ab e^{ax+by} f'(ax-by) + abu - ab e^{ax+by} f'(ax-by) + abu$$

$$= 2abu = \text{R.H.S}$$

$$\text{Thus } b \frac{\partial u}{\partial x} + a \frac{\partial u}{\partial y} = 2 a b u$$

3. If $u = \log \sqrt{x^2 + y^2 + z^2}$, show that $(x^2 + y^2 + z^2) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = 1$

Solution : By data $u = \log \sqrt{x^2 + y^2 + z^2} = \frac{1}{2} \log (x^2 + y^2 + z^2)$

The given u is a symmetric function of x, y, z ,

(It is enough if we compute only one of the required partial derivative)

$$\frac{\partial u}{\partial x} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2 + z^2} \cdot 2x = \frac{x}{x^2 + y^2 + z^2}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} \right) = \frac{\partial}{\partial x} \left(\frac{x}{x^2 + y^2 + z^2} \right)$$

$$\text{ie., } \frac{(x^2 + y^2 + z^2)1 - x \cdot 2x}{(x^2 + y^2 + z^2)^2} = \frac{y^2 + z^2 - x^2}{(x^2 + y^2 + z^2)^2}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{y^2 + z^2 - x^2}{(x^2 + y^2 + z^2)^2} \quad \dots(1)$$

$$\text{Similarly } \therefore \frac{\partial^2 u}{\partial y^2} = \frac{z^2 + x^2 - y^2}{(x^2 + y^2 + z^2)^2} \quad \dots(2)$$

$$\frac{\partial^2 u}{\partial z^2} = \frac{x^2 + y^2 - z^2}{(x^2 + y^2 + z^2)^2} \quad \dots(3)$$

Adding (1), (2) and (3) we get,

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^2} = \frac{1}{x^2 + y^2 + z^2}$$

$$\text{Thus } (x^2 + y^2 + z^2) \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) = 1$$

4. If $u = \log (\tan x + \tan y + \tan z)$, show that,
 $\sin 2x u_x + \sin 2y u_y + \sin 2z u_z = 2$

Solution : $u = \log (\tan x + \tan y + \tan z)$ is a symmetric function.

$$u_x = \frac{\sec^2 x}{\tan x + \tan y + \tan z}$$

$$\sin 2x u_x = \frac{(2 \sin x \cos x) \sec^2 x}{\tan x + \tan y + \tan z}$$

$$\text{Or } \sin 2x u_x = \frac{2 \tan x}{\tan x + \tan y + \tan z} \quad \dots(1)$$

$$\text{Similarly } \sin 2y u_y = \frac{2 \tan y}{\tan x + \tan y + \tan z} \quad \dots(2)$$

$$\sin 2z u_z = \frac{2 \tan z}{\tan x + \tan y + \tan z} \quad \dots(3)$$

Adding (1), (2) and (3) we get,

$$\sin 2x u_x + \sin 2y u_y + \sin 2z u_z = \frac{2(\tan x + \tan y + \tan z)}{(\tan x + \tan y + \tan z)} = 2$$

$$\text{Thus } \sin 2x u_x + \sin 2y u_y + \sin 2z u_z = 2$$

5. If $u = \log (x^3 + y^3 + z^3 - 3xyz)$ then prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$ and hence show

$$\text{that } \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x + y + z)^2}$$

Solution : $u = \log (x^3 + y^3 + z^3 - 3xyz)$ is a symmetric function

$$\frac{\partial u}{\partial x} = \frac{3x^2 - 3yz}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(1)$$

$$\frac{\partial u}{\partial y} = \frac{3y^2 - 3zx}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(2)$$

$$\frac{\partial u}{\partial z} = \frac{3z^2 - 3xy}{x^3 + y^3 + z^3 - 3xyz} \quad \dots(3)$$

Adding (1), (2) and (3) we get,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x^3 + y^3 + z^3 - 3xyz)}$$

Recalling a standard elementary result,

$$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

We have,

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)}$$

$$\text{Thus } \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x + y + z}$$

$$\begin{aligned} \text{Further } \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) u \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} \right) \\ &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) \left(\frac{3}{x + y + z} \right), \text{ by using the earlier result.} \\ &= \frac{\partial}{\partial x} \left(\frac{3}{x + y + z} \right) + \frac{\partial}{\partial y} \left(\frac{3}{x + y + z} \right) + \frac{\partial}{\partial z} \left(\frac{3}{x + y + z} \right) \\ &= \frac{-3}{(x + y + z)^2} + \frac{-3}{(x + y + z)^2} + \frac{-3}{(x + y + z)^2} = \frac{-9}{(x + y + z)^2} \end{aligned}$$

$$\text{Thus } \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x + y + z)^2}$$

6. If $u = f(r)$ and $x = r \cos \theta$, $y = r \sin \theta$,

prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

Solution : Observing the required partial derivative we conclude that u must be a function of x, y . But

$u = f(r)$ by data and hence we need to have r as a function of x, y . Since $x = r \cos \theta$, $y = r \sin \theta$ we have $x^2 + y^2 = r^2$.

$\therefore$ we have $u = f(r)$ where $r = \sqrt{x^2 + y^2}$

$$\frac{\partial^2 u}{\partial x^2} = \frac{f'(r)}{r^3} (r^2 - x^2) + \frac{f''(r)}{r^2} \cdot x^2 \text{ and}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{f'(r)}{r^3} (r^2 - y^2) + \frac{f''(r)}{r^2} \cdot y^2$$

Adding these results we get,

$$\begin{aligned} \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} &= \frac{f'(r)}{r^3} \{2x^2 - (x^2 + y^2)\} + \frac{f''(r)}{r^2} (x^2 + y^2) \\ &= \frac{f'(r)}{r^3} \cdot r^2 + \frac{f''(r)}{r^2} \cdot r^2 = \frac{1}{r} f'(r) + f''(r) \end{aligned}$$

Thus $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = f''(r) + \frac{1}{r} f'(r)$

7. Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$

Proof : Since $u = f(x, y)$ is a homogeneous function of degree n we have by the definition,

$$u = x^n g(y/x) \quad \dots(1)$$

Let us differentiate this w.r.t x and also w.r.t y

$$\therefore \frac{\partial u}{\partial x} = x^n \cdot g'(y/x) \cdot \left(-\frac{y}{x^2}\right) + nx^{n-1} g(y/x)$$

$$\text{ie., } \frac{\partial u}{\partial x} = x^{n-2} y g'(y/x) + nx^{n-1} g(y/x) \quad \dots(2)$$

$$\text{Also } \frac{\partial u}{\partial y} = x^n \cdot g'(y/x) \cdot \left(\frac{1}{x}\right)$$

$$\text{ie., } \frac{\partial u}{\partial y} = x^{n-1} \cdot g'(y/x) \quad \dots(3)$$

Now consider $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$ as a consequence of (2) and (3)

$$= x \left[-x^{n-2} y g'(y/x) + n x^{n-1} g(y/x) \right] + y \left[x^{n-1} g'(y/x) \right]$$

$$= -x^{n-1} y g'(y/x) + n x^n g(y/x) + x^{n-1} y g'(y/x)$$

$$= n \cdot x^n g(y/x)$$

$$= n u, \text{ by using (1)}$$

Thus we have proved Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u ; x u_x + y u_y = n u$$

8. **Prove that** $x^2 \frac{\partial^2 u}{\partial x^2} + 2 x y \frac{\partial^2 u}{\partial x \partial y} = n(n-1)u$

Proof : Since $u = f(x, y)$ is a homogeneous function of degree n , we have Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = n u \quad \dots(1)$$

Differentiating (1) partially w.r.t. x and also w.r.t. y we get,

$$\left(x \frac{\partial^2 u}{\partial x^2} + 1 \cdot \frac{\partial u}{\partial x} \right) + y \frac{\partial^2 u}{\partial x \partial y} = n \frac{\partial u}{\partial x} \quad \dots(2)$$

$$\text{Also, } x \frac{\partial^2 u}{\partial y \partial x} + \left(y \frac{\partial^2 u}{\partial y^2} + 1 \cdot \frac{\partial u}{\partial y} \right) = n \frac{\partial u}{\partial y} \quad \dots(3)$$

We shall now multiply (2) by x and (3) by y .

$$x^2 \frac{\partial^2 u}{\partial x^2} + x \frac{\partial u}{\partial x} + x y \frac{\partial^2 u}{\partial x \partial y} = n x \frac{\partial u}{\partial x} \text{ and}$$

$$x y \frac{\partial^2 u}{\partial y \partial x} + y^2 \frac{\partial^2 u}{\partial y^2} + y \frac{\partial u}{\partial y} = n y \frac{\partial u}{\partial y}$$

Adding these using the fact that $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$ we get,

$$\left(x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} \right) + \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = n \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right)$$

$$\text{ie., } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + n u = n (n u), \text{ by using (1)}$$

$$\text{or } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + n (nu) - nu = n (n-1)u$$

$$\text{Thus } x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} + n (n-1)u$$

$$\text{ie., } x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = n (n-1)u$$

9. If $u = \frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$

Solution : (Observe that the degree is 0 in every term)

$$u = \frac{x}{y+z} + \frac{y}{z+x} + \frac{z}{x+y}$$

We shall divide both numerator and denominator of every term by x.

$$u = \frac{1}{y/x + z/x} + \frac{y/x}{z/x + 1} + \frac{z}{1 + y/x} = x^0 \{g(y/x, z/x)\}$$

$$\Rightarrow u \text{ is homogeneous of degree } 0. \therefore n = 0$$

$$\text{We have Euler's theorem, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = n u$$

$$\text{Putting } n = 0 \text{ we get, } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

10. If $u = \log \left(\frac{x^4 + y^4}{x+y} \right)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

Solution : we cannot put the given u in the form $x^n g(y/x)$

$$\therefore e^u = \frac{x^4 + y^4}{x+y} = \frac{x^4(1 + y^4/x^4)}{x(1 + y/x)} = x^3 \left\{ \frac{1 + (y/x)^4}{1 + (y/x)} \right\}$$

$$\text{ie., } e^u = x^3 g(y/x) \Rightarrow e^u \text{ is homogeneous of degree } 3 \therefore n = 3$$

Now applying Euler's theorem for the homogeneous function e^u

We have $x \frac{\partial(e^u)}{\partial x} + y \frac{\partial(e^u)}{\partial y} = n e^u$

ie., $x e^u \frac{\partial u}{\partial x} + y e^u \frac{\partial u}{\partial y} = 3e^u$

Dividing by e^u we get $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

11. If $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ show that

(i) $x u_x + y u_y = \sin 2 u$

(ii) $x^2 u_{xx} + 2 x y u_{xy} + y^2 u_{yy} = \sin 4 u - \sin 2 u$

Solution : (i) $u = \tan^{-1} \left(\frac{x^3 + y^3}{x - y} \right)$ by data

$$\Rightarrow \tan u = \frac{x^3 + y^3}{x - y} = \frac{x^3(1 + y^3/x^3)}{x(1 - y/x)} = x^2 \left\{ \frac{1 + (y/x)^3}{1 - (y/x)} \right\}$$

ie., $\tan u = x^2/g(y/x) \Rightarrow \tan u$ is homogeneous of degree 2.

Applying Euler's theorem for the function $\tan u$ we have,

$$x \frac{\partial(\tan u)}{\partial x} + y \frac{\partial(\tan u)}{\partial y} = n \cdot \tan u ; n = 2$$

ie., $x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} = 2 \tan u$

or $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{2 \tan u}{\sec^2 u} = 2 \cos^2 u \frac{\sin u}{\cos u} = 2 \cos u \sin u = \sin 2 u$

$\therefore x u_x + y u_y = \sin 2u$

(ii) We have $x u_x + y u_y = \sin 2 u$... (1)

Differentiating (1) w.r.t x and also w.r.t y partially we get

$x u_{xx} + 1 \cdot u_x + y u_{xy} = 2 \cos 2u \cdot u_x$... (2)

And $x u_{yx} + y u_{yy} + 1 \cdot u_y = 2 \cos 2u \cdot u_y$... (3)

Multiplying (2) by x and (3) by y we get,

$x^2 u_{xx} + x u_x + xy u_{xy} = 2 \cos 2 u \cdot x u_x$

$$xy u_{yx} + y^2 u_{yy} + y u_y = 2 \cos 2u \cdot y u_y$$

Adding these by using the fact that $u_{yx} = u_{xy}$, we get

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} + (xy_x + yu_y) = 2 \cos 2u (xu_x + y u_y)$$

By using (1) we have,

$$x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = 2 \cos 2u \sin 2u - \sin 2u$$

(since $\sin 2\theta = 2 \cos \theta \sin \theta$, first term in the R.H.S becomes $\sin 4u$)

$$\text{Thus } x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \sin 4u - \sin 2u$$

12. If $u = f\left(\frac{x}{y}, \frac{y}{z}, \frac{z}{x}\right)$ Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$

>> here we need to convert the given function u into a composite function.

$$\text{Let } u = f(p, q, r) \text{ where } p = \frac{x}{y}, q = \frac{y}{z}, r = \frac{z}{x}$$

$$\text{ie., } \{u \rightarrow (p, q, r) \rightarrow (x, y, z)\} \Rightarrow u \rightarrow x, y, z$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x}$$

$$\text{ie., } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \cdot \frac{1}{y} + \frac{\partial u}{\partial q} \cdot 0 + \frac{\partial u}{\partial r} \cdot \left(-\frac{z}{x^2}\right)$$

$$\therefore x \frac{\partial u}{\partial x} = \frac{x}{y} \frac{\partial u}{\partial p} - \frac{z}{x} \frac{\partial u}{\partial r} \quad \dots(1)$$

Similarly by symmetry we can write,

$$y \frac{\partial u}{\partial y} = \frac{y}{z} \frac{\partial u}{\partial q} - \frac{x}{y} \frac{\partial u}{\partial p} \quad \dots(2)$$

$$z \frac{\partial u}{\partial z} = \frac{z}{x} \frac{\partial u}{\partial r} - \frac{y}{z} \frac{\partial u}{\partial q} \quad \dots(3)$$

$$\text{Adding (1), (2) and (3) we get } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$$

13. If $u = f(x - y, y - z, z - x)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

>> Let $u = f(p, q, r)$ where $p = x - y, q = y - z, r = z - x$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x}$$

$$\text{ie., } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \cdot 1 + \frac{\partial u}{\partial q} \cdot 0 + \frac{\partial u}{\partial r} (-1)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} - \frac{\partial u}{\partial r} \quad \dots(1)$$

Similarly we have by symmetry

$$\frac{\partial u}{\partial y} = \frac{\partial u}{\partial q} - \frac{\partial u}{\partial p} \quad \dots(2)$$

$$\frac{\partial u}{\partial z} = \frac{\partial u}{\partial r} - \frac{\partial u}{\partial q} \quad \dots(3)$$

Adding (1), (2) and (3) we get, $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$

14. If $z = f(x, y)$ where $x = r \cos \theta$ and $y = r \sin \theta$

$$\text{Show that } \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2 = \left(\frac{\partial z}{\partial r} \right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta} \right)^2$$

Solution : $\{z \rightarrow (x, y) \rightarrow (r, \theta)\} \Rightarrow z \rightarrow (r, \theta)$

$$\therefore \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial r}; \frac{\partial z}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial \theta}$$

$$\text{ie., } \frac{\partial z}{\partial r} = \frac{\partial z}{\partial x} \cos \theta + \frac{\partial z}{\partial y} \sin \theta \quad \dots(1)$$

$$\text{and } \frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} (-r \sin \theta) + \frac{\partial z}{\partial y} (r \cos \theta) = r \left[-\frac{\partial z}{\partial x} \sin \theta + \frac{\partial z}{\partial y} \cos \theta \right]$$

$$\text{or } \frac{1}{r} \frac{\partial z}{\partial \theta} = -\frac{\partial z}{\partial x} \sin \theta + \frac{\partial z}{\partial y} \cos \theta$$

squaring and adding (1), (2) and collecting suitable terms have,

$$\begin{aligned} \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 &= \left(\frac{\partial z}{\partial x}\right)^2 [\cos^2 \theta + \sin^2 \theta] \\ &+ \left(\frac{\partial z}{\partial y}\right)^2 [\sin^2 \theta + \cos^2 \theta] + 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \cos \theta \sin \theta - 2 \frac{\partial z}{\partial x} \frac{\partial z}{\partial y} \sin \theta \cos \theta \\ \therefore \left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial z}{\partial \theta}\right)^2 &= \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 \text{ ie., R.H.S = L.H.S} \end{aligned}$$

15. If $z = f(x, y)$ where $x = e^u + e^{-v}$, $y = e^{-u} - e^v$

Prove that $x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v}$

Solution : $\{z \rightarrow (x, y) \rightarrow (u, v)\} \Rightarrow z \rightarrow (u, v)$

$$\therefore \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}; \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v}$$

$$\text{ie., } \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot e^u + \frac{\partial z}{\partial y} (-e^{-u}) \quad \dots(1)$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} (-e^{-v}) + \frac{\partial z}{\partial y} (-e^v) \quad \dots(2)$$

Consider R.H.S = $\frac{\partial z}{\partial u} - \frac{\partial z}{\partial v}$ and (1) - (2) yields

$$\frac{\partial z}{\partial x} (e^u + e^{-v}) - \frac{\partial z}{\partial y} (e^{-u} - e^v) = \frac{\partial z}{\partial x} \cdot x - \frac{\partial z}{\partial y} \cdot y$$

$$\text{Thus } \frac{\partial z}{\partial u} - \frac{\partial z}{\partial v} = x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} \quad \text{ie., R.H.S = L.H.S}$$

16. Find $\frac{\partial(u, v, w)}{\partial(x, y, z)}$ where $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$

Solution : The definition of $J = \frac{\partial(u, v, w)}{\partial(x, y, z)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix}$

But $u = x^2 + y^2 + z^2$, $v = xy + yz + zx$, $w = x + y + z$

Substituting for the partial derivatives we get

$$J = \begin{vmatrix} 2x & 2y & 2z \\ y+z & x+z & y+x \\ 1 & 1 & 1 \end{vmatrix}$$

Expanding by the first row,

$$\begin{aligned} J &= 2x \{(x+z) - (y+x)\} - 2y \{(y+z) - (y+x)\} \\ &\quad + 2z \{(y+z) - (x+z)\} \\ &= 2x(z-y) - 2y(z-x) + 2z(y-x) \\ &= 2(xz - xy - yz + xy + yz - xz) = 0 \quad \text{Thus } J = 0 \end{aligned}$$

17. If $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$, show that $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$

Solution : by data $u = \frac{yz}{x}$, $v = \frac{zx}{y}$, $w = \frac{xy}{z}$

$$\begin{aligned} \frac{\partial(u, v, w)}{\partial(x, y, z)} &= \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{vmatrix} = \begin{vmatrix} -\frac{yz}{x^2} & \frac{z}{x} & \frac{y}{x} \\ \frac{z}{y} & -\frac{zx}{y^2} & \frac{x}{y} \\ \frac{y}{z} & \frac{x}{z} & -\frac{xy}{z^2} \end{vmatrix} \\ &= \frac{-yz}{x^2} \left\{ \left(-\frac{zx}{y^2} \right) \left(-\frac{xy}{z^2} \right) - \left(\frac{x}{z} \right) \left(\frac{x}{y} \right) \right\} \\ &\quad - \frac{z}{x} \left\{ \frac{z}{y} \left(-\frac{zx}{y^2} \right) - \frac{y}{z} \cdot \frac{x}{y} \right\} + \frac{y}{x} \left\{ \frac{z}{y} \cdot \frac{x}{z} - \frac{y}{z} \left(-\frac{zx}{y^2} \right) \right\} \\ &= \frac{-yz}{x^2} \left\{ \frac{x^2}{yz} - \frac{x^2}{yz} \right\} - \frac{z}{x} \left\{ \frac{-x}{z} - \frac{x}{z} \right\} + \frac{y}{x} \left\{ \frac{x}{y} + \frac{x}{y} \right\} \\ &= 0 + 1 + 1 + 1 = 4 \end{aligned}$$

Thus $\frac{\partial(u, v, w)}{\partial(x, y, z)} = 4$

18. If $u + v = e^x \cos y$ and $u - v = e^x \sin y$ find the jacobian of the functions u and v w.r.t x and y .

Solution : we have to find $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}$

Using the given data we have to solve for u and v in terms of x and y .

By data $u + v = e^x \cos y$ (1)

$u - v = e^x \sin y$ (2)

(1) + (2) gives : $2u = e^x (\cos y + \sin y)$

(2) - (1) gives : $2v = e^x (\cos y - \sin y)$

ie., $u = \frac{e^x}{2} (\cos y + \sin y)$; $v = \frac{e^x}{2} (\cos y - \sin y)$

$\therefore \frac{\partial u}{\partial x} = \frac{e^x}{2} (\cos y + \sin y)$, $\frac{\partial v}{\partial x} = \frac{e^x}{2} (-\sin y - \cos y)$

Now $\frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{e^x}{2} (\cos y + \sin y) & \frac{e^x}{2} (-\sin y + \cos y) \\ \frac{e^x}{2} (\cos y - \sin y) & \frac{-e^x}{2} (\sin y + \cos y) \end{vmatrix}$

$= \frac{e^x}{2} \cdot \frac{e^x}{2} \{ -(\cos y + \sin y)^2 - (\cos y - \sin y)^2 \}$

$= \frac{-e^{2x}}{4} \{ 1 + \sin 2y + (1 - \sin 2y) \} = \frac{-e^{2x}}{2}$

Thus $\frac{\partial(u, v)}{\partial(x, y)} = \frac{-e^{2x}}{2}$

19. (a) If $x = r \cos \theta$, $y = r \sin \theta$ find the value of $\frac{\partial(r, \theta)}{\partial(x, y)}$

(b) Further verify that $\frac{\partial(x, y)}{\partial(r, \theta)} \cdot \frac{\partial(r, \theta)}{\partial(x, y)} = 1$

(a) **Solution :** We shall first express r, θ in terms of x and y .

We have $x = r \cos \theta$, $y = r \sin \theta$ by data.

$$\therefore x^2 + y^2 = r^2 \text{ and } \frac{y}{x} = \tan \theta \text{ or } \theta = \tan^{-1}(y/x)$$

$$\text{Consider } r^2 = x^2 + y^2$$

Differentiating partially w.r.t x and also w.r.t y we get,

$$2r \frac{\partial r}{\partial x} = 2x \quad \text{and} \quad 2r \frac{\partial r}{\partial y} = 2y$$

$$\therefore \frac{\partial r}{\partial x} = \frac{x}{r} \quad \text{and} \quad \frac{\partial r}{\partial y} = \frac{y}{r}$$

$$\text{Also consider } \theta = \tan^{-1}(y/x)$$

$$\therefore \frac{\partial \theta}{\partial x} = \frac{1}{1 + (y/x)^2} \cdot \frac{-y}{x^2} \quad \text{and} \quad \frac{\partial \theta}{\partial y} = \frac{1}{1 + (y/x)^2} \cdot \frac{1}{x}$$

$$\text{i.e.,} \quad \frac{\partial \theta}{\partial x} = \frac{-y}{x^2 + y^2} \quad \text{and} \quad \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2}$$

$$\text{Now } \frac{\partial(r, \theta)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{vmatrix} = \begin{vmatrix} \frac{x}{r} & \frac{y}{r} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} \end{vmatrix}$$

$$\text{i.e.,} \quad = \frac{x^2}{r(x^2 + y^2)} + \frac{y^2}{r(x^2 + y^2)} = \frac{(x^2 + y^2)}{r(x^2 + y^2)} = \frac{1}{r}$$

$$\therefore \frac{\partial(r, \theta)}{\partial(x, y)} = \frac{1}{r}$$

Solution (b) : Consider $x = r \cos \theta, y = r \sin \theta$

$$\frac{\partial(x, y)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r(\cos^2 \theta + \sin^2 \theta) = r$$

$$\therefore \frac{\partial(x, y)}{\partial(r, \theta)} = r$$

$$\text{From (1) and (2) : } \frac{\partial(x, y)}{\partial(x, \theta)} \cdot \frac{\partial(r, \theta)}{\partial(x, y)} = r \cdot \frac{1}{r} = 1$$

20. If $x = u(1-v), y = uv$ then show that $JJ' = 1$

Solution : $x = u(1-v); y = uv$

$$\frac{\partial x}{\partial u} = (1-v), \quad \frac{\partial y}{\partial u} = v \quad \frac{\partial x}{\partial v} = -u, \quad \frac{\partial y}{\partial v} = u$$

$$J = \frac{\partial(x, y)}{\partial(u, v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} (1-v) & -u \\ v & u \end{vmatrix}$$

$$= (1-v)u + uv = u \quad \therefore J = u$$

Next we shall express u and v in terms of x and y .

By data $x = u - uv$ and $y = uv$

$$\text{Hence } x + y = u. \text{ Also } v = \frac{y}{u} = \frac{y}{x+y}$$

$$\text{Now we have, } u = x + y; v = \frac{y}{x+y} \quad \therefore \frac{\partial u}{\partial x} = 1, \frac{\partial u}{\partial y} = 1,$$

$$\frac{\partial v}{\partial x} = \frac{(x+y) \cdot 0 - y \cdot 1}{(x+y)^2} = -\frac{y}{(x+y)^2}$$

$$\begin{aligned} \therefore J' &= \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ -y & x \end{vmatrix} \\ &= \frac{x}{(x+y)^2} + \frac{y}{(x+y)^2} = \frac{x+y}{(x+y)^2} = \frac{1}{x+y} = \frac{1}{u} \end{aligned}$$

$$\text{Thus } J' = \frac{1}{u} \text{ Hence } J \cdot J' = u \cdot \frac{1}{u} \text{ Thus } JJ' = 1$$

21. State Taylor's Theorem for Functions of Two Variables.

Statement: Considering $f(x+h, y+k)$ as a function of a single variable x , we have by Taylor's Theorem

$$f(x+h, y+k) = f(x, y+k) + h \frac{\partial f(x, y+k)}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 f(x, y+k)}{\partial x^2} + \dots \quad (1)$$

Now expanding $f(x, y+k)$ as function of y only,

$$f(x, y+k) = f(x, y) + k \frac{\partial f(x, y)}{\partial y} + \frac{k^2}{2!} \frac{\partial^2 f(x, y)}{\partial y^2} + \dots$$

$$\therefore \text{(i) takes the form } f(x+h, y+k) = f(x, y) + h \frac{\partial f(x, y)}{\partial x} + \frac{h^2}{2!} \frac{\partial^2 f(x, y)}{\partial x^2} + \dots +$$

$$+ h \frac{\partial}{\partial x} \left\{ f(x, y) + k \frac{\partial f(x, y)}{\partial y} + \frac{k^2}{2!} \frac{\partial^2 f(x, y)}{\partial y^2} + \dots \right\}$$

$$+ \frac{h^2}{2!} \frac{\partial^2}{\partial x^2} \left\{ f(x, y) + k \frac{\partial f(x, y)}{\partial y} + \dots \right\}$$

$$\text{Hence } f(x+h, y+k) = f(x, y) + h \frac{\partial f}{\partial x} + k \frac{\partial f}{\partial y} + \frac{1}{2!} \left(h^2 \frac{\partial^2 f}{\partial x^2} + 2hk \frac{\partial^2 f}{\partial x \partial y} + k^2 \frac{\partial^2 f}{\partial y^2} \right) + \dots \quad (1)$$

In symbol we write it as

$$F(x+h, y+k) = f(x,y) + \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right) f + \frac{1}{2!} \left(h \frac{\partial}{\partial x} + k \frac{\partial}{\partial y} \right)^2 f + \dots$$

Taking $x = a$ and $y = b$, (1) becomes

$$f(a+h, b+k) = f(a,b) + [h f_x(a,b) + k f_y(a,b)] + \frac{1}{2!} [h^2 f_{xx}(a,b) + 2hkf_{xy}(a,b) + k^2 f_{yy}(a,b)] + \dots$$

Putting $a+h = x$ and $b+k = y$ so that $h = x-a$, $k = y-b$, we get

$$\begin{aligned} F(x,y) &= f(a,b) + [(x-a) f_x(a,b) + (y-b) f_y(a,b)] \\ &= \frac{1}{2!} [(x-a)^2 f_{xx}(a,b) + 2(x-a)(y-b) f_{xy}(a,b) + (y-b)^2 f_{yy}(a,b)] + \dots \quad (2) \end{aligned}$$

This is Taylor's expansion of $f(x,y)$ in powers of $(x-a)$ and $(y-b)$. It is used to expand $f(x,y)$ in the neighborhood of (a,b)

corollary, putting $a = 0$, $b = 0$ in (2), we get

$$f(x,y) = f(0,0) + [x f_x(0,0) + y f_y(0,0)] + \frac{1}{2!} [x^2 f_{xx}(0,0) + 2xy f_{xy}(0,0) + y^2 f_{yy}(0,0)] + \dots \quad (3)$$

This is Maclaurin's Expansion of $f(x,y)$

22. Expand $e^x \log(1+y)$ in powers of x and y up to terms of third degree.

Solution: Here

$$f(x,y) = e^x \log(1+y) \quad \therefore f(0,0) = 0$$

$$f_x(x,y) = e^x \log(1+y) \quad \therefore f_x(0,0) = 0$$

$$f_y(x,y) = e^x \frac{1}{1+y} \quad \therefore f_y(0,0) = 1$$

$$f_{xx}(x,y) = e^x \log(1+y) \quad \therefore f_{xx}(0,0) = 0$$

$$f_{xy}(x,y) = e^x \frac{1}{1+y} \quad \therefore f_{xy}(0,0) = 1$$

$$f_{yy}(x,y) = -e^x (1+y)^{-2} \quad \therefore f_{yy}(0,0) = -1$$

$$f_{xxx}(x,y) = e^x \log(1+y) \quad \therefore f_{xxx}(0,0) = 0$$

$$f_{xyy}(x,y) = e^x \frac{1}{1+y} \quad \therefore f_{xyy}(0,0) = 1$$

$$f_{xyy}(x,y) = -e^x (1+y)^{-2} \quad \therefore f_{xyy}(0,0) = -1$$

$$f_{yyy}(x,y) = 2e^x (1+y)^{-3} \quad \therefore f_{yyy}(0,0) = 2$$

Now, Maclaurin's expansion of $f(x,y)$ gives

$$f(x,y) = f(0,0) + x(f_x(0,0) + y f_y(0,0)) + \frac{1}{2!} \{x^2 f_{xx}(0,0) + 2xy f_{xy}(0,0) + y^2 f_{yy}(0,0)\} + \frac{1}{3!} \{x^3 f_{xxx}(0,0) + 3x^2 y f_{xxy}(0,0) + 3xy^2 f_{xyy}(0,0) + y^3 f_{yyy}(0,0)\} + \dots$$

$$\begin{aligned} \therefore e^x \log(1+y) &= 0 + x.0 + y(1) + \frac{1}{2!} \{x^2.0 + 2xy(1) + y^2(-1)\} \\ &\quad + \frac{1}{3!} \{x^3.0 + 3x^2y(1) + 3xy^2(-1) + y^3(2)\} + \dots \\ &= y + xy - \frac{1}{2} y^2 + \frac{1}{2} (x^2y - xy^2) + \frac{1}{2} y^3 + \dots \end{aligned}$$

23. Expand $f(x,y) = e^x \cos y$ by Taylor's Theorem about the point $\left(1, \frac{\pi}{4}\right)$ up to the Second degree terms.

Solution: $f(x,y) = e^x \cos y$ and $a = 1, b = \frac{\pi}{4} \quad \therefore f = \left(1, \frac{\pi}{4}\right) = \frac{e}{\sqrt{2}}$

$$f_x(x,y) = e^x \cos y \quad \therefore f\left(1, \frac{\pi}{4}\right) = \frac{e}{\sqrt{2}}$$

$$f_y(x,y) = -e^x \sin y \quad \therefore f_y\left(1, \frac{\pi}{4}\right) = -\frac{e}{\sqrt{2}}$$

$$f_{xx}(x,y) = e^x \cos y \quad \therefore f_{xx}\left(1, \frac{\pi}{4}\right) = \frac{e}{\sqrt{2}}$$

$$f_{xy}(x,y) = -e^x \sin y \quad \therefore f_{xy}\left(1, \frac{\pi}{4}\right) = -\frac{e}{\sqrt{2}}$$

$$f_{yy}(x,y) = -e^x \cos y \quad \therefore f_{yy}\left(1, \frac{\pi}{4}\right) = -\frac{e}{\sqrt{2}}$$

Hence by Taylor's Theorem, we obtain

$$\begin{aligned}
 f(x,y) &= f\left(1, \frac{\pi}{4}\right) + \left[(x-1)f_x + \left(y - \frac{\pi}{4}\right)f_y\right] + \\
 &\quad \frac{1}{2!} \left[(x-1)^2 f_{xx} + 2(x-1)\left(y - \frac{\pi}{4}\right) f_{xy} + \left(y - \frac{\pi}{4}\right)^2 f_{yy}\right] + \dots \\
 \text{i.e., } e^x \cos y &= \frac{e}{\sqrt{2}} + \left[(x-1)\frac{e}{\sqrt{2}} + \left(y - \frac{\pi}{4}\right)\left(-\frac{e}{\sqrt{2}}\right)\right] + \frac{1}{2!} \\
 &\quad \left[(x-1)^2 \frac{e}{\sqrt{2}} + 2(x-1)\left(y - \frac{\pi}{4}\right)\left(-\frac{e}{\sqrt{2}}\right) + \left(y - \frac{\pi}{4}\right)^2 \left(\frac{-e}{\sqrt{2}}\right)\right] + \dots \\
 e^x \cos y &= \frac{e}{\sqrt{2}} \left[1 + (x-1) - \left(y - \frac{\pi}{4}\right)\right] + \frac{1}{2!} \left[(x-1)^2 - 2(x-1)\left(y - \frac{\pi}{4}\right) - \left(y - \frac{\pi}{4}\right)^2\right] + \dots
 \end{aligned}$$

Exercise:

- 1) Expand e^{xy} up to Second degree terms by using Maclaurin's theorem
- 2) Expand $\log(1 - x - y)$ up to Third degree terms by using Maclaurin's theorem
- 3) Expand x^2y about the point $(1, -2)$ by Taylor's expansion
- 4) Obtain the Taylor's expansion of $e^x \sin y$ about the point $\left(0, \frac{\pi}{2}\right)$ up to Second degree terms
- 5) Expand $e^{\sin x}$ up to the term containing x^4

Maxima and Minima:-

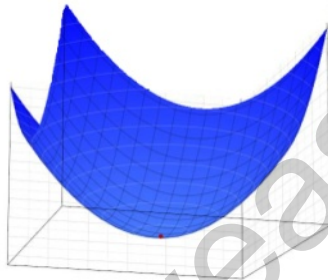
In mathematics, the maximum and minimum (plural: maxima and minima) of a function, known collectively as extrema (singular: extremum), are the largest and smallest value that the function takes at a point within a given neighborhood.

A function $f(x, y)$ is said to have a Maximum value at (a, b) if there exists a neighborhood point of (a, b) (say $(a+h, b+k)$) such that $f(a, b) > f(a+h, b+k)$.

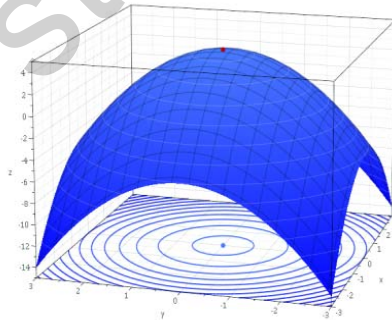
Similarly,

Minimum value at (a, b) if there exists a neighborhood point of (a, b) (say $(a+h, b+k)$) such that $f(a, b) < f(a+h, b+k)$.

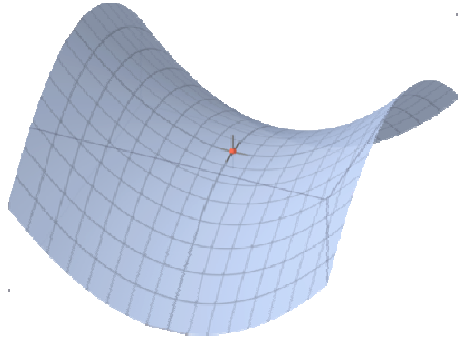
A Minimum point on the graph (in red) $f(x, y) = x^2 + y^2(1-x)^3$



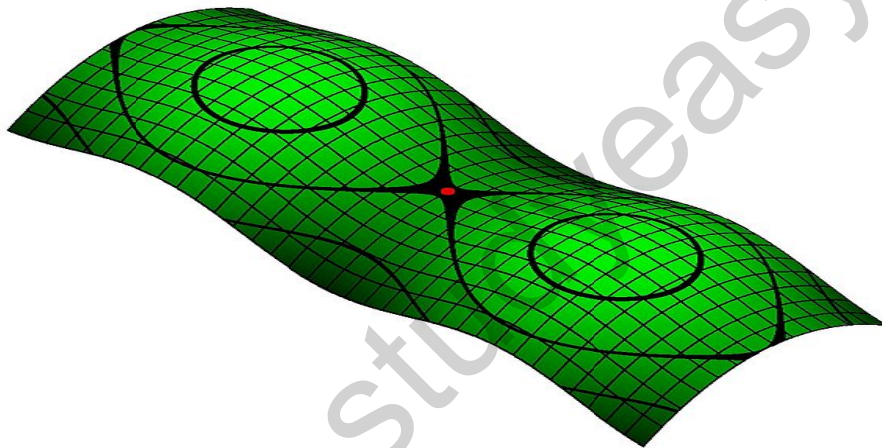
A Maximum point on the graph is at the top (in red)



A saddle point on the graph of $z=x^2-y^2$ (in red)



Saddle point between two hills.



Necessary and Sufficient Condition:-

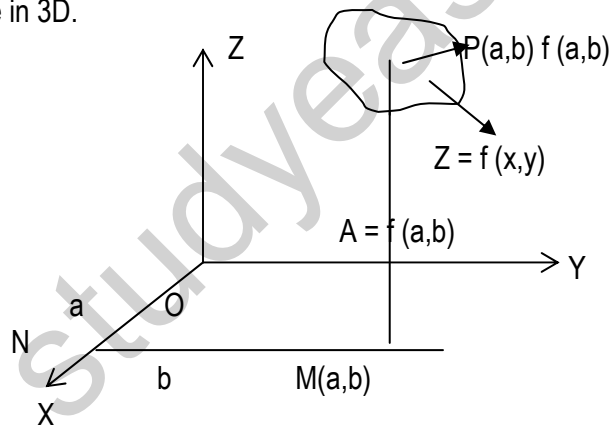
- If $f_x = 0$ and $f_y = 0$ (Necessary Condition)
 - Function will be minimum if $AC - B^2 > 0$ and $A > 0$
 - Function will be maximum if $AC - B^2 > 0$ and $A < 0$
 - Function will be neither maxima nor minima if $AC - B^2 < 0$
 - If $AC - B^2 = 0$ we cannot make any conclusion without any further analysis
- where $A = f_{xx}$, $B = f_{xy}$, $C = f_{yy}$

Working Procedure:-

- First we find Stationary points by considering $f_x = 0$ and $f_y = 0$.
- Function will be minimum if $AC - B^2 > 0$ and $A > 0$ at that stationary point
- Function will be maximum if $AC - B^2 > 0$ and $A < 0$ at that stationary point
- Function will be neither maximum nor minimum if $AC - B^2 < 0$ at that stationary point and it is called as *SADDLE POINT*.

25. Explain Maxima & Minima for Functions of Two Variables & hence obtain the Necessary Conditions for Maxima, Minima.

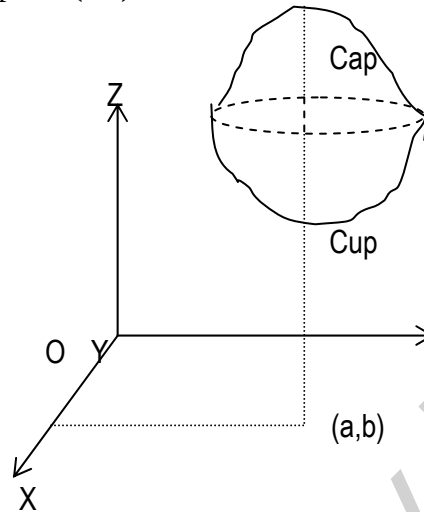
Solution: Let $Z = f(x, y)$ be a given function of two independent variables x & y . The above equation represents a surface in 3D.



A given points (a, b) on the surface has Co-ordinates $[a, b, f(a, b)]$

Definition:

The function $Z = f(x,y)$ is said to be a maximum at the point (a,b) if $f(x,y) < f(a,b)$ in the neighborhood of the point (a,b)



Definition:

The function $Z = f(x,y)$ is said to possess a minimum at the point (a,b) if $f(x,y) > f(a,b)$ in the neighborhood of the point (a,b)

Necessary Condition for Maxima, Minima:

If $Z = f(x,y)$ has a max or min at (a,b) then $f_x(a,b) = 0$, $f_y(a,b) = 0$

Sufficient Conditions for Maxima, Minima:

Put $R = f_{xx}(a,b)$, $S = f_{xy}(a,b)$, $T = f_{yy}(a,b)$

(1) Suppose $S^2 - RT > 0$

There is no maxima or minima at (a,b)

(2) Suppose $S^2 - RT < 0$

Thus there is maxima or minima at (a,b) according as $R < 0$ Or $R > 0$

(3) Suppose $S^2 - RT = 0$, Then there is a saddle point at (a,b)

26. Find the maxima and minima of the functions $f(x,y) = x^3 + y^3 - 3axy$, $a > 0$ is constant.

Solution: Given $f(x,y) = x^3 + y^3 - 3axy$

$$f_x = 3x^2 - 3ay, f_y = 3y^2 - 3ax$$

$$f_{xx} = 6x, f_{yy} = 6y.$$

Put $f_x = 0, f_y = 0$ and solve

$$\text{i.e., } 3x^2 - 3ay = 0 \text{ \& } 3y^2 - 3ax = 0$$

$$\text{i.e., } x^2 = ay \text{ \& } y^2 = ax$$

$$\Rightarrow y = \frac{x^2}{a} \quad \therefore \Rightarrow \left(\frac{x^2}{a} \right)^2 = ax \quad (\because x^2 = ay)$$

$$\therefore \frac{x^4}{a^2} = ax$$

$$\therefore x^4 = a^3x$$

$$\text{i.e., } x(x^3 - a^3) = 0$$

$$\therefore x = 0, x = a$$

$$\Rightarrow y = 0, y = \pm a$$

$\therefore$ The critical Or stationary points are $(0,0)$, (a,a) and $(a,-a)$

(1) At $(0,0)$

$$R = f_{xx}(0,0) = 0$$

$$S = f_{xy}(0,0) = -3a$$

$$T = f_{yy}(0,0) = 0$$

$$\therefore S^2 - RT = 9a^2 - 0 = 9a^2 > 0$$

$\therefore$ There is neither a maximum or a minimum at $(0,0)$

27. Examine the following functions for extreme values $f = x^4 + y^4 - 2x^2 + 4xy - 2y^2$

Solution:

$$f_x = 4x^3 - 4x + 4y$$

$$f_y = 4y^3 - 4x - 4y$$

$$f_{xy} = 4, \quad f_{xx} = 12x^2 - 4, \quad f_{yy} = 12y^2 - 4$$

Put $f_x = 0, f_y = 0$ and solve

$$\text{i.e., } 4x^3 - 4x + 4y = 0 \quad \rightarrow (1)$$

$$4y^3 + 4x - 4y = 0 \quad \rightarrow (2)$$

Adding (1) & (2), we get

$$4(x^3 + y^3) = 0$$

$$\text{i.e., } x^3 + y^3 = 0$$

$$\text{i.e., } y = -x$$

Substitute $y = -x$ in (1), we get

$$4x^3 - 4x - 4x = 0$$

$$\text{i.e., } 4x^3 - 8x = 0$$

$$\text{i.e., } x^3 - 2x = 0 \Rightarrow x(x^2 - 2) = 0$$

$$\text{i.e., } x = 0 \text{ \& } x^2 - 2 = 0$$

$$\text{i.e., } x = 0 \text{ \& } x = \pm \sqrt{2}$$

$$x = \sqrt{2}, -\sqrt{2}$$

$$\therefore x = 0, \sqrt{2}, -\sqrt{2} \text{ and corresponding values of } y \text{ are } y = 0, -\sqrt{2}, \sqrt{2}$$

$$\therefore \text{ The critical points are } (0,0), \left(\sqrt{2}, -\sqrt{2} \right), \left(-\sqrt{2}, \sqrt{2} \right)$$

(1) at (0,0)

$$R = f_{xx}(0,0) = -4$$

$$S = f_{xy}(0,0) = 4$$

$$T = f_{yy}(0,0) = -4$$

$$\therefore S^2 - RT = 16 - (-4)(-4) = 16 - 16 = 0$$

i.e., $S^2 - RT = 0$, These is a saddle point at $(0,0)$

(2) at $(\sqrt{2}, -\sqrt{2})$

$$R = f_{xx}(\sqrt{2}, -\sqrt{2}) = 24 - 4 = 20$$

$$S = f_{xy}(\sqrt{2}, -\sqrt{2}) = 4$$

$$T = f_{yy}(\sqrt{2}, -\sqrt{2}) = 20$$

$$\therefore S^2 - RT = 16 - (20)(20) = 16 - 400 = -384 < 0$$

Thus these is neither maximum nor minimum according to $R < 0$ or $R > 0$ at $(\sqrt{2}, -\sqrt{2})$

Hence $R = 20 > 0$

$\therefore$ There is a minimum at $(\sqrt{2}, -\sqrt{2})$

$$\begin{aligned} \therefore f_{\min} &= (\sqrt{2})^4 + (-\sqrt{2})^4 - 2(\sqrt{2})^2 + 4\sqrt{2}(-\sqrt{2}) - 2(-\sqrt{2})^2 \\ &= 4 + 4 - 4 - 8 - 4 \\ &= -8 \end{aligned}$$

(3) at $(-\sqrt{2}, \sqrt{2})$

$$R = f_{xx}(-\sqrt{2}, \sqrt{2}) = 20 > 0$$

$$S = f_{xy}(-\sqrt{2}, \sqrt{2}) = 4$$

$$T = f_{yy}(-\sqrt{2}, \sqrt{2}) = 20$$

$$\therefore S^2 - RT = 16 - 400 = -384 < 0$$

Since $R > 0$, $\therefore$ There is minima at $(-\sqrt{2}, \sqrt{2})$

$\therefore f_{\min} = -8$ at $(-\sqrt{2}, \sqrt{2})$

$\therefore$ Extreme Value = - 8 at $(-\sqrt{2}, \sqrt{2})$ & $(-\sqrt{2}, \sqrt{2})$

Exercise:

- 1) Find the extreme values of $f = x^3 y^2 (1 - x - y)$
- 2) Determine the maxima or minima of the function $\sin x + \sin y + \sin (x + y)$
- 3) Examine the function $f(x, y) = 1 + \sin(x^2 + y^2)$ for extremum.

28. If $PV^2 = K$ and if the relative errors in P is 0.05 and in V is 0.025 show that the error in K is 10%.

Solution : $PV^2 = K$ by data. Also $\frac{\delta P}{P} = 0.05$ and $\frac{\delta V}{V} = 0.025$

$$\Rightarrow \log P + 2 \log V = \log K$$

$$\Rightarrow \delta(\log P) + 2\delta(\log V) = \delta(\log K)$$

$$\text{i.e., } \frac{1}{P} \delta P + 2 \cdot \frac{1}{V} \delta V = \frac{1}{K} \delta K$$

$$\text{i.e., } 0.05 + 2(0.025) = \frac{\delta K}{K} \text{ or } \frac{\delta K}{K} = 0.1$$

$$\therefore \frac{\delta K}{K} \times 100 = (0.1) \times 100 = 10$$

Thus the error in K is 10%.

29. The time T of a complete oscillation of a simple pendulum is given by the formula

$$T = 2\pi\sqrt{l/g}.$$

- (i) If g is a constant find the error in the calculated value of T due to an error of 3% in the value of l .
- (ii) Find the maximum error in T due to possible errors upto 1% in l and 3% in g .

Solution :

$$(i) \quad T = 2\pi\sqrt{l/g}, \quad g = \text{Constant}, \quad \frac{\delta l}{l} \times 100 = 3$$

$$\Rightarrow \log T = \log 2\pi + \frac{1}{2}(\log l - \log g)$$

$$\Rightarrow \delta(\log T) = \delta(\log 2\pi) + \frac{1}{2}\delta(\log g)$$

$$\text{i.e., } \frac{\delta T}{T} = 0 + \frac{1}{2} \frac{\delta l}{l} - 0$$

$$\text{or } \frac{\delta T}{T} \times 100 = \frac{1}{2} \left(\frac{\delta l}{l} \times 100 \right) = \frac{1}{2} (3) = 1.5$$

$\therefore$ the error in $T = 1.5\%$.

(ii) If g is not a constant we have,

$$\frac{\delta T}{T} \times 100 = \frac{1}{2} \left(\frac{\delta l}{l} \times 100 \right) - \frac{1}{2} \left(\frac{\delta g}{g} \times 100 \right)$$

The error in T will be maximum if the error in l is positive and the error in g is negative (or vice-versa) as the difference in errors converts in to a sum.

$$\therefore \max \left(\frac{\delta T}{T} \times 100 \right) = \frac{1}{2} (+1) - \frac{1}{2} (-3) = 2$$

$\therefore$ the maximum error in T is 2%.

30. The current measured by a tangent galvanometer is given by the relation $c = k \tan \theta$ where θ is the angle of deflection. Show that the relative error in c due to a given error in θ is minimum when $\theta = 45^\circ$.

Solution : Consider $c = k \tan \theta$. k is taken as a constant.

$$\Rightarrow \log c = \log k + \log (\tan \theta)$$

$$\Rightarrow \delta(\log c) = \delta(\log k) + \delta \log (\tan \theta)$$

$$\text{i.e., } \frac{1}{c} \delta c = 0 + \frac{\sec^2 \theta}{\tan \theta} \delta \theta$$

$$\text{i.e., } \frac{\delta c}{c} = \frac{\cos \theta}{\sin \theta} \cdot \frac{1}{\cos^2 \theta} \delta \theta \quad \text{or} \quad \frac{\delta \theta}{c} = \frac{\delta \theta}{\sin \theta \cos \theta}$$

$$\text{i.e., } \frac{\delta c}{c} = \frac{2}{\sin 2\theta} \delta \theta$$

The relative error in c being $\delta c / c$ minimum when the denominator of the R.H.S. is maximum and the maximum value of a sine function is 1.

$$\therefore \sin 2\theta = 1 \Rightarrow 2\theta = 90^\circ \text{ or } \theta = 45^\circ.$$

Thus the relative error in c is minimum when $\theta = 45^\circ$

31. If $T = \frac{1}{2} mv^2$ is the kinetic energy, find approximately the change in T as m changes

from 49 to 49.5 and v changes from 1600 to 1590. 6 Marks

Solution : We have by data $T = \frac{1}{2} mv^2$ and

$$m = 49, m + \delta m = 49.5 \quad \therefore \delta m = 0.5$$

$$v = 1600, v + \delta v = 1590 \therefore \delta v = -10$$

We have to find δT . (logarithm is not required)

$$\therefore \delta T = \frac{1}{2} \delta (mv^2)$$

$$\frac{1}{2} \{m(2v\delta v) + \delta m \cdot v^2\}$$

$$\text{i.e., } = \frac{1}{2} \{(49) (2) (1600) (-10) + (0.5) (1600)^2\} = 1,44,000$$

Thus the change in $T = \delta T = -1,44,000$

32. The pressure p and the volume v of a gas are connected by the relation $p v^{1.4} = \text{constant}$. Find the percentage increase in pressure corresponding to a diminution of $\frac{1}{2}\%$ in volume.

Solution :

$$p v^{1.4} = \text{Constant} = c(\text{say}), \text{ by data.}$$

$$\Rightarrow \log p + 1.4 \log v = \log c$$

$$\Rightarrow \delta(\log p) + 1.4\delta(\log v) = \delta(\log c)$$

$$\text{i.e., } \frac{\delta p}{p} + 1.4 \left(\frac{\delta v}{v} \right) = 0; \text{ But } \frac{\delta v}{v} \times 100 = -\frac{1}{2}, \text{ by data.}$$

$$\therefore \frac{\delta p}{p} \times 100 + 1.4 \left(\frac{\delta v}{v} \times 100 \right) = 0 \text{ or } \frac{\delta p}{p} \times 100 = +0.7.$$

Thus the percentage increase in pressure = 0.7

33. Find the percentage error in the area of an ellipse when an error of $\pm 1\%$ is made in measuring the major and minor axis.

Solution : For the ellipse $x^2/a^2 + y^2/b^2 = 1$ the area (A) is given by πab where $2a$ and $2b$ are the lengths of the major and minor axis.

Let $2a = x$ and $2b = y$.

$$\text{By data } \frac{\delta x}{x} \times 100 = 1, \frac{\delta y}{y} \times 100 = 1.$$

$$A = \pi ab = \pi \cdot \frac{x}{2} \cdot \frac{y}{2} = \frac{\pi}{4} xy$$

$$\therefore \log A = \log (\pi / 4) + \log x + \log y$$

$$\Rightarrow \delta(\log A) = \delta \log (\pi / 4) + \delta(\log x) + \delta(\log y)$$

$$\text{i.e., } \frac{\delta A}{A} = 0 + \frac{\delta x}{x} + \frac{\delta y}{y} \text{ or } \frac{\delta A}{A} \times 100 = \frac{\delta x}{x} \times 100 + \frac{\delta y}{y} \times 100$$

$$\therefore \frac{\delta A}{A} \times 100 = 1 + 1 = 2$$

Thus error in the area = 2%

27. If the sides and angles of a triangle ABC vary in such way that the circum radius remains constant, prove that

$$\frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 0$$

Solution : If the triangle ABC is inscribed in a circle of radius r and if a,b,c respectively denotes the sides opposite to the angles A,B,C we have the sine rule (formula) given by

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2r$$

$$\text{or } a = 2r \sin A, b = 2r \sin B, c = 2r \sin C$$

$$\Rightarrow \delta a = 2r \delta (\sin A), \delta b = 2r \delta (\sin B), \delta c = 2r \delta (\sin C)$$

$$\text{i.e., } \delta a = 2r \cos A \delta A, \delta b = 2r \cos B \delta B, \delta c = 2r \cos C \delta C$$

$$\text{or } \frac{\delta a}{\cos A} = 2r \delta A, \frac{\delta b}{\cos B} = 2r \delta B, \frac{\delta c}{\cos C} = 2r \delta C$$

Adding all these results we get,

$$\frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 2r(\delta A + \delta B + \delta C) = 2r \delta (A + B + C)$$

But $A+B+C = 180 = \pi$ radians = constant.

$$\Rightarrow \delta (A + B + C) = \delta (\text{constant}) = 0$$

$$\text{Thus } \frac{\delta a}{\cos A} + \frac{\delta b}{\cos B} + \frac{\delta c}{\cos C} = 0$$

Multiple choice Questions:

DIFFERENTIAL CALCULUS

1) The radius of curvature at any point of catenary $S=C \tan \Phi$ is

- a) $c \sec^2 \Phi$ b) $c \cos^2 \Phi$
c) $c \tan^2 \Phi$ d) none

2) Stationary points of $f(x,y) = \sin x + \sin y + \sin (x + y)$ is

- a) $(\pi/3, \pi/3)$ b) $(\pi/6, \pi/6)$
c) $(\pi/4, \pi/4)$ d) none

3) If the curvature is zero, that point is known as-----

- a) Point of inflection b) Stationary point
c) (a) or (b) d) none

4) The radius of curvature of the curve $y = 4 \sin x - \sin 2x$ at $x=\pi/2$ is

- a) $5\sqrt{5}/4$ b) $-5\sqrt{5}/4$
c) $5\sqrt{5}/2$ d) none

5) The function $x^2+2xy+2y^2+2x+2y$ has a minimum value at

- a) $(-3/2, 1/2)$ b) $(3/2, 1/2)$ c) $(3/2, -1/2)$ d) none

6) The stationary point of $f(x,y,z)=x^2+y^2+z^2$ where x,y,z are connected by $x+y+z=a$ is

- a) (a,a) b) $(-a,-a)$ c) $(2a, 0)$ d) none

7) The radius of curvature of the curve $\sqrt{x}+\sqrt{y}=1$ at the point $(1/4, 1/4)$ is -----

- a) $\rho=1/\sqrt{2}$ b) $\rho=\sqrt{2}$ c) $\rho=-1/\sqrt{2}$ d) none)

8) The expression for the derivative of arc length in Cartesian form is given by

- a) $ds/d\theta = \sqrt{r^2 + (dr/d\theta)^2}$ b) $ds/dx = \sqrt{1 + (dy/dx)^2}$
c) $ds/dr = \sqrt{1 + (d\theta/dr)^2}$ d) $ds/dt = 1 + (dy/dx)^2$

9) The formulae for radius of curvature in Cartesian form is

a) $\rho = (x^1)^2 + (y^1)^2)^{3/2} / x^1 y^{11} + x^{11} y^1$

b) $\rho = \frac{(1 + y_1^2)^{3/2}}{y_2}$

c) $\rho = [(x^1)^2 + (y^1)^2]^{3/2}$

d) $\rho = (1+(y')^2)^{3/2}$

10) The function for which Rolle's theorem is true is:

- a) $f(x)=\log x$ in the interval $[1/2,2]$
- b) $f(x)=|x+1|$ in the interval $[-2,2]$
- c) $f(x)=|x|$ in the interval $[-1,1]$
- d) Non of the above

11) The value of 'c' in Rolle's theorem, where $-\pi/2 < c < \pi/2$ and $f(x)=\cos x$ is equal to:

- a) $\pi/4$
- b) $\pi/3$
- c) π
- d) 0

12). The expansion of $\tan x$ in powers of x by Maclaurin's theorem is valid in the interval :

- a) $(-\infty, \infty)$
- b) $(-3\pi/2, 3\pi/2)$
- c) $(-\pi, \pi)$
- d) $(-\pi/2, \pi/2)$

13) The value of 'c' in Lagrange's mean value theorem, where $[1,2]$ and $f(x)=x(x-1)$ is :

- a) $5/4$
- b) $3/2$
- c) $7/4$
- d) $11/6$

14) The value of 'c' in Rolle's theorem, where $[0, \pi]$ and $f(x)=\sin x$ is equal to:

- a) $\pi/6$
- b) $\pi/3$
- c) $\pi/2$
- d) Non of these

15) The maximum value of $\log x/x$ is:

- a) 1
- b) e
- c) $2/e$
- d) $1/e$

16) The maximum value of $(1/x)^x$ is equal to:

- a) e
- b) 1
- c) $e^{1/e}$
- d) $(1/e)^e$

17) The difference between the maximum and minimum values of the function $a \sin x + b \cos x$ is:

- a) $2\sqrt{a^2 + b^2}$
- b) $2(a^2 + b^2)$
- c) $a^2 + b^2$
- d) $-\sqrt{a^2 + b^2}$

18) Which one of the following statement is correct for the function $f(x)=x^3$

- a) $f(x)$ has a maximum value at $x=0$
- b) $f(x)$ has a minimum value at $x=0$
- c) $f(x)$ has neither a maximum nor a minimum value at $x=0$
- d) $f(x)$ has no point of inflexion

19) which one of the following is not an indeterminate form

- a) $\infty + \infty$
- b) $\infty - \infty$
- c) ∞ / ∞
- d) $0 \times \infty$

20) In Lagrange's mean value theorem, $f'(c)=$

- a) $f(a) - f(b) / b - a$
- b) $f(b) - f(a) / a - b$
- c) $f(b) - f(a) / b - a$
- d) none

21) In Cauchy's mean value theorem , $f'(c) / g'(c) =$

- a) $f(a) - f(b) / g(b) - g(a)$ b) $f(b) - f(a) / g(b) - g(a)$
 c) $f(b) + f(a) / g(b) + g(a)$ d) none

22) $\lim_{x \rightarrow 0} \frac{a^x - b^x}{x}$ is equal to :

- a) 0 b) ∞ c) $\log(a/b)$ d) $\log(a-b)$

23) The value of $\lim_{x \rightarrow 0} \frac{\log x}{x - 1}$ is equal to:

- a) -1 b) ∞ c) 1 d) 0

24) The value of $\lim_{x \rightarrow 0} (1 - \cos x / 3x^2)$ is equal to:

- a) 3 b) 1/3 c) 1/6 d) 1/9

25) $\lim_{x \rightarrow 0} (e^x + e^{-x} - 2) / x^2$ is equal to :

- a) 1 b) -1 c) 1/2 d) -1/2

26) The value of $\lim_{x \rightarrow 0} (1 + x)^{1/x}$ is :

- a) 1 b) -1 c) 1/e d) e

27) The value of $\lim_{x \rightarrow 0} \frac{\sin x}{x} = \dots$

- a) 1 b) 2 c) 3 d) 0

28) The formulae for radius of curvature in polar form is

- a) $\rho = r dr / dp$ b) $\rho = [(r^2 + (r_1')^2)^{3/2}] / [r^2 + 2(r_1')^2 - r r_2']$
 c) $\rho = (r^2 + (r_1')^2)^{3/2}$ d) None

29) The value of $\lim_{x \rightarrow 0} 1^{1/x} = \dots$

- a) 1 b) ∞ c) $-\infty$ d) 0
-

KEY ANSWERS:

1-a	2-a	3-a	4-a	5-a	6-a	7-a	8-b	9-b	10-a
11-d	12-d	13-b	14-c	15-d	16-c	17-a	18-c	19-a	20-c
21-b	22-c	23-c	24-c	25-a	26-c	27-a	28-b	29-b	--

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DIFFERENTIATION UNDER THE INTEGRAL SIGN:

Consider an integral involving one parameter α and denote it as

$$I(\alpha) = \int_a^b f(x, \alpha) dx$$

where a and b may be constants or functions of α . To find the derivative of $I(\alpha)$ when it exists it is not possible to first evaluate this integral and then to find the derivative, such problems are solved by using the following rules.

(A) Leibnitz's Rule for Constant limits of Integration:

Let $f(x, \alpha)$ and $\frac{\partial f}{\partial \alpha}(x, \alpha)$ be continuous functions of x and α then

$$\frac{d}{d\alpha} \int_a^b f(x, \alpha) dx = \int_a^b \frac{\partial f}{\partial \alpha}(x, \alpha) dx,$$

where a, b are constants and independent of parameter α

(B) Leibnitz's Rule for Variable Limits of Integration:

If in the integral $I(\alpha) = \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha) dx$

$f(x, \alpha)$ satisfies the same conditions, and are functions of the parameter α , then

$$\frac{d}{d\alpha} \int_{a(\alpha)}^{b(\alpha)} f(x, \alpha) dx = \int_{a(\alpha)}^{b(\alpha)} \frac{\partial f}{\partial \alpha}(x, \alpha) dx + f(b, \alpha) \frac{db}{d\alpha} - f(a, \alpha) \frac{da}{d\alpha}$$

 **

Example 1: Evaluate $\int_0^\infty \frac{e^{-ax} \sin x}{x} dx$ and hence show that $\int_0^\infty \frac{\sin x}{x} dx = \frac{\pi}{2}$ by using differentiation under integral sign.

Solution: Let $I(\alpha) = \int_0^\infty \frac{e^{-ax} \sin x}{x} dx$

Differentiate w.r.t α by Leibnitz's Rule under integral sign.

$$\begin{aligned} \therefore \frac{d}{d\alpha} (I(\alpha)) &= \int_0^\infty \frac{\partial}{\partial \alpha} \frac{e^{-ax} \sin x}{x} dx \\ &= \int_0^\infty e^{-ax} \sin x dx \\ &= - \left[e^{-ax} \left(\frac{-\alpha \sin x - \cos x}{\alpha^2 + 1} \right) \right]_0^\infty \text{ then} \end{aligned}$$

$$I(\alpha) = -\tan^{-1} \alpha + k \dots \dots \dots (1)$$

To find the constant k, put $\alpha = \infty$ in (i)

$$\therefore I(\infty) = -\tan^{-1} \infty + k = -\frac{\pi}{2} + k \quad \therefore k = \frac{\pi}{2}$$

$$\therefore I(\alpha) = \frac{\pi}{2} - \tan^{-1} \alpha \quad \text{put } \alpha = 0 \text{ we get } \int_0^{\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}$$

Example 2: using differentiation under the integral sign, evaluate

$$\int_0^1 \frac{x^{\alpha}-1}{\log x} dx, \alpha \geq 0$$

Solution: Let $I(\alpha) = \int_0^1 \frac{x^{\alpha}-1}{\log x} dx \dots \dots \dots (1)$

$$\text{then } \frac{d}{d\alpha}(I(\alpha)) = \int_0^1 \frac{\partial}{\partial \alpha} \frac{x^{\alpha}-1}{\log x} dx = \int_0^1 \frac{x^{\alpha} \log x}{\log x} dx = \int_0^1 x^{\alpha} dx = \frac{1}{1+\alpha}$$

Integrating both sides w.r.t α

$$\therefore I(\alpha) = \log(1+\alpha) + C \dots \dots \dots (2)$$

From (1) when $\alpha = 0$ $I(0) = 0$

From (2) when $I(0) = \log(1) + C \therefore C = 0$

The solution is $I(\alpha) = \log(1+\alpha)$

Example 3: Evaluate $\int_0^{\infty} \frac{\tan^{-1} ax}{x(1+x^2)} dx$ using the method of differentiation under integral sign.

Solution: Let $I = \int_0^{\infty} \frac{\tan^{-1} ax}{x(1+x^2)} dx$

Differentiate w.r.t a by Leibnitz's rule under integral sign

$$\begin{aligned} \therefore \frac{d}{da}(I(a)) &= \int_0^{\infty} \frac{\partial}{\partial a} (\tan^{-1} ax) \frac{1}{x(1+x^2)} dx \\ &= \int_0^{\infty} \frac{x}{(1+a^2x^2)} \cdot \frac{1}{x(1+x^2)} dx \end{aligned}$$

Let $\frac{1}{(1+a^2x^2)(1+x^2)} = \frac{Ax+B}{(1+a^2x^2)} + \frac{Cx+D}{(1+x^2)}$ by partial fractions

Solving, $A = 0, B = \frac{a^2}{a^2-1}, C = 0, D = \frac{-1}{a^2-1}$

$$\therefore \frac{d}{da}(I(a)) = \frac{1}{a^2-1} \int_0^{\infty} \left\{ \frac{a^2}{(1+a^2x^2)} - \frac{1}{(1+x^2)} \right\} dx = \frac{\pi}{2(a+1)}$$

Integrating w.r.t $a \therefore I(a) = \int \frac{\pi}{2} \left(\frac{1}{a+1} \right) da = \frac{\pi}{2} \log(a+1) + C$

Then $C = 0$ by putting $a=0$

Example 4: Evaluate $\int_0^\pi \log(1 + a \cos x) dx$ using the method of differentiation under integral sign

Reduction formulae:

$$\begin{aligned} \text{I)} & \int \sin^n \theta \, d\theta \\ \text{II)} & \int \cos^n \theta \, d\theta \\ \text{III)} & \int \sin^m \theta \cos^n \theta \, d\theta \end{aligned}$$

And to evaluate

$$\begin{aligned} \text{I)} & \int_0^{\frac{\pi}{2}} \sin^n \theta \, d\theta \\ \text{II)} & \int_0^{\frac{\pi}{2}} \cos^n \theta \, d\theta \\ \text{III)} & \int_0^{\pi/2} \sin^m \theta \cos^n \theta \, d\theta \end{aligned}$$

$$\begin{aligned} (a) \int \sin^n x \, dx &= \int \sin^{n-1} x \cdot \sin x \, dx \quad \text{With } \theta = x \\ &= \sin^{n-1} x \cdot (-\cos x) - \int (n-1) \sin^{n-2} x \cdot \cos x \cdot (-\cos x) \, dx \\ &= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \cdot (1 - \sin^2 x) \, dx \\ &= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \, dx - (n-1) \int \sin^n x \, dx \\ n \int \sin^n x \, dx &= -\sin^{n-1} x \cdot \cos x + (n-1) \int \sin^{n-2} x \, dx \end{aligned}$$

Or

$$\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cdot \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx \quad \dots\dots\dots (1)$$

Similarly, (b) $\int \cos^n x \, dx = -\frac{\sin x \cdot \cos^{n-1} x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx \quad \dots\dots\dots (2)$

Thus (1) and (2) are the required reduction formulae

•Reduction formulae for $\int \sin^m x \cos^n x \, dx$

Let $I_{m,n} = \int \sin^m x \cos^n x \, dx \dots\dots\dots (1)$

Where m and n are positive integers. Writing

$$I_{m,n} = \int (\sin^{m-1} x)(\sin x \cos^n x) \, dx$$

Shifting the last term on the right hand side to the left hand side and multiplying by (n+1)/(m+n), we get

$$I_{m,n} = - \frac{(\sin^{m-1} x)(\cos^{n+1} x)}{m+n} + \frac{m-1}{m+n} I_{m-2,n}$$

Similarly, by starting with

$$I_{m,n} = \int \cos^{n-1} x (\cos x \sin^m x) \, dx \dots\dots\dots (2)$$

And integrating by parts, we get

$$I_{m,n} = \frac{(\sin^{m+1} x)(\cos^{n-1} x)}{m+n} +$$

TO Evaluate

Then

etc

i)

ii)

(put $x = a \sin \theta$, so that $dx = a \cos \theta \, d\theta$

Also when $x = 0$, $\theta = 0$, when $x = a$, $\theta = \pi/2$)

=

=

=

iii) Evaluate $\int_0^{\pi/2} \frac{1}{(a^2 + x^2)^n} dx$ (put $x = a \tan \theta$, so that $dx = a \sec^2 \theta d\theta$)

Also when $x = 0$, $\theta = 0$, when $x = \infty$, $\theta = \pi/2$

$$\begin{aligned}
 &= \int_0^{\pi/2} \frac{a \sec^2 \theta}{a^{2n} \sec^{2n} \theta} d\theta \\
 &= \frac{1}{a^{2n-1}} \int_0^{\pi/2} \cos^{2n-2} \theta d\theta \\
 &= \frac{1}{a^{2n-1}} \cdot \frac{(2n-3)(2n-5)\dots\dots\dots 3.1}{(2n-2)(2n-4)\dots\dots\dots 4.2} \cdot \frac{\pi}{2}
 \end{aligned}$$

iv) Evaluate $\int \sin^4 x \cos^2 x dx$

$$\begin{aligned}
 &= \int \sin^2 x (\sin x \cos x)^2 dx \\
 &= \int \frac{1-\cos 2x}{2} \left(\frac{\sin 2x}{2}\right)^2 dx \\
 &= \frac{1}{8} \int (\sin^2 2x - \sin^2 2x \cos 2x) dx \\
 &= \frac{1}{8} \left[\int \frac{1-\cos 4x}{2} dx - \frac{1}{2} \int \sin^2 2x (\cos 2x \cdot 2) dx \right] \\
 &= \frac{1}{16} \left[\int dx - \int \cos 4x dx - \frac{\sin^3 2x}{3} dx \right] \\
 &= \frac{1}{16} \left[x - \frac{1}{4} \sin 4x - \frac{1}{3} \sin^3 2x \right]
 \end{aligned}$$

v) Evaluate $\int_0^{\pi/6} \cos^4 \theta \sin^3 6\theta d\theta$

$$\begin{aligned}
 &= \int_0^{\pi/6} \cos^4 \theta (2 \sin 3\theta \cos 3\theta)^3 d\theta \\
 &= 8 \int_0^{\pi/6} \sin^3 \theta \cos^7 3\theta d\theta \\
 &= \frac{8}{3} \int_0^{\pi/2} \sin^3 x \cos^7 x dx \\
 &= \frac{8}{3} \cdot \frac{2 \times 6 \cdot 4 \cdot 2}{10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} = \frac{1}{15}
 \end{aligned}$$

(Put $3\theta = x$; so that $3d\theta = dx$. Also when $\theta = 0$, $x = 0$)

When $\theta = \pi/6$, $x = \pi/2$

vi) Evaluate $\int_0^1 x^4 (1-x^2)^{3/2} dx$

$$\begin{aligned}
 &= \int_0^{\pi/2} \sin^4 t (\cos^2)^{3/2} \cos t dt = \\
 &= \int_0^{\pi/2} \sin^4 t \cos^4 t dt
 \end{aligned}$$

(put $x = \sin t$ so that $dx = \cos t \, dt$, when $x = 0$, $t = 0$ when $x = 1$, $t = \pi/2$)

Tracing of Curves:

For the evaluation of Mathematical quantities such as Area, Length, Volume and Surface area we need the rough graph of the equation in either Cartesian or parametric or polar form depending on the statement of the problem. We use the following theoretical steps to draw the rough graph.

A) Cartesian Curves: $y = f(x)$

a) Symmetry:

- i) If the power of y in the equation is even, the curve is symmetric about x - axis
- ii) If the power of x in the equation is even, the curve is symmetric about y - axis
- iii) If both the powers x and y are even then the curve is symmetric about both the axis.
- iv) If the interchange of x and y leaves the equation unaltered then the curve is symmetric about the line $y = x$
- v) Replacing x by $-x$ and y by $-y$ leaves the equation unchanged the curve has a symmetry in opposite quadrants.

b) Curve through the origin:

The curve passes through the origin, if the equation does not contain constant term.

- c) Find the origin, is on the curve. If it is, find the tangents at 0, by equating the lowest degree terms to zero.

- i) Find the points of intersections with the coordinate axes and the tangents at these points.

For, put $x = 0$ find y ; and put $y = 0$,

find x . At these points, find $\frac{dy}{dx}$.

If $\frac{dy}{dx} = \infty$, then the tangent is parallel to y axis.

If $\frac{dy}{dx} = 0$, then the tangent is parallel to x axis.

- d) **Asymptotes:** express the equation of the curve in the form $y = f(x)$. Equate the denominator to zero. If the denominator contains x , then there is an asymptote.

- e) Find the region in which the curve lies.
f) Find the interval in which the curve is increasing or decreasing.
-

B) Parametric Form: $x=f(t)$, $y=g(t)$

In this case we try to convert the parametric form into Cartesian form by eliminating the parameter (if possible). Otherwise we observe the following

- I) Find dy/dt and dx/dt and hence dy/dx .
 - II) Assign a few values for t and find the corresponding value for x, y, y' .
 - III) Mark the corresponding points, observing the slope at these points.
-

C) Polar curves: $r = f(\theta)$

- a) **Symmetry:**
 1. If the substitution of $-\theta$ for θ in the equation, leaves the equation unaltered, the curve is symmetrical about the initial line.
 2. If the power of r is even, the curve is symmetrical about the pole.
 - b) Form the table, the value of r , for both positive and negative values of θ and hence note how r varies with θ . Find in particular the value of θ which gives $r = 0$ and $r = \infty$.
 - c) Find $\tan \phi$. This will indicate the direction of the tangent.
 - d) Sometimes by the nature of the equation it is possible to ascertain the value of r and θ that are contained between certain limits.
 - e) Transform into Cartesian, if necessary and adopt the method given before.
 - f) Sketch the figure.
-

PROBLEMS FOR TRACING THE CURVES

1. **Astroid** : $x = a \cos^3 t, y = a \sin^3 t$ (or $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$)

It is symmetrical about the x-axis

Limits $|x| \leq a$

and $|y| \leq a$

The curve lies entirely within the square bounded by the lines $x = \pm a, y = \pm a$

Points: we have — [Compiled by www.studyeasy.in](http://www.studyeasy.in)

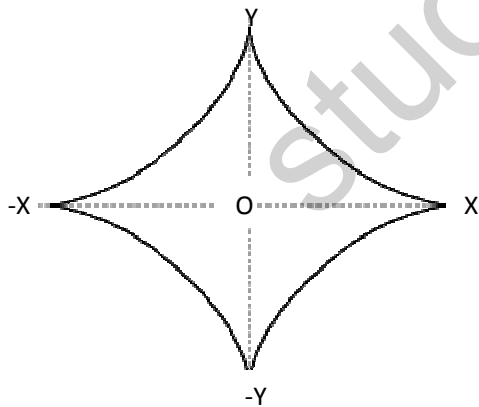
— when $t = 0$ or

— , when $t = -$

As t increases	x	Y	—	Portion traced
From 0 to $-$	+ve and decreases from a to 0	+ve and increases from 0 to a	From 0 to ∞	A to B
From $-$ to π	-ve and increases numerically from 0 to $-a$	+ve and decreases from a to 0	From ∞ to 0	B to C

As t increases from π to 2π , we get the reflection of the curve ABC in the $x -$ axis. The values of $t > 2\pi$ give no new points.

Hence the shape of the curve is as shown in the fig.



Here $ox = oy = a$

It is symmetrical about the y axis. As such we may consider the curve only for positive values of x or θ .

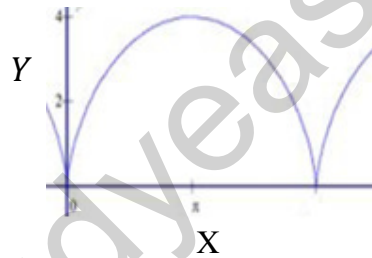
Limits: The greatest value of y is 2a and the least value is zero. Therefore the curve lies entirely between the lines $y = 0$ and $y = 2a$.

Points: We have

— — — — —
— — — — —

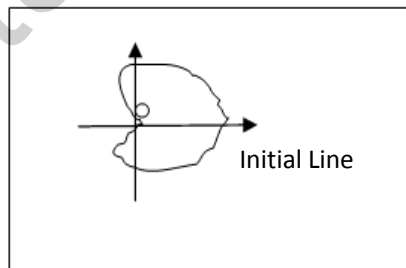
As θ increases	x	Y	—	Portion traced
From 0 to π	increases from 0 to $a\pi$	increases from 0 to 2π	From ∞ to 0	0 to A
From π to 2π	increases from $a\pi$ to $2a\pi$	decreases from $2a$ to 0	From 0 to ∞	A to B

As θ decreases from 0 to -2π , we get the reflection of the arch OAB in the y-axis. Hence the shape of the curve is as in the fig.



2. Cardioid: $r = a(1 + \cos \theta)$

Fig.,



A cardioid is symmetrical about the initial line and lies entirely within the circle $r = 2a$. Its name has been derived from the Latin word 'Kardia'- meaning heart. Because it is a heart shaped curve.

- I) Length
- II) Area
- III) Volume
- IV) Surface area

Table to find the values: Area, Length, Volume, The surface area

Quantity Coordinate system	Area (A)	Length (S)	By revolving about the axis of rotation to form solid	
			Volume (V)	Surface area (SA)
Cartesian form $y = f(x)$	$\int_a^b f(x) dx$	$\int_a^b \sqrt{1 + y_1^2} dx$	$\int_a^b \pi y^2 dx$	$\int_a^b 2\pi y \frac{ds}{dx} dx$ Where $\frac{ds}{dx} = \sqrt{1 + y_1^2}$
Parametric form $x = x(t)$ $y = y(t)$	$\int_a^b f(x) \frac{dx}{dt} dt$	$\int_a^b \sqrt{(x^1)^2 + (y^1)^2} dt$	$\int_a^b \pi y^2 \frac{dx}{dt} dt$	$\int_a^b 2\pi y \frac{ds}{dt} dt$ Where $\frac{ds}{dt} = \sqrt{(x^1)^2 + (y^1)^2}$
Polar form $r = f(\theta)$	$\int_a^b \frac{1}{2} r^2 d\theta$	$\int_a^b \sqrt{r^2 + r_1^2} d\theta$	$\int_a^b \pi y^2 \frac{dx}{d\theta} d\theta$	$\int_a^b 2\pi y \frac{ds}{d\theta} d\theta$ Where $\frac{ds}{d\theta} = \sqrt{r^2 + r_1^2}$

3. Find the entire length of the cardioid $r = a(1 + \cos \theta)$, Also show that the upper half is bisected by $\theta = \frac{\pi}{3}$. The cardioid is symmetrical about the initial line and for its upper half, increases from 0 to π .

Also Also, $\frac{dr}{d\theta} = -a \sin \theta$.

$$\begin{aligned}
 \therefore \text{Length of the curve} &= 2 \int_0^\pi \sqrt{r^2 + \left[\frac{dr}{d\theta}\right]^2} d\theta \\
 &= 2 \int_0^\pi \sqrt{[a(1 + \cos \theta)]^2 + (-a \sin \theta)^2} d\theta \\
 &= 2 \int_0^\pi \sqrt{2(1 + \cos \theta)} d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= 4a \left| \frac{\sin \theta / 2}{1/2} \right|_0^{\pi} \\
 &= 8a (\sin \pi/2 - \sin 0) = 8a
 \end{aligned}$$

$\therefore$ Length of upper half of the curve of the curve is $4a$. Also length of the arch AP from 0 to $\pi/3$

$$\begin{aligned}
 &= 2 \int_0^{\pi/3} \sqrt{2(1 + \cos \theta)} d\theta = 2 \int_0^{\pi/3} \cos \frac{\theta}{2} \cdot d\theta \\
 &= 4a \left| \sin \theta / 2 \right|_0^{\pi/3} \\
 &= 4a \left[\frac{1}{2} - 0 \right] \\
 &= 2a \text{ (half the length of upper half of the cardioids)}
 \end{aligned}$$

Fig.

1. Find the entire length of the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Solution:

The equation to the curve is $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$, the curve is symmetrical about the axis and it meets the 'x' - axis at $x = a$

Fig.

If S_1 = the length of the curve AB

Then required length is $4S_1$

$$S = 4S_1 = \int_0^a \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

Now,

$$x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}} \quad \therefore \frac{dy}{dx} = -\left(\frac{y}{x}\right)^{1/3}$$

$$\begin{aligned}
 S &= 4 \int_0^a \sqrt{1 + \left(\frac{y}{x}\right)^{2/3}} dx \\
 &= 4 \int_0^a \sqrt{\frac{x^{2/3} + y^{2/3}}{x^{2/3}}} dx
 \end{aligned}$$

$$\begin{aligned}
&= 4 \int_0^a \sqrt{\frac{a^{2/3}}{x^{2/3}}} dx \\
&= 4 \int_0^a a^{1/3} x^{-1/3} dx = 4 a^{1/3} \left[\frac{x^{2/3}}{2/3} \right]_0^a \\
s &= 6a \text{ units}
\end{aligned}$$

Fig.

2. Find the perimeter of cardioid $r = a(1 + \cos\theta)$.

Solution:

The equation to the curve is symmetrical about the initial line.

Fig.

$\therefore$ The required length of the curve is twice the length of the curve OPA

At O, $\theta = \pi$ and at A $\theta = 0$

$$\text{Now, } r = a(1 + \cos\theta) \quad \therefore \frac{dr}{d\theta} = -a \sin\theta$$

$$s = 2 \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$s = 2 \int_0^\pi \sqrt{a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta} d\theta$$

$$s = 2 \int_0^\pi 2a \cos \frac{\theta}{2} d\theta$$

$$s = 4a \left[\frac{\sin \frac{\theta}{2}}{\frac{1}{2}} \right]_0^\pi = 8a \text{ units}$$

3. Find the area of the $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

Solution: The parametric equation to the curve $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ is given by :

$$x = a \cos^3\theta, y = a \sin^3\theta$$

$$\text{Area} = 4 \int_0^a y dx$$

$$\text{Put } x = a \cos^3\theta, y = a \sin^3\theta$$

$$\therefore dx = -3a \cos^2\theta \sin\theta d\theta$$

$$\text{when } x = 0, \theta = \pi/2; \text{ when } x = a, \theta = 0$$

$$A = 4 \int_0^{\pi/2} a \sin^3\theta (-3a \cos^2\theta \sin\theta d\theta)$$

$$= 12a^2 \int_0^{\pi/2} \sin^4\theta \cos^2\theta d\theta$$

$$= 12a^2 \left(\frac{3}{4+2} \cdot \frac{1}{2+2} \right) \left(\frac{1}{2} \cdot \frac{\pi}{2} \right)$$

$$= \frac{3\pi a^2}{8} \text{ Sq. units}$$

4. Find the area of the cardioid $r = a(1 + \cos\theta)$.

Solution: The curve is symmetrical about the initial line.

$\therefore$ Total area = 2 $\times$ area above the line $\theta = 0$

$$A = 2 \int_0^{\pi} \frac{1}{2} r^2 d\theta \quad \text{is the formula for area in polar curves}$$

$$A = 2 \int_0^{\pi} \frac{1}{2} a^2 (1 + \cos\theta)^2 d\theta$$

$$A = \int_0^{\pi} a^2 \left(2\cos^2 \frac{\theta}{2} \right)^2 d\theta$$

$$A = 4a^2 \int_0^{\pi} \cos^4 \frac{\theta}{2} d\theta$$

Put $\theta/2 = t \therefore d\theta = 2dt$

$$= 4a^2 \int_0^{\pi/2} \cos^4 t \cdot 2dt$$

$$= 8a^2 \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

$$= \frac{3\pi a^2}{2} \text{ Sq. units}$$

5. Find the area bounded by an arch of the cycloid

$x = a(t - \sin t)$, $y = a(1 - \cos t)$, $0 \leq t \leq 2\pi$ and its base.

Solution: $x = a(t - \sin t)$, $y = a(1 - \cos t)$ for this arch 't' varies from 0 to 2π .

$$\begin{aligned} \therefore \text{Area} &= \int_{t=0}^{2\pi} y \, dx \\ &= \int_{t=0}^{2\pi} a(1 - \cos t) \cdot a(1 - \cos t) \, dt \quad \text{since } dx = a(1 - \cos t) \, dt \\ &= a^2 \int_{t=0}^{2\pi} (1 - \cos t)^2 \, dt \\ &= 4a^2 \int_{t=0}^{2\pi} \sin^4 \frac{t}{2} \, dt \\ &= 8a^2 \int_{t=0}^{\pi} \sin^4 t \, dt \\ &= 16a^2 \int_{t=0}^{\pi/2} \sin^4 t \, dt \\ &= 16a^2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} \\ &= 3\pi a^2 \text{ Sq. units} \end{aligned}$$

6. Find the volume generated by revolving the cardioid $r = a(1 + \cos\theta)$ about the initial line.

Solution: For the curve, varies from 0 to

Find the volume of the solid obtained by revolving the Astroid $x^{2/3} + y^{2/3} = a^{2/3}$

Volume is obtained by revolving the curve from $x = 0$ to $x = a$ about x-axis and taking two times the result.

$$\begin{aligned} V &= 2 \int_0^a \pi y^2 dx \\ &= 2 \int_{\frac{\pi}{2}}^0 \pi (a \sin^3 \theta)^2 (-3a \cos 2\theta \sin \theta) d\theta \\ &= \frac{32\pi a^3}{105} \text{ cubic units} \end{aligned}$$

Problems for practice:

1. Find the surface area of $r = a(1 - \cos \theta)$
2. Find the volume of the solid obtained by revolving the cissoid $y^2(2a - x) = x^3$ about its asymptote.
3. Find the length between $[0, 2\pi]$ of the curve $x = a(\theta + \sin \theta)$, $y = a(1 - \cos \theta)$.

Find the surface area of solid generated by revolving the astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ about the axis.

Solution: The required surface area is equal to twice the surface area generated by revolving the part of the astroid in the first quadrant about the axis.

Taking $x = a \cos^3 t$, $y = a \sin^3 t$ we have,

$$\begin{aligned} \text{Surface area} &= 2 \int_0^{\pi/2} 2\pi y dx = 4 \int_0^{\pi/2} y \frac{dx}{dt} dt \\ &= 4 \int_0^{\pi/2} y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt \\ &= 4\pi \int_0^{\pi/2} (a \sin^3 t) \{ (-3a \cos^2 t \sin t)^2 + (3a \cos^2 t \sin t)^2 \}^{1/2} dt \\ &= 12a^2 \int_0^{\pi/2} \sin^4 t \cos t dt \quad \text{Put } z = \sin t \\ &= \frac{12}{5} \pi a^2 \text{ Sq. units} \end{aligned}$$

7. Find the surface area of the solid generated when the cardioid $r = a(1 + \cos \theta)$ revolves about the initial line.

Solution: The equation to the curve is $r = a(1 + \cos\theta)$. For the upper part of the curve, θ varies from 0 to π

Put $x = r \cos\theta$, $y = r \sin\theta$

$$\begin{aligned}\therefore \text{Surface area} &= \int_0^\pi 2\pi y ds \\ &= 2\pi \int_0^\pi (r \sin\theta) \frac{ds}{d\theta} d\theta \\ &= 2\pi \int_0^\pi (r \sin\theta) \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= 2\pi \int_0^\pi a(1 + \cos\theta) \cdot \sin\theta \sqrt{a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta} d\theta \\ &= 16\pi a^2\end{aligned}$$

UNIT IV: VECTOR CALCULUS

Scalar and Vector point functions:

(I) If to each point $p(R)$ of a region E in space there corresponds a definite scalar denoted by $f(R)$, then $f(R)$ is called a 'scalar point function' in E . The region E so defined is called a scalar field.

Ex: a) The temperature at any instant

b) The density of a body and potential due to gravitational matter.

(II) If to each point $p(R)$ of a region E in space there corresponds a definite vector denoted by $F(R)$, then it is called the vector point function in E . The region E so defined is called a vector field.

Ex: a) The velocity of a moving fluid at any instant

b) The gravitational intensity of force.

Note: Differentiation of vector point functions follows the same rules as those of ordinary calculus.

If $F(x, y, z)$ be a vector point function then

$$\frac{dF}{dt} = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt}$$

$$\frac{dF}{dt} = \frac{\partial F}{\partial x} \frac{dx}{dt} + \frac{\partial F}{\partial y} \frac{dy}{dt} + \frac{\partial F}{\partial z} \frac{dz}{dt}$$

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy + \frac{\partial F}{\partial z} dz$$

$$dF = \left(\frac{\partial}{\partial x} dx + \frac{\partial}{\partial y} dy + \frac{\partial}{\partial z} dz \right) F$$

(1)

Vector operator del (∇)

The operator is of the form

$$\nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z}$$

GRADIENT, DIVERGENCE, CURL (G D C)

Gradient of the scalar point function:

It is the vector point function ∇f defined as the gradient of the scalar point function f and is written as $\text{grad } f$, then

$$\begin{aligned} \text{grad } f &= \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f \\ &= i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \end{aligned}$$

DIVERGENCE OF A VECTOR POINT FUNCTION

The divergence of a continuously differentiable vector point function F ($\text{div } F$) is defined by the equation

$$\text{div } F = \nabla \cdot F = i \frac{\partial F}{\partial x} + j \frac{\partial F}{\partial y} + k \frac{\partial F}{\partial z}$$

If $F = f i + \phi j + \varphi k$ then

$$\operatorname{div} F = \nabla \cdot F = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \cdot (f i + \phi j + \varphi k)$$

$$\nabla \cdot F = \frac{\partial f}{\partial x} + \frac{\partial \phi}{\partial y} + \frac{\partial \varphi}{\partial z}$$

CURL OF A VECTOR POINT FUNCTION

The curl of a continuously differentiable vector point function F is defined by the equation

$$\operatorname{curl} F = \nabla \times F = i \times \frac{\partial F}{\partial x} + j \times \frac{\partial F}{\partial y} + k \times \frac{\partial F}{\partial z}$$

$$\operatorname{curl} F = \nabla \times F = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times (f i + \phi j + \varphi k)$$

$$\operatorname{curl} F = \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ f & \phi & \varphi \end{vmatrix}$$

$$\nabla \times F = i \left(\frac{\partial \varphi}{\partial y} - \frac{\partial \phi}{\partial z} \right) - j \left(\frac{\partial \varphi}{\partial x} - \frac{\partial f}{\partial z} \right) + k \left(\frac{\partial \phi}{\partial x} - \frac{\partial f}{\partial y} \right)$$

$$\nabla f \text{ and } \nabla \times f$$

$$\nabla \cdot F$$

DEL APPLIED TWICE TO POINT FUNCTIONS

Let F and ϕ being vector point functions, we can form their divergence and curl, whereas f being a scalar point function, we can have its gradient only. Then we have Five formulas:

$$\operatorname{div} \operatorname{grad} f = \nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$\operatorname{curl} \operatorname{grad} f = \nabla \times \nabla f = 0$$

$$3. \operatorname{div} \operatorname{curl} F = \nabla \cdot \nabla \times F = 0 \quad \operatorname{div} \operatorname{curl} F = \nabla \cdot \nabla \times F = 0$$

$$\operatorname{curl} \operatorname{curl} F = \operatorname{grad} \operatorname{div} F - \nabla^2 F$$

$$\text{grad div } F = \text{curl curl } F + \nabla^2 F$$

$$\nabla(\nabla \cdot F) = \nabla \times (\nabla \times F) + \nabla^2 F$$

PROOF:

(I) To prove that $\nabla^2 f = \nabla \cdot \nabla f = \nabla \cdot \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) f$

$$= \nabla \cdot \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right)$$

$$= \text{curl grad } f = \nabla \times \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right)$$

$$= \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

$$= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) f$$

$$= \nabla^2 f$$

where $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$ is called Laplacian operator

and $\nabla^2 f = 0$ is called Laplace's equation

(II) To prove that $\text{curl grad } f = 0$

$$\text{curl grad } f = \nabla \times \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) \times \left(i \frac{\partial f}{\partial x} + j \frac{\partial f}{\partial y} + k \frac{\partial f}{\partial z} \right)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix}$$

$$= i \left(\frac{\partial f}{\partial y \partial z} - \frac{\partial f}{\partial y \partial z} \right) - j \left(\frac{\partial f}{\partial x \partial z} - \frac{\partial f}{\partial x \partial z} \right) + k \left(\frac{\partial f}{\partial x \partial y} - \frac{\partial f}{\partial x \partial y} \right)$$

$$\nabla \times \nabla f = 0$$

(III) To prove that $\nabla \cdot \nabla \times F = \left(\sum i \frac{\partial}{\partial x} \right) \cdot \left(i \times \frac{\partial F}{\partial x} + j \times \frac{\partial F}{\partial y} + k \times \frac{\partial F}{\partial z} \right)$

$$= \sum i \cdot \left(i \times \frac{\partial^2 F}{\partial x^2} + j \times \frac{\partial^2 F}{\partial y \partial x} + k \times \frac{\partial^2 F}{\partial z \partial x} \right)$$

$$= \sum \left(i \times i \cdot \frac{\partial^2 F}{\partial x^2} + i \times j \cdot \frac{\partial^2 F}{\partial x \partial y} + i \times k \cdot \frac{\partial^2 F}{\partial x \partial z} \right)$$

$$= \sum \left(k \cdot \frac{\partial^2 F}{\partial x \partial y} - j \cdot \frac{\partial^2 F}{\partial x \partial z} \right)$$

$$= 0$$

(IV) To prove that $\nabla \times \nabla \times F = \left(\sum i \frac{\partial}{\partial x} \right) \times \left(i \times \frac{\partial F}{\partial x} + j \times \frac{\partial F}{\partial y} + k \times \frac{\partial F}{\partial z} \right)$

$$= \sum i \times \left(i \times \frac{\partial^2 F}{\partial x^2} + j \times \frac{\partial^2 F}{\partial x \partial y} + k \times \frac{\partial^2 F}{\partial x \partial z} \right)$$

$$= \sum \left[\left(i \cdot \frac{\partial^2 F}{\partial x^2} \right) i - (i \cdot i) \frac{\partial^2 F}{\partial x^2} \right] + \left[\left(i \cdot \frac{\partial^2 F}{\partial x \partial y} \right) j - (i \cdot j) \frac{\partial^2 F}{\partial x \partial y} \right] \\ + \left[\left(i \cdot \frac{\partial^2 F}{\partial x \partial z} \right) k - (i \cdot k) \frac{\partial^2 F}{\partial x \partial z} \right]$$

$$= \sum i \cdot \frac{\partial^2 F}{\partial x^2} i + i \cdot \frac{\partial^2 F}{\partial x \partial y} j + i \cdot \frac{\partial^2 F}{\partial x \partial z} k - \sum \frac{\partial^2 F}{\partial x^2}$$

$$= \sum i \frac{\partial}{\partial x} \left(i \cdot \frac{\partial F}{\partial x} + j \cdot \frac{\partial F}{\partial y} + k \cdot \frac{\partial F}{\partial z} \right) - \sum \frac{\partial^2 F}{\partial x^2}$$

$$= \nabla(\nabla \cdot F) - \nabla^2 F$$

$$\text{curl curl } F = \nabla(\nabla \cdot F) - \nabla^2 F$$

(V) To prove that

We have by (IV) which implies $\nabla(\nabla \cdot F) = \text{curl curl } F + \nabla^2 F$

$$\nabla(\nabla \cdot F) = \nabla \times (\nabla \times F) + \nabla^2 F$$

$$i + 3j + 2k$$

Problem: Show that $\nabla^2 r^n = n(n+1)r^{n-2}$

Solution: $\nabla^2(r^n) = \nabla \cdot (\nabla r^n)$

$$\begin{aligned} &= \nabla \cdot \left(n r^{n-1} \frac{R}{r} \right) = n \nabla \cdot (r^{n-2} R) = n[(\nabla r^{n-2}) \cdot R + r^{n-2}(\nabla \cdot R)] \\ &= n \left[(n-2)r^{n-3} \frac{R}{r} \cdot R + r^{n-2}(3) \right] \\ &= n[(n-2)r^{n-4}(r^2) + 3r^{n-2}] \\ &= n(n+1)r^{n-2} \end{aligned}$$

Otherwise: $\nabla^2(r^n) = \frac{\partial^2(r^n)}{\partial x^2} + \frac{\partial^2(r^n)}{\partial y^2} + \frac{\partial^2(r^n)}{\partial z^2}$

Now $\frac{\partial(r^n)}{\partial x} = n r^{n-1} \frac{\partial r}{\partial x} = n r^{n-1} \frac{x}{r} = n r^{n-2} x$

$$\begin{aligned} (\partial^2(r^n))/(\partial x^2) &= n[r^{n-2} + (n-2)r^{n-4}x^2] \dots \dots \dots (1) \\ \partial x x] &= n[r^{n-2} + (n-2)r^{n-4}x^2] \dots \dots \dots (1) \end{aligned}$$

Similarly, $\frac{\partial^2(r^n)}{\partial y^2} = n[r^{n-2} + (n-2)r^{n-4}y^2] \dots \dots \dots (2)$

$$\frac{\partial^2(r^n)}{\partial z^2} = n[r^{n-2} + (n-2)r^{n-4}z^2] \dots \dots \dots (3)$$

Adding equations (1), (2) and (3), gives

$$\begin{aligned} \nabla^2(r^n) &= n[3r^{n-2} + (n-2)r^{n-4}(x^2 + y^2 + z^2)] \\ &= n[3r^{n-2} + (n-2)r^{n-4}r^2] = n(n+1)r^{n-2} \end{aligned}$$

In particular $\nabla^2 \left(\frac{1}{r} \right) = 0$

components of velocity and acceleration at $t=2$ in the direction of $i + 3j + 2k$

Solution: *The position vector of the particle at time t is,*

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$$\vec{R} = (t^3 + 1)i + t^2 j + (t + 5) k$$

The velocity $\vec{V} = \frac{d\vec{R}}{dt}$ is given by

$$\vec{V} = \frac{d\vec{R}}{dt} = 3t^2 i + 2 t j + k$$

The acceleration $\vec{a} = \frac{d\vec{V}}{dt} = \frac{d^2\vec{R}}{dt^2}$ is given by

$$\vec{a} = \frac{d^2\vec{R}}{dt^2} = 6 t i + 2 j$$

Then at time $t = 2$, $\vec{V}$ and $\vec{a}$ are given by

$$\vec{V} = \frac{d\vec{R}}{dt} = 12 i + 4 j + k$$

$$\vec{a} = \frac{d^2\vec{R}}{dt^2} = 12 i + 2 j$$

Velocity vector in the direction of $i + 3j + 2k$ is

$$= \vec{V} \cdot \left(\frac{i+3j+2k}{\sqrt{14}} \right) = (12 i + 4 j + k) \cdot \left(\frac{i+3j+2k}{\sqrt{14}} \right) = \frac{26}{\sqrt{14}}$$

Acceleration in the direction of $i + 3j + 2k$ is

$$= \vec{a} \cdot \left(\frac{i+3j+2k}{\sqrt{14}} \right) = (12 i + 2 j) \cdot \left(\frac{i+3j+2k}{\sqrt{14}} \right) = \frac{18}{\sqrt{14}}$$

Problem: Find the unit vector normal to the surface $xy^3z^2 = 4$ at the point $(-1, -1, 2)$

Solution: Vector normal to the given surface is $\nabla(xy^3z^2)$

$$\begin{aligned} \nabla(xy^3z^2) &= \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (xy^3z^2) \\ &= y^3z^2 i + 3xy^2z^2 j + 2xy^3z k \end{aligned}$$

At $(-1, -1, 2)$

$$= 4 i - 12 j + 4 k$$

Hence the desired unit normal to the surface is

$$= \frac{4 i - 12 j + 4 k}{\sqrt{16 + 144 + 16}} = \frac{1}{\sqrt{176}} (4i - 12j + 4k)$$

yz^3 at the point $(2, -1, 1)$ in the direction of $i + 2j + 2k$

Solution: Vector normal to the surface $f(x, y, z)$ is ∇f

$$\text{i.e. } \nabla f = \left(i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right) (xy^2 + yz^3)$$

$$\nabla f = y^2 i + (2xy + z^3) j + (3yz^2) k$$

At $(2, -1, 1)$

$$\nabla f = i - 3j - 3k$$

Directional derivative in the direction of $i + 2j + 2k$ is

$$= (\nabla f)_{(2, -1, 1)} \cdot \frac{i + 2j + 2k}{\sqrt{1 + 4 + 4}} = (i - 3j - 3k) \cdot \frac{i + 2j + 2k}{\sqrt{9}}$$

$$= \frac{1 - 6 - 6}{3} = \frac{-11}{3}$$

$$\nabla^2(r^n) = n(n + 1)r^{n-1}$$

$$\nabla^2(r^n) = \nabla(\nabla r^n)$$

=

Problem: Find the constants a, b, c such that the vector

$$\vec{F} = (x + y + az)\hat{i} + (x + cy + 2z)\hat{j} + (bx + 2y - z)\hat{k} \text{ is irrotational}$$

Solution:

$$\text{Given } \vec{F} = (x + y + az)\hat{i} + (x + cy + 2z)\hat{j} + (bx + 2y - z)\hat{k}$$

is irrotational

Since the vector field is irrotational

$$\text{Therefore } \nabla \times \vec{F} = 0$$

$$\text{Curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + y + az & x + cy + 2z & bx + 2y - z \end{vmatrix}$$

$$= (c + 1)\hat{i} - (1 - a)\hat{j} + (b - 1)\hat{k}$$

$$\text{i.e., } (c + 1)\hat{i} - (1 - a)\hat{j} + (b - 1)\hat{k} = 0$$

This is possible only when, $c - 1 = 0, 1 - a = 0, b - 1 = 0$

which implies $a = 1, b = 1, c = 1$

Orthogonal curvilinear co-ordinates

Let the rectangular co-ordinates (x,y,z) of Any point be expressed as function of (u,v,w) ,

$$\text{So that } x = x(u,v,w), y = y(u,v,w), z = z(u,v,w) \dots (1)$$

Suppose that (1) can be solved for u,v,w in terms of x,y,z

$$\text{i.e } u = u(x,y,z), v = v(x,y,z), w = w(x,y,z) \dots (2)$$

We assume that the functions in (1) and (2) are single valued functions and have continuous partial derivatives so that the correspondence between (x,y,z) and (u,v,w) is unique. Then (u,v,w) are called curvilinear co-ordinates of (x,y,z) . Each of u,v,w has a level of surface through an arbitrary point .

The surface $u = u_0, v = v_0, w = w_0$ are called co-ordinate surface through $p(u_0, v_0, w_0)$

Each pair of these co-ordinate surface intersects in curves called the co-ordinate curves. The curve of intersection of $u = u_0$ will be called the w -curve, for only w changes along this curve. Similarly we define u and v -curves.

In vector notation, (1) can be written as

$$\mathbf{R} = x(u,v,w)\mathbf{I} + y(u,v,w)\mathbf{J} + z(u,v,w)\mathbf{K}$$

$$\therefore d\mathbf{R} = \frac{\partial \mathbf{R}}{\partial u} du + \frac{\partial \mathbf{R}}{\partial v} dv + \frac{\partial \mathbf{R}}{\partial w} dw$$

Then $\frac{\partial R}{\partial u}$ is a tangent vector to the u-curve at p.

If A is a unit vector at p in this direction, then

$$\frac{\partial R}{\partial u} = h_1 A$$

Where $h_1 = \left| \frac{\partial R}{\partial u} \right|$

similarly if B and C be the tangent vector to the v-curve and w-curve at p, then

$$h_2 B = \frac{\partial R}{\partial v} \text{ and } h_3 C = \frac{\partial R}{\partial w}$$

Where $h_2 = \left| \frac{\partial R}{\partial v} \right|$ and $h_3 = \left| \frac{\partial R}{\partial w} \right|$

then (3) can be written as

$$dR = h_1 du A + h_2 dv B + h_3 dw C \dots \dots \dots (4)$$

Since ∇_u is normal to the surface $u = u_0$ at p, therefore, a unit vector in this direction is

given by $A' = \frac{\nabla_u}{|\nabla_u|}$

similarly the unit vector $B' = \frac{\nabla_v}{|\nabla_v|}$ and $C' = \frac{\nabla_w}{|\nabla_w|}$

Are normal to the surface $u = u_0$ and $v = v_0$ at p

Respectively. Thus at each point p of a curvilinear coordinate system there exists two traids of unit vector: A, B, C tangents to the coordinate surfaces

In particular, when the coordinate surfaces intersect at right angles, the three coordinate Curves are also mutually orthogonal and u, v, w are called the orthogonal curvilinear coordinates. In this case A, B, C and A', B', C' are mutually perpendicular unit vector triads and hence become identical. Henceforth, we shall refer to orthogonal curvilinear coordinates only.

Multiplying (3) scalarly by ∇_u , we get

$$\nabla_u \cdot dR = du = \left(\nabla_u \frac{\partial R}{\partial u} \right) du + \left(\nabla_u \frac{\partial R}{\partial v} \right) dv + \left(\nabla_u \frac{\partial R}{\partial w} \right) dw$$

Hence $\nabla_u \frac{\partial R}{\partial u} = 1, \nabla_u \frac{\partial R}{\partial v} = 0, \nabla_u \frac{\partial R}{\partial w} = 0$

Similarly $\nabla_v \frac{\partial R}{\partial u} = 0, \nabla_v \frac{\partial R}{\partial v} = 1, \nabla_v \frac{\partial R}{\partial w} = 0$

And $\nabla_w \frac{\partial R}{\partial u} = 0, \nabla_w \frac{\partial R}{\partial v} = 0, \nabla_w \frac{\partial R}{\partial w} = 1$

These relations show that the sets $\frac{\partial R}{\partial u}, \frac{\partial R}{\partial v}, \frac{\partial R}{\partial w}$ and $\nabla_u, \nabla_v, \nabla_w$ constitute reciprocal system of vectors.

$$\begin{aligned}\nabla_u &= \frac{\frac{\partial R}{\partial v} \times \frac{\partial R}{\partial w}}{\left[\frac{\partial R}{\partial u}, \frac{\partial R}{\partial v} \times \frac{\partial R}{\partial w} \right]} = \frac{(h_2 B) \times (h_3 C)}{[(h_1 A) \cdot (h_2 B) \times (h_3 C)]} \\ &= \frac{h_2 h_3 B \times C}{h_1 h_2 h_3 [ABC]} = \frac{A}{h_1} \\ A &= h_1 \nabla_u \dots \dots \dots (5)\end{aligned}$$

$$\text{Also } A = B \times C = h_2 h_3 \nabla_v \times \nabla_w$$

$$\text{similarly } B = h_2 \nabla_v \text{ and } C = h_3 \nabla_w$$

$$\text{similarly } B = h_1 h_3 \nabla_w \times \nabla_u \text{ and } C = h_1 h_2 \nabla_u \times \nabla_v \dots \dots (6)$$

Arc length and volume elements.

The element of arc length ds is determined from (4).

$$\therefore ds^2 = dR \cdot dR = h_1^2 du^2 + h_2^2 dv^2 + h_3^2 dw^2 \dots \dots (7)$$

The element of arc length ds_1 along u -curve at p is for v and w being constants $ds_1 = h_1 du$

Similarly elements of arc length along v and w

Curves at p are $ds_2 = h_2 dv$ and $ds_3 = h_3 dw$. The

element of arc length ds , therefore, corresponds

to the length of the diagonal of the Parellelopiped.

Also the volume element is given by

$$\begin{aligned} dv &= [(h_1 duA). (h_2 dvB) \times (h_3 dwC)] \\ &= h_1 h_2 h_3 dudvdw \end{aligned}$$

This can also be written as

$$\begin{aligned} dv &= \left[\frac{\partial R}{\partial u} \cdot \frac{\partial R}{\partial v} \times \frac{\partial R}{\partial w} \right] dudvdw \\ &= \frac{\partial(x, y, z)}{\partial(u, v, w)} dudvdw \end{aligned}$$

This is $\frac{\partial(x, y, z)}{\partial(u, v, w)}$ is called Jacobian .

Two special curvilinear systems:

1)Cylindrical co-ordinates:

Any point P(x,y,z) whose projection on the xy-plane is Q(x,y) has the cylindrical coordinates (ρ, ϕ, z) , where $\rho = OQ$, $\phi = XOQ$ and $z = QP$.

The level surfaces $\rho = \rho_0, \phi = \phi_0, z = z_0$ are respectively cylinders about Z-axis ; planes through the Z-axis and planes perpendicular to the Z-axis.

The co-ordinate curves for ρ are rays perpendicular to the Z-axis; for ϕ horizontal circles with centers on the Z-axis; for z lines parallels to the Z-axis.

$$x = \rho \cos \phi,$$

$$\begin{aligned}\therefore (ds)^2 &= (dx)^2 + (dy)^2 + (dz)^2 \\ &= (d\rho)^2 + \rho^2(d\phi)^2 + (dz)^2\end{aligned}$$

So that scale factors are $h_1=1$, $h_2 = \rho$, $h_3= 1$. Also the volume element

$$dV=\rho \, d\rho \, d\phi \, dz.$$

2) Spherical polar co-ordinates:

Let $p(x,y,z)$ be any point whose projection on the xy -plane is $Q(x,y)$.

Then the Spherical polar co-ordinates of p are (r, θ, ϕ) such that $r = op$, $\theta = \angle OPZ$ and $\phi = \angle POQ$.

The level surfaces $r = r_0$, $\theta = \theta_0$, $\phi = \phi_0$ are respectively spheres about O , cones about the Z -axis with vertex at O and planes through the Z -axis. The co-ordinate curves for r are rays from the origin; for θ , vertical circles with centre at O (called meridians); for ϕ , horizontal circles with centres on the Z -axis.

$$\begin{aligned}x &= OQ \cos\phi \\ &= OP \cos(90-\theta)\cos\phi \\ &= r \sin\theta \cos\phi,\end{aligned}$$

$$y = OQ \sin\phi = r \sin\theta \sin\phi$$

$$z = r \cos\theta$$

$$\begin{aligned}\therefore (ds)^2 &= (dx)^2 + (dy)^2 + (dz)^2 \\ &= (dr)^2 + r^2(d\theta)^2 + (r\sin\theta)^2(d\phi)^2\end{aligned}$$

So that the scale factors are $h_1 = 1$, $h_2 = r$, $h_3 = r\sin\theta$

Also the volume element $dv = r^2 \sin\theta dr d\theta d\phi$

UNIT - 6

DIFFERENTIAL EQUATIONS

#####

Text Book:

- ❖ Higher Engineering Mathematics B S Grewal

Reference:

- ❖ Higher Engineering Mathematics John Bird
- ❖ Advanced Engineering Mathematics. Zill and Cullion
- ❖ Internet

Unit –VI:

Goal:

- ❖ One of our goal in this course is to solve or find solutions of differential equations.
- ❖ Sketch the family of curves given by the Differential equation
- ❖ A particular curve of a family may be determined when a point on the curve is specified

Notation:

Leibnitz notation $\frac{dy}{dx} \cdot \frac{d^2y}{dx^2} \cdot \frac{d^3y}{dx^3}$

Prime notation: $y', y'', y''' \dots y^n$

Newton's dot notation: $\frac{d^2s}{dt^2} = -32$ becomes $\ddot{s} = -32$

Learning outcomes

Upon successful completion of ordinary differential equation, it is expected that a student will be able to do the following.

- ❖ Identify an ordinary differential equation & its order
- ❖ Verify whether a given function is a solution of a given ordinary differential equation
- ❖ Classify ordinary differential equations as variable separable, homogenous, linear, exact and reducible to these forms.
- ❖ Find solutions of separable differential equations
- ❖ To be able to derive an ordinary differential equations as the mathematical model for a physical phenomenon.
- ❖ Solve first order homogeneous differential equations.

Introduction to Differential Equations:

- ❖ In Mathematics, a differential equation is an equation in which the derivative of a function appear as variables.
- ❖ Many of the fundamental laws of Physics, Chemistry, Biology and economics can be formulated as differential equations
- ❖ Definition: The derivative of a function $y = \Phi(x)$ is itself another function $\Phi'(x)$ found by an appropriate rule.

For example: The function $y = e^{0.1x^2}$ is differentiable on the interval $(-\infty, \infty)$, and its derivative is $\frac{dy}{dx} = 0.2x e^{0.1x^2}$

If we replace $e^{0.1x^2}$ by the symbol y , we obtain $\frac{dy}{dx} = 0.2xy$

Definition:

A differential equation is an equation which involves differential coefficients or the differentials.

Definition:

A differential equation is simply an equation involving an unknown function and its derivatives

$$1. e^x dx + e^y dy = 0$$

$$2. \frac{d^2x}{dt^2} + n^2x = 0$$

$$3. y = x \frac{dx}{dy} + \frac{x}{\frac{dx}{dy}}$$

$$4. \frac{d^4y}{dt^4} + \frac{d^2x}{dt^2} + \left(\frac{dx}{dt}\right)^5 = e^t$$

$$5. k \frac{d^2y}{dx^2} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}$$

$$6. \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

Definition ODE:

A differential equation involving derivatives with respect to a single independent variable is called an ordinary differential equation.

Partial Differential Equations:

A differential equation involving partial derivatives is called a partial differential equation.

Order of a Differential Equation:

The order of a differential equation is the order of the highest order derivative occurring in it.
Degree of a Differential Equation:

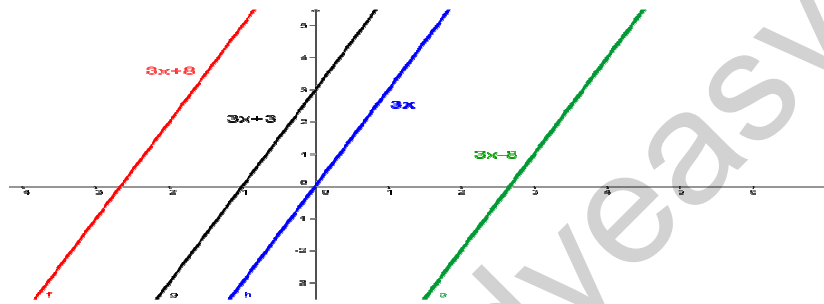
The degree of a differential equation is the degree of the highest order derivative occurring in it, when the derivatives are free from radicals and fractions.

Examples 1, 2, 3, 4 and 6 are of first degree Ex: 5 is of second degree because it can be written as

$$\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^3 = \left[k \frac{d^2 y}{dx^2} \right]^2$$

Family of Curves: Integrating both sides of the derivative $\frac{dy}{dx} = 3$ gives $y = 3x + c$,

Where c is an arbitrary constant $y = 3x + c$, represents a family of curves, each of the curves in the family depending on the value of c . Examples include $y = 3x + 8$, $y = 3x + 3$, $y = 3x$ and $y = 3x - 8$ and these are shown in the following figure.



A **particular curve** of a family may be determined when a **point** on the curve is specified. Thus, if $y = 3x + c$ passes through the point (1, 2), from which $c = -1$. The equation of the curve passing through (1,2) is therefore $y = 3x - 1$

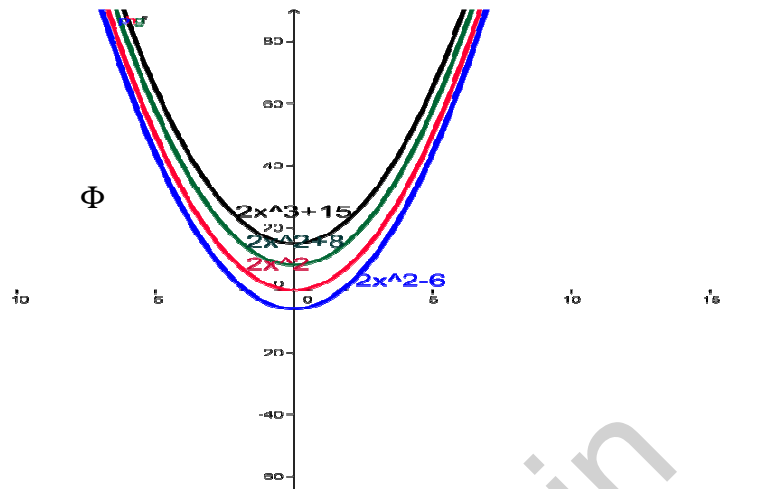
Example: Sketch the family of curves given by the equation $\frac{dy}{dx} = 4x$ and determine the equation of one of these curves which passes through the point (2,3).

Solution: Integrating both sides of $\frac{dy}{dx} = 4x$ with respect to x gives

$$\int \frac{dy}{dx} dx = \int 4x dx \text{ i.e. } y = 2x^2 + c$$

To determine the equation of the curve passing through the point (2, 3), from which $c = -5$

Hence equation of the Curve passing through the point (2,3) is $y = 2x^2 - 5$.



Concept of a Solution of an ODE:

Any function Φ , defined on an interval I and possessing n derivatives that are continuous on I which when substituted into an n^{th} order ordinary differential equation reduces the equation to an identity, is said to be a solution of the equation on the interval

How do you Solve a Differential Equation for Unknown Function: $y = \Phi(x)$?

The problem is loosely equivalent to the familiar reverse problem of differential calculus: Given a derivative, find an antiderivative

For example: In our initial discussion we have already seen that $y = e^{0.1x^2}$ is a solution of $\frac{dy}{dx} = 0.2xy$ on the interval $(-\infty, \infty)$

Solution Curve Or Integral Curve: A solution curve is the graph of a differentiable function

Solution of a Differential Equation:

A solution or Integral of a differential equation is a relation between the independent and the dependent variables which satisfies the given differential equation.

General Solution or Complete Integral:

A solution containing the number of arbitrary constants equal to the order of the equation is called the general solution or complete integral.

Particular Solution:

Any solution obtained from the general solution by giving **specific values** to one or more of the arbitrary constants is called a particular solution.

For example: $y^2 = a(x + b)$ is the general solution of the second order

differential equation $y \frac{d^2 y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$, as it contains two arbitrary constants. If we put **a =**

1, b = 0 then $y^2 = x$ particular solution

The Solution of Equations of The Form : $\frac{dy}{dx} = f(x)$

A differential equation of the form $\frac{dy}{dx} = f(x)$ is solved by direct integration, i.e.

$$y = \int f(x) dx + c$$

Example: Determine the general solution of $x \frac{dy}{dx} = 2 - 4x^3$

Solution: Rearranging $x \frac{dy}{dx} = 2 - 4x^3$ gives:

$$\frac{dy}{dx} = \frac{2 - 4x^3}{x} = \frac{2}{x} - \frac{4x^3}{x} = \frac{2}{x} - 4x^2$$

Integrating both sides gives: $y = 2 \log x - \frac{4x^3}{3} + c$

Problem: Find the particular solution of the differential equation $5 \frac{dy}{dx} + 2x = 3$, given

the boundary conditions $y = 1 \frac{2}{5}$ when $x = 2$.

Solution: $5 \frac{dy}{dx} + 2x = 3$, then $\frac{dy}{dx} = \frac{3 - 2x}{5} = \frac{3}{5} - \frac{2x}{5}$

On integration we get $y = \frac{3x}{5} - \frac{x^2}{5} + c$

Substituting the boundary conditions $y = 1\frac{2}{5}$ and $x = 2$ to evaluate c, gives c Hence the particular solution is $y = \frac{3x}{5} - \frac{x^2}{5} + 1$

Example: The bending moment M of the beam is given by $\frac{dM}{dx} = -w(l - x)$, where w and l are constants.

Determine M in terms of x given: $M = \frac{1}{2} w l^2$ when $x = 0$.

Solution: $\frac{dM}{dx} = -w(l - x) = -wl + wx$:

Integrating with respect to x gives $M = -wlx + \frac{wx^2}{2} + c$ When

$$M = \frac{1}{2} w l^2, \quad x = 0$$

Thus $\frac{1}{2} w l^2 = -wl(0) + \frac{w(0)^2}{2} + c$ From which, $c = \frac{1}{2} w l^2$

Hence the particular solution is $M = -wlx + \frac{w(x)^2}{2} + \frac{1}{2} w l^2$

$$\text{i.e } M = \frac{1}{2} w (l^2 - 2lx + x^2)$$

The Solution of Equations of the Form $\frac{dy}{dx} = f(y)$:

Or $dx = \frac{dy}{f(y)}$ and then the solution is obtained by direct integration i.e

$$\int dx = \int \frac{dy}{f(y)}$$

Example: Find the general solution of $\frac{dy}{dx} = 3 + 2y$

Solution: $dx = \frac{dy}{3 + 2y}$,

Integrating both sides gives $\int dx = \int \frac{dy}{3 + 2y}$

$x = \frac{1}{2} \log(3 + 2y) + c$, General Solution

Example: Determine the particular solution of $(y^2 - 1)\frac{dy}{dx} = 3y$

Given that $y = 1$ when $x = 2\frac{1}{6}$.

Solution: Rearranging $dx = \left(\frac{y^2 - 1}{3y} \right) dy$
 $= \left(\frac{y}{3} - \frac{1}{3y} \right) dy$

Integrating gives: $\int dx = \int \left(\frac{y}{3} - \frac{1}{3y} \right) dy$ i.e. $x = \frac{y^2}{6} - \frac{1}{3} \log y + c$,

which is the General Solution.

When $y = 1$, $x = 2\frac{1}{6}$, thus $2\frac{1}{6} = \frac{1}{6} - \frac{1}{3} \log 1 + c$, from which, $C=2$.

Hence the particular solution is $x = \frac{y^2}{6} - \frac{1}{3} \log y + 2$.

Example: The variation of resistance, R ohms, of an aluminium conductor with temperature $\theta^\circ C$ is given by $\frac{dR}{d\theta} = \alpha R$, where α is the temperature coefficient of resistance of aluminium.

If $R = R_0$ when $\theta = 0^\circ C$. Solve the equation for R .

Solution: $\frac{dR}{d\theta} = \alpha R$ of the form $\frac{dy}{dx} = f(y)$, Rearranging gives: $d\theta = \frac{dR}{\alpha R}$

Integrating both sides gives: $\int d\theta = \int \frac{dR}{\alpha R}$

$$i.e. \theta = \frac{1}{\alpha} \log R + c$$

Which is the general solution.

Substituting the boundary conditions $R = R_0$ when $\theta = 0$ gives:

$$0 = \frac{1}{\alpha} \log R_0 + c \text{ for which } c = -\frac{1}{\alpha} \log R_0$$

Hence the particular solution is

$$\begin{aligned} \theta &= \frac{1}{\alpha} \log R - \frac{1}{\alpha} \log R_0 \\ &= \frac{1}{\alpha} (\log R - \log R_0) \\ \theta &= \frac{1}{\alpha} \log \frac{R}{R_0} \\ \alpha \theta &= \log \frac{R}{R_0} \end{aligned}$$

Hence $e^{\alpha \theta} = \frac{R}{R_0} \Rightarrow R = R_0 e^{\alpha \theta}$

The Solution of Equation of The Form $\frac{dy}{dx} = f(x) \cdot f(y)$:

A differential equation of the form $\frac{dy}{dx} = f(x) \cdot f(y)$: where $f(x)$ is a function of

x only and $f(y)$ is a function of y only, may be rearranged as $\frac{dy}{f(y)} = f(x) dx$,

And then the solution is obtained by direct integration i.e. $\int \frac{dy}{f(y)} = \int f(x) dx$.

Example: Solve the equation $4xy \frac{dy}{dx} = y^2 - 1$

Solution: Separating the variables gives $\left(\frac{4y}{y^2 - 1} \right) dy = \frac{1}{x} dx$

Integrating both sides gives: $\int \left(\frac{4y}{y^2 - 1} \right) dy = \int \frac{1}{x} dx$

On integration, the general solution is $2\log(y^2 - 1) = \log x + c$

$$\text{or } \log(y^2 - 1)^2 - \log x = c$$

$$\text{or } \log \left\{ \frac{(y^2 - 1)^2}{x} \right\} = c \text{ or } \log \frac{(y^2 - 1)^2}{x} = e^c.$$

Example: Determine the particular solution of $\frac{d\theta}{dt} = 2e^{3t-2\theta}$, given that $t=0$ when $\theta=0$.

Solution: $\frac{d\theta}{dt} = 2e^{3t-2\theta}$

$$= 2(e^{3t})(e^{-2\theta})$$

Separating the variables gives:

$$\frac{d\theta}{e^{-2\theta}} = 2e^{3t} dt$$

i.e. $e^{2\theta} d\theta = 2e^{3t} dt,$

Integrating both sides gives $\int e^{2\theta} d\theta = \int 2e^{3t} dt,$

Thus the general solution is $\frac{e^{2\theta}}{2} = \frac{2}{3} e^{3t} + c$

When $t = 0, \theta = 0$, thus: $\frac{1}{2} e^{\theta} = \frac{2}{3} e^{\theta} + c$ from which

$$c = -\frac{1}{6}.$$

Hence the particular solution is: $\frac{1}{2} e^{2\theta} = \frac{2}{3} e^{3t} - \frac{1}{6}$

or $3e^{2\theta} = 4e^{3t} - 1.$

Example: Find the curve which satisfies the equation $xy = (1 + x^2) \frac{dy}{dx}$ and passes through the point $(0, 1)$.

Solution: Separating the variables gives: $\frac{x}{(1 + x^2)} dx = \frac{dy}{y}$

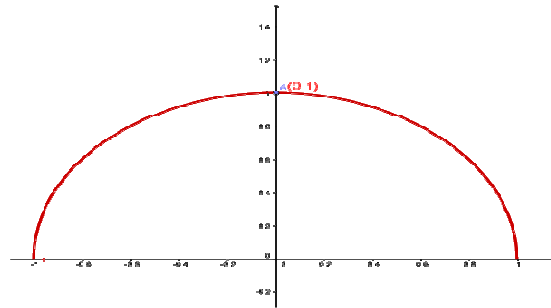
Integrating both sides gives $\frac{1}{2} \log(1 + x^2) = \log y + c$

when $x = 0, y = 1$ thus $\frac{1}{2} \log 1 = \log 1 + c$

from which $c = 0$. Hence, the particular solution is $\frac{1}{2} \log(1 + x^2) = \log y$ i.e.

$$\log(1 + x^2)^{\frac{1}{2}} = \log y \text{ from which } y = (1 + x^2)^{\frac{1}{2}}$$

Hence the equation of the curve is $y = \sqrt{1 - x^2}$



Example: When a cake is removed from an oven, its temperature is measured at $300^\circ F$. Three minutes later its temperature is $200^\circ F$. How long will it take for the cake to cool off to a room temperature of $70^\circ F$?

Solution: Given $T_m = 70$,

By Newton's Law of cooling, we have $\frac{dT}{dt} = k(T - T_m)$, $T(0) = 300$

i.e. $\frac{dT}{dt} = k(T - 70)$ ----(1) and determine the value of k so that $T(3) = 200$,

separating the variables gives: $\frac{dT}{T - 70} = k dt$ Integrating both sides gives

$$\log(T - 70) = kt + c_1$$

$$\Rightarrow T = 70 + c_2 e^{kt} \text{ -----(2)}$$

Using initial conditions:

when $t = 0$, $T = 300$

$$T(0) = 70 + c_2 e^{k \cdot 0}$$

$$300 = 70 + c_2 \therefore c_2 = 230,$$

$$\text{equation (2) becomes } T = 70 + 230e^{kt} \text{ -----(3)}$$

At $t = 3$: $T(3)=200$, equation (3) becomes

$$T = 70 + 230e^{3k}$$

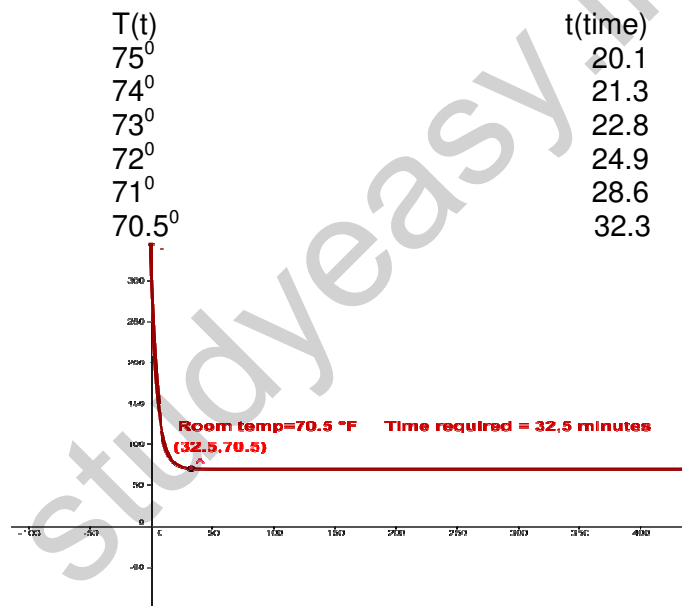
$$200 = 70 + 230e^{3k}$$

$$\Rightarrow 130 = 230e^{3k}$$

$$\Rightarrow \frac{130}{230} = e^{3k} \text{ or } k = \frac{1}{3} \log \frac{13}{23} = -0.19018$$

$$\text{thus } T(t) = 70 + 230e^{-0.19018t} \text{ -----(4)}$$

We expect the cake to reach the room temperature after a reasonably long period of time



Example:

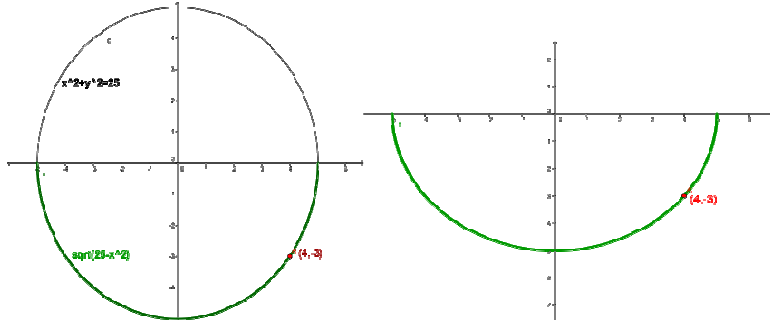
Solve the initial value problem $\frac{dy}{dx} = -\frac{x}{y}$, $y(4) = -3$.

Solution: By rewriting the equation as $y \, dy = -x \, dx$ we get

$$\int y \, dy = -\int x \, dx \text{ and } \frac{y^2}{2} = -\frac{x^2}{2} + c_1$$

$$\Rightarrow x^2 + y^2 = c^2, \quad c^2 = 2c_1$$

This solution of the differential equation represents a family of concentric circles centered at the origin. When $x = 4$, $y = -3$ so that $c^2 = 25$. Thus the initial value problem determines the circle $x^2 + y^2 = 25$ with radius 5. Particular solution is $y = -\sqrt{25 - x^2}$, $-5 < x < 5$. The solution curve is the lower semicircle, shown in green color, that contains the point (4, -3).



Example: Weight Reduction Model:

- Weight of person = x kg
- Tries to reduce weight
- Weight loss per month = 10% of weight
- Starting weight = 100kg

$$\frac{dx}{dt} = -0.1x$$

Initial conditions $x = 100$ at $t = 0$. Determine $x(t)$ as a function of t .

Analytical Solution of Simple model:

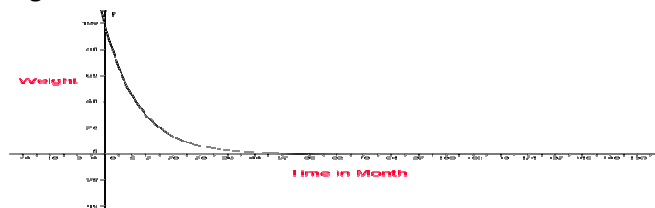
Recall the model: $\frac{dx}{dt} = -0.1x$ ($t = 0$) = 100

Cross multiplying, $\frac{dx}{x} = -0.1 dt$

Integrating both sides from 0 to t $\int \frac{dx}{x} = -0.1 \int dt$

$C + \log x(t) = -0.1t$. Using initial conditions $C = -\log 100$. Thus the final solution is

$$\log \frac{x(t)}{100} = -0.1t \quad \text{or} \quad x(t) = 100 e^{-0.1t}$$



Example: Solve the differential equation $(x + y)^2 \frac{dy}{dx} = a^2$

$$\Rightarrow \left(\frac{x + y}{a} \right)^2 = \frac{dx}{dy} \quad \text{----- (1)}$$

$$\text{Let } \frac{x + y}{a} = t \quad \text{----- (2)}$$

Differentiating equation (2) with respect to y we get

$$\frac{1}{a} \left[1 + \frac{dx}{dy} \right] = \frac{dt}{dy}$$

$$\Rightarrow \frac{dx}{dy} = a \frac{dt}{dy} - 1$$

$$\Rightarrow t^2 = a \frac{dt}{dy} - 1$$

$$\Rightarrow t^2 + 1 = a \frac{dt}{dy}$$

$$\Rightarrow \frac{dy}{a} = \frac{dt}{t^2 + 1} \quad (\text{variable separable})$$

on integration

$$\frac{y}{a} = \tan^{-1}(t) + c$$

$$\Rightarrow a \tan^{-1} \left(\frac{x + y}{a} \right) = y + c'$$

Homogeneous Equation:

Definition: A function is said to be homogeneous of the n^{th} degree in x and y if it can be put in the form $x^n f(y/x)$.

These equations can be put in the form $\frac{dy}{dx} = \frac{f_1(x, y)}{f_2(x, y)}$ ----- (1)

where f_1 and f_2 are expressions homogeneous of the same degree in x and y .

The solution is obtained as follows

Put $y = v x$ in (1) we get

$$\begin{aligned}\frac{dy}{dx} &= \frac{f_1(x, y)}{f_2(x, y)} \text{ from (1)} \\ &= F(v), \quad \text{a function of } v\end{aligned}$$

$$v + x \frac{dv}{dx} = F(v)$$

$$x \frac{dv}{dx} = F(v) - v$$

$$\frac{dv}{F(v) - v} = \frac{dx}{x} \text{ (v.s.)} \text{ By direct integration we get the solution of the given equation.}$$

Example:

Solve $(x^2 + y^2) dx - 2xy dy = 0$.

The given equation can be put in the form

$$\frac{dy}{dx} = \frac{x^2 + y^2}{2xy} \text{ -----(1).}$$

This is homogeneous in x & y

Solution: Put $y = vx$ in equation (1) we get

$$\begin{aligned}v + x \frac{dv}{dx} &= \frac{1 + v^2}{2v} \\ \Rightarrow x \frac{dv}{dx} &= \frac{1 - v^2}{2v} \\ \frac{2v dv}{1 - v^2} &= \frac{dx}{x} \text{ (v.s.)}\end{aligned}$$

$$\begin{aligned}\text{on integration we get } -\log(1 - v^2) - \log x &= \log c \\ \Rightarrow \log(1 - v^2) + \log x &= -\log c \\ \log(1 - v^2) x &= K'\end{aligned}$$

$$\Rightarrow (1 - v^2) \cdot x = C$$

$$\Rightarrow \left(1 - \frac{y^2}{x^2}\right) \cdot x = C$$

$$\Rightarrow x^2 - y^2 = xC$$

Example: Solve $y^2 dx + (xy + x^2) dy = 0$ -----(1)

Equation (1) can be put in the form

$$\frac{dy}{dx} = -\frac{y^2}{xy + x^2}, \text{ Which is homogeneous in } x \text{ and } y$$

Solution: Put $y = vx$ and we obtain $v + x \frac{dv}{dx} = -\frac{v^2}{v+1}$

$$\Rightarrow x \cdot \frac{dv}{dx} = \frac{-2v^2 + v}{v+1}$$

$$\Rightarrow \frac{(v+1)}{v(2v+1)} + \frac{dx}{x} = 0 \text{ ----- (3)}$$

Now $\frac{(v+1)}{v(2v+1)} = \frac{v}{v(2v+1)} + \frac{1}{v(2v+1)}$ -----(4)

Again, $\frac{1}{v(2v+1)} = \frac{A}{v} + \frac{B}{2v+1}$

$$1 = A(2v+1) + Bv$$

$$A=1, \quad B=-2$$

$$\frac{1}{v(2v+1)} = \frac{1}{v} - \frac{2}{2v+1}$$

Equation (4) becomes

$$\begin{aligned} \frac{v+1}{v(2v+1)} &= \frac{1}{(2v+1)} + \frac{1}{v} - \frac{2}{2v+1} \\ &= \frac{1}{v} - \frac{1}{2v+1} \end{aligned}$$

Equation (3) becomes $\frac{1}{v} - \frac{1}{2v+1} + \frac{dx}{x} = 0$

on integration we get $\log v - \frac{1}{2} \log(2v+1) + \log x = c$

$$\log v - \log(2v+1)^{\frac{1}{2}} + \log x = c$$

$$\log \left\{ \frac{vx}{(2v+1)^{\frac{1}{2}}} \right\} = c$$

$$\left\{ \frac{vx}{(2v+1)^{\frac{1}{2}}} \right\} = k$$

Non-homogeneous equation of the first degree in x and y:

These equations of the form

$$\frac{dy}{dx} = \frac{ax+by+c}{a'x+b'y+c'} \text{-----(1)}$$

put $x = x' + h$ and $y = y' + k$, where h and k are some constants to be determined.

Then

$$\begin{aligned} dx &= dx' \text{ and } dy = dy' \\ \therefore \frac{dy}{dx} &= \frac{dy'}{dx'} = \frac{a(x'+h) + b(y'+k) + c}{a'(x'+h) + b'(y'+k) + c'} \\ &= \frac{ax' + by' + ah + bk + c}{a'x' + b'y' + a'h + b'k + c'} \end{aligned}$$

If h and k are determined so that $ah + bk + c = 0$ & $a'h + b'k + c' = 0$

Then $\frac{dy'}{dx'} = \frac{ax' + by'}{a'x' + b'y'}$, which is homogeneous in x' and y' , which can be

solved by putting $y' = vx'$.

This method fails when $a : b = a' : b'$

$$i.e. \text{ when } \frac{a}{a'} = \frac{b}{b'}$$

Because the h and k become indeterminate

$$\text{Suppose } \frac{a}{a'} = \frac{b}{b'} = \frac{1}{m}$$

Then Equation (1) can be written as

$$\frac{dy}{dx} = \frac{ax + by + c}{m(ax + by) + c'} \quad \text{put } ax + by = v \text{ then}$$

$$a + b \frac{dy}{dx} = \frac{dv}{dx}$$

$$a + b \cdot \frac{v + c}{m v + c'} = \frac{dv}{dx}$$

Then by separating the variables and integrating we get the solution.

Example: Solve $(2x - 4y + 5)dy = (x - 2y + 3)dx$.

Solution: The given equation can be written as

$$\frac{dy}{dx} = \frac{x - 2y + 3}{2x - 4y + 5} \left(\text{Here } \frac{a}{a'} = \frac{b}{b'} \right)$$

$$= \frac{(x - 2y) + 3}{2(x - 2y) + 5} \quad \text{--- (1)}$$

Put $v = x - 2y$

$$\frac{dv}{dx} = 1 - 2 \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{1}{2} \left(1 - \frac{dv}{dx} \right)$$

From equation (1),

$$\frac{v + 3}{2v + 5} = \frac{1}{2} \left(1 - \frac{dv}{dx} \right)$$

$$\Rightarrow \frac{2v + 6}{2v + 5} = 1 - \frac{dv}{dx}$$

$$\begin{aligned}
 \frac{dv}{dx} &= 1 - \frac{2v+6}{2v+5} \\
 &= \frac{2v+5-2v-6}{2v+5} \\
 &= -\frac{1}{2v+5}
 \end{aligned}$$

On integration we get

$$v^2 + 5v = -x + c$$

$$i.e. (x-2y)^2 + 5(x-2y) = -x + c.$$

Example: Solve $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$ (Here $\frac{a}{a'} \neq \frac{b}{b'}$)

Solution: Given equation is non-homogeneous in x and y

Put $x = x' + h, \quad y = y' + k$

$$\therefore \frac{dy}{dx} = \frac{x' + 2y' + h + 2k - 3}{2x' + y' + 2h + k - 3} \quad \text{----- (1)}$$

We find h, k such that $\left. \begin{aligned} h + 2k - 3 &= 0 \\ 2h + k - 3 &= 0 \end{aligned} \right\} \text{ solve by cross multiplication}$

$$\frac{h}{-6+3} = \frac{k}{-6+3} = \frac{1}{1-4} \therefore h=1, k=1$$

Equation (1) becomes $\frac{dy'}{dx'} = \frac{x' + 2y'}{2x' + y'}$

This is homogeneous in x' and y'

Put $y' = vx'$

$$\begin{aligned}
 \therefore \frac{dy'}{dx'} &= v + x' \frac{dv}{dx'} \\
 &= \frac{1 + 2v}{2 + v}
 \end{aligned}$$

On simplification we get

$$\frac{2+v}{1-v^2} dv = \frac{dx'}{x'} \quad (\text{variables are separable})$$

$$\int \frac{2+v}{1-v^2} dv = \int \frac{dx'}{x'}$$

$$\int \frac{2+v}{(1+v)(1-v)} dv = \log x' + c \quad \text{----- (2)}$$

$$\text{now } \frac{2+v}{(1+v)(1-v)} = \frac{A}{(1+v)} + \frac{B}{(1-v)}$$

$$A = 1/2, \quad B = 3/2$$

Equation (2) becomes

$$\int \left[\frac{1}{2(1+v)} + \frac{3}{2(1-v)} \right] dv = \log x' + c$$

$$\Rightarrow \frac{1}{2} \log(1+v) - \frac{3}{2} \log(1-v) = \log x' + c,$$

On simplification we get

$$\frac{1+v}{(1-v)^3} = \log x'^2 + c.$$

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Differential Equations

Unit-7

Exact Differential Equations: $M dx + N dy = 0$

Verify the condition $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Then integrate $M dx$ with respect to x as if y were constants,

then integrate the terms in $N dy$ which do not contain terms in x and equate sum of these two integrals to a constant.

Example: Solve

$$(a^2 - 2xy - y^2)dx - (x + y)^2 dy = 0$$

$$M = a^2 - 2xy - y^2; \quad N = -(x + y)^2 = -x^2 - y^2 - 2xy$$

$$\therefore \frac{\partial M}{\partial y} = -2x - 2y; \quad \frac{\partial N}{\partial x} = -2x - 2y \quad \therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ The given equation is an exact differential equation. $\therefore$ Solution is given by

$$\int (a^2 - 2xy - y^2) dx + \int (-y^2) dy = 0$$

$$\Rightarrow 3a^2x - 3x^2y - 3y^2x - y^3 = 3c$$

Example:

$$\text{Solve } (e^y + 1) \cos x dx + e^y \sin x dy = 0$$

$$\frac{\partial M}{\partial y} = e^y \cos x; \quad \frac{\partial N}{\partial x} = e^y \cos x$$

$$\text{Solution is : } \int (e^y + 1) \cos x dx = c$$

$$(e^y + 1) \sin x = c$$

Integrating Factors:

Sometimes a differential equations which is **not exact** may become **exact** on multiplication by a suitable function known as an Integrating Factor.

For example the differential equation $ydx - xdy = 0$

is not exact, but when multiplied by $\frac{1}{y^2}$ or $\frac{1}{x^2}$ or by $\frac{1}{xy}$

it becomes an exact equation. The number of integrating factor is **infinite** for a given equation.

Rules For Finding The Integrating Factors::

Rule I: when $Mx + Ny \neq 0$ and the equation is homogeneous then $\frac{1}{Mx + Ny}$ is an Integrating factor of $Mdx + Ndy = 0$

Example: Solve: $x^2 ydx - (x^3 + y^3)dy = 0$

Solution: $Mx + Ny = -y^4$ I.F. = $-1/y^4$

Multiply Given equation By I.F. $-\frac{x^2}{y^3}dx + \left(\frac{x^3}{y^4} + \frac{1}{y}\right)dy = 0$

On integration we have $-\frac{x^3}{3y^3} + \log y = c$

Rule II: When $Mx - Ny \neq 0$ and the equation has the form $f_1(xy)ydx + f_2(xy)x dy = 0$

then Integrating factor is $\frac{1}{Mx - Ny}$

Example: Solve $(1 + xy)ydx + (1 - xy)x dy = 0$

$Mx - Ny = 2x^2y^2$ I.F. = $1/2x^2y^2$

Multiply given equation by I.F. we get $\frac{dx}{x} - \frac{dy}{y} + \frac{ydx + xdy}{x^2 y^2}$ on integration

$\log \frac{x}{y} - \frac{1}{xy} = \log c$

Rule III: When $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$ is a function x alone say f(x) then $e^{\int f(x) dx}$ is an Integrating factor

Problem:

Solve $(x^2 + y^2 + 2x)dx + 2ydy = 0$

$$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = \frac{\frac{2y}{2y} - 1}{2y} = 1 \quad \text{I.F.} = e^{\int 1 dx} = e^x$$

Multiplying given equation by I.F. and it becomes exact

Solution of given equation is given by $\int e^x (x^2 + y^2 + 2x) dx = c$

$$x^2 e^x + y^2 e^x = c$$

Rule IV: When $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$ is a function y alone say F(y) then $e^{\int F(y) dy}$ is an Integrating factor

Example: solve $(y^4 + 2y)dx + (xy^3 + 2y^4 - 4x)dy = 0$

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{-3(y^2 + 2)}{y(y^2 + 2)} = \frac{-3}{y}$$

$$I.F. = e^{\int F(y) dy} = \frac{1}{y^{-3}}$$

Multiply given equation by I.F. and it becomes

$$\left(y + \frac{2}{y^2}\right)dx + \left(x + 2y - \frac{4x}{y^3}\right)dy = 0$$

Solution is given by $\left(y + \frac{2}{y^2} \right) x + y^2 = c$

Linear Differential Equation:

A differential equation is said to be linear if the dependent variable and its derivatives appear only in the first degree.

The first order linear differential equation is of the form $\frac{dy}{dx} + Py = Q$ where P and Q are functions of x only or constants.

Solution of Linear differential equation:

$$y (I.F.) = \int Q (I.F.) dx + c$$

where $I.F. = e^{\int P dx}$

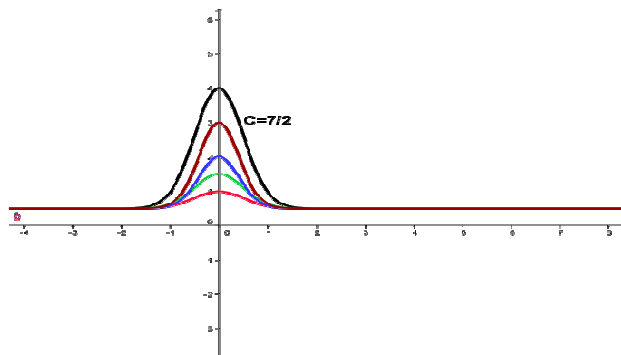
Example: Solve $\frac{1}{x} \frac{dy}{dx} + 4y = 2$ given the boundary conditions x = 0 when y = 4.

Solution : Rearranging gives $\frac{dy}{dx} + 4xy = 2x$

Integrating factor $e^{\int p dx} = e^{2x^2}$.

Solution gives: $ye^{2x^2} = \int e^{2x^2} (2x) dx = \frac{1}{2} e^{2x^2} + c$.

Using B.C. Obtain C=7/2. Hence particular solution gives $y = \frac{1}{2} (1 + 7e^{-2x^2})$



Example: Solve the differential equation $\frac{dy}{d\theta} = \sec \theta + y \tan \theta$

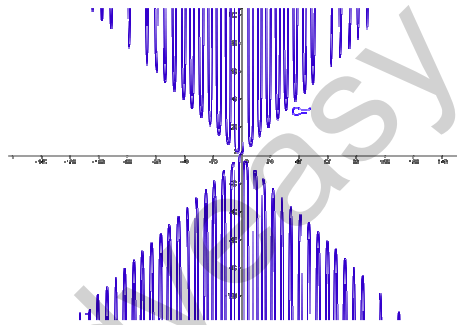
given the boundary conditions $y = 1$ when $\theta = 0$.

Solution: Rearranging gives $\frac{dy}{d\theta} - (\tan \theta)y = \sec \theta$

Integrating factor $e^{\int p dx} = e^{\ln(\cos \theta)} = \cos \theta$.

Solution gives: $y \cos \theta = \int \cos \theta (\sec \theta) d\theta = \theta + c$.

When $\theta = 0$, $y = 1$, thus $c=1$. Hence particular solution is $y = \frac{(\theta + 1)}{\cos \theta}$.



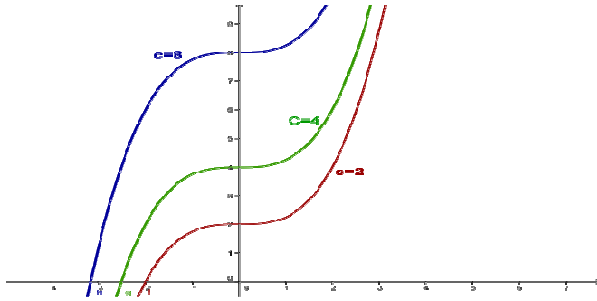
Example: Solve the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$

Solution:

Integrating factor $e^{\int p dx} = e^{\ln x} = x$.

$$y x = \int x^2 x dx$$

Solution gives: $y = \frac{x^3}{4} + c$



Equations Reducible to linear form Or (Bernoulli's equation):

$$\frac{dy}{dx} + Py = Qy^{-n} \quad \text{Dividing by } y^n$$

$$y^{-n} \frac{dy}{dx} + Py^{-n+1} = Q \quad \text{----- (1)}$$

$$\text{put } v = y^{-n+1} \text{ we get } (-n+1)y^{-n+1-1} \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{i.e. } y^{-n} \frac{dy}{dx} = \frac{1}{(-n+1)} \frac{dv}{dx}$$

$$\frac{1}{(-n+1)} \frac{dv}{dx} + Pv = Q$$

$$\frac{dv}{dx} + (-n+1)Pv = (-n+1)Q$$

Which is linear in v

$$\text{Example: Solve } \frac{dy}{dx} + \frac{y}{x} = y^{-2}$$

$$\text{Solution: } y^{-2} \frac{dy}{dx} + y^{-1} \frac{1}{x} = 1 \quad \text{--- (1)}$$

$$\text{Put } y^{-1} = v, \quad -1 y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$$

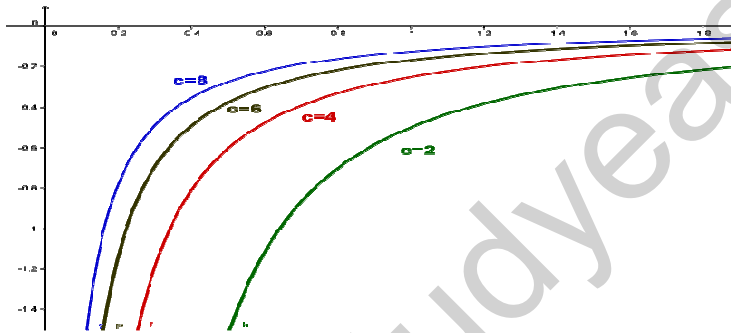
$$- \frac{dv}{dx} + v \frac{1}{x} = 1$$

From (1) $i.e. \frac{dv}{dx} - v \frac{1}{x} = -1$ which is linear in v

Integrating factor $e^{\int p dx} = e^{\int -\frac{1}{x} dx} = \frac{1}{x}$

Solution is $v \frac{1}{x} = \int -1 \frac{1}{x} dx + c$

$$\frac{1}{xy} = -\log x + c$$



Example: solve $\frac{dy}{dx} - 2y \tan x = y^2 \tan^2 x$

Solution: $y^{-2} \frac{dy}{dx} - 2y^{-1} \tan x = \tan^2 x \quad (1)$

Put $y^{-1} = v$ $- y^{-2} \frac{dy}{dx} = \frac{dv}{dx}$

From (1) $\frac{dv}{dx} + 2v \tan x = -\tan^2 x$ which is linear in v

Integrating factor $e^{\int p dx} = e^{\int 2 \tan x dx} = e^{2 \log \sec x} = \sec^2 x$

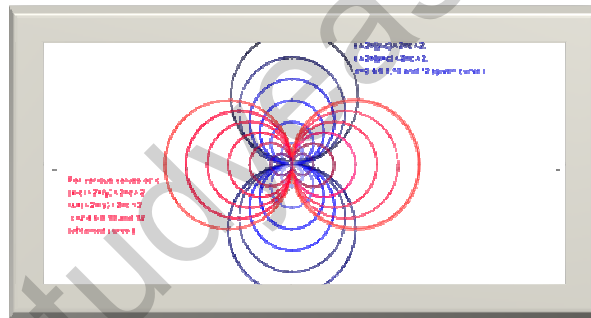
∴ Solution is $y^{-1} \sec^2 x = \int -\tan^2 x \sec^2 x dx + c$

$$y^{-1} \sec^2 x = -\frac{\tan^3 x}{3} + c$$

Orthogonal trajectory

Introduction:

The word “Orthogonal” comes from the Greek “right angle” and the word “trajectory” comes from Latin “cut-across”. Hence the curve that cut across the other at right angle is called orthogonal trajectory.



Rules to find the equation of orthogonal trajectories of a family of cartesian curves:

- Diff $f(x, y, c)=0$ and eliminate c .
- Resulting equation gives the DE of the family $f(x, y, c)=0$
- i.e. $F\left\{x, y, \frac{dy}{dx}\right\} = 0$
- Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ in $F\left\{x, y, \frac{dy}{dx}\right\} = 0$
- The differential equation of orthogonal trajectory is $F\left\{x, y, -\frac{dx}{dy}\right\} = 0$
- Integrate $F\left\{x, y, -\frac{dx}{dy}\right\} = 0$ to get the equation of the required orthogonal trajectory

Rules to find the equation of orthogonal trajectories of a family of polar curves:

Let the equation of given family of curves be $f(r, \theta, c) = 0$

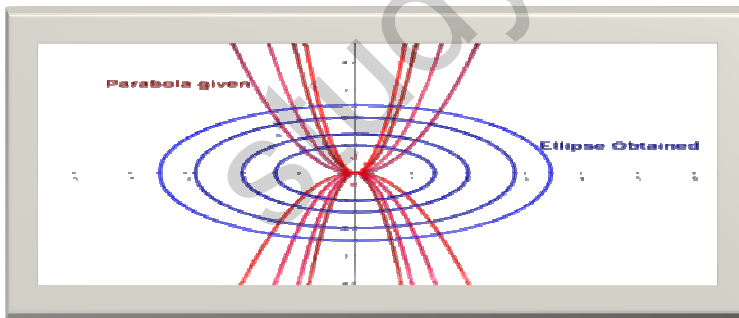
- Diff $f(r, \theta, c) = 0$ and eliminate c .
- Resulting equation gives the DE of the family $f(r, \theta, c) = 0$
- i.e. $F\left\{r, \theta, \frac{dr}{d\theta}\right\} = 0$
- Replace $\frac{dr}{d\theta}$ by $-r^2 \frac{d\theta}{dr}$ in $F\left\{r, \theta, \frac{dr}{d\theta}\right\} = 0$
- The DE of OT is $F\left\{r, \theta, -r^2 \frac{d\theta}{dr}\right\} = 0$
- Integrate $F\left\{r, \theta, -r^2 \frac{d\theta}{dr}\right\} = 0$ to get the equation of the required orthogonal trajectory

Example :Find the orthogonal trajectories of parabolas $y = cx^2$

Solution: Differentiate given equation we get $\frac{dy}{dx} = 2Cx = 2\left(\frac{y}{x^2}\right)x = \frac{2y}{x}$

Replace $\frac{dy}{dx} = -\frac{dx}{dy}$ we get $\frac{dy}{dx} = -\frac{x}{2y}$ is the DE of OT.

$2y dy = -x dx$ (V.S) On integration we get $y^2 + \frac{x^2}{2} = c$



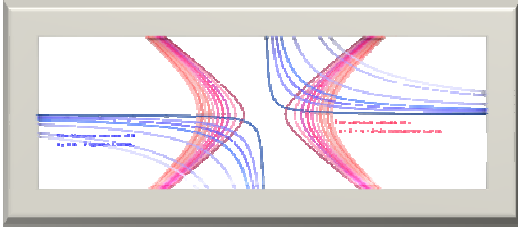
Example: Find the orthogonal trajectories of the series of hyperbolas $xy = k^2$

Solution: Differentiation given equation w.r.t x we get $y + x y_1 = 0$ Replace y_1 by $-\frac{1}{y_1}$

The differential equation of the OT is

$$y - x \frac{dx}{dy} = 0 \quad \text{or} \quad -x \frac{dx}{dy} = -y \Rightarrow -x dx = -y dy \Rightarrow x dx - y dy = 0$$

$$\frac{x^2}{2} - \frac{y^2}{2} = c \Rightarrow x^2 - y^2 = 2c = c_1 \quad \text{Which the required OT}$$



Example: Find the orthogonal trajectories of circles

$$x^2 + (y-c)^2 = c^2 \quad (1)$$

solution: Differentiation w.r.t x, $2x + 2(y-c)y_1 = 0$

$$\text{or } x + (y-c)y_1 = 0 \quad (2)$$

From (1) $x^2 + y^2 - 2cy = 0$ or $c = (x^2 + y^2) / 2y$ Substitute this in (2) we get

$$x + \left(y - \frac{x^2 + y^2}{2y} \right) y_1 = 0 \quad \text{or} \quad x + \left(\frac{y^2 - x^2}{2y} \right) y_1 = 0$$

$$\left(\frac{y^2 - x^2}{2y} \right) y_1 = -x \Rightarrow \left(\frac{x^2 - y^2}{2y} \right) y_1 = x \Rightarrow y_1 = \frac{2xy}{x^2 - y^2} \quad (3)$$

Replace y_1 by $-\frac{1}{y_1}$ The differential equation of the OT is

$$-\frac{1}{y_1} = \frac{2xy}{x^2 - y^2} \Rightarrow \frac{y^2 - x^2}{2xy} = y_1 \quad \text{or} \quad 2xyy_1 - y^2 + x^2 = 0 \quad (4)$$

$$2xy \frac{dy}{dx} - y^2 + x^2 = 0 \quad \text{dividen by } 2xy \text{ or } 2x \Rightarrow \frac{dy}{dx} - \frac{y}{2x} = \frac{-x}{2y}$$

$$y \frac{dy}{dx} - \frac{y^2}{2x} = \frac{-x}{2}$$

(5) Put $y^2/2 = z$ Diff w. r. t. x

$$y \frac{dy}{dx} = \frac{dz}{dx}$$

Equation (5) becomes $\frac{dz}{dx} - \frac{z}{x} = \frac{-x}{2}$

linear in z I.F.=1/x solution is

$$z \frac{1}{x} = \int -\frac{1}{2} dx + c = -\frac{x}{2} + c \quad \text{or} \quad \frac{y^2}{2x} = -\frac{x}{2} + c \quad \text{or} \quad y^2 + x^2 - 2cx = 0$$

-
- $x^2 + y^2 = r^2$
 $x^2 + y^2 = 4r^2$
 $x^2 + y^2 = 9r^2$
 $x^2 + y^2 = 16r^2$
- The circles intersect at $x = 0$ and $y = 0$
 The circles intersect at $x = 0$ and $y = 0$
 The circles intersect at $x = 0$ and $y = 0$
 The circles intersect at $x = 0$ and $y = 0$

Solution: Given equation $r = a(1+\cos(\theta))$ Diff. w.r.t. θ

$$r = -\frac{1}{\sin \theta} \frac{dr}{d\theta} (1 + \cos \theta) \Rightarrow \frac{dr}{d\theta} = \frac{-r \sin \theta}{1 + \cos \theta} = \frac{-2r \sin(\theta/2) \cos(\theta/2)}{2 \cos^2(\theta/2)} = -r \tan(\theta/2)$$

Replace $\frac{dr}{d\theta} = \cot\left(\frac{\theta}{2}\right) \Rightarrow \frac{dr}{r} = \cot\left(\frac{\theta}{2}\right) d\theta$ **Integrate this we get**

A diagram showing two overlapping ellipses. The left ellipse is blue and the right ellipse is green. They intersect at a single point, which is marked with a small red dot. The entire diagram is set against a white background within a grey border.

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Unit-VII: Linear Algebra-I

❖ Purpose of lesson :

- ❖ To show what are the matrices , why they are useful, how they are classified as various types and how they are solved.
- ❖ **Introduction:** Matrices is a powerful tool of modern Mathematics and its study is becoming important day by day due to its wide applications in almost every branch of science and especially in Physics and Engineering. Matrices are used by Sociologists in the study of dominance within a group, by Demographers in the study of births and deaths, etc, by Economist in the study of inter-industry economists, input-output tables and for various practical business purposes, by statisticians in the study of 'design of experiments and for various practical business purposes, by statisticians in the study of 'design of experiments and multivariate analysis by Engineers in the study of 'network analysis' which is used in electrical and communication engineering.

Application:

The following table shows the number of transistors and resistors purchased by a manufacturer from suppliers A and B for the first week of January

	A	B
Transistor	400	800
Resistors	600	500

- a) Write the data in the table as 2×2 matrix S_1
- b) Use scalar multiplication to find a matrix S_2 whose entries are all 10 % large than the corresponding entries of S_1
- c) Suppose that S_2 is the supply matrix for the second week of January. Find $S_1 + S_2$ and explain what its entries represent.

Solution: a) The supply table can be written as the following 2×2 matrix

$$S_1 = \begin{bmatrix} 400 & 800 \\ 600 & 500 \end{bmatrix}$$

b) If the entries of S_2 are 10 % large than the entries of S_1 , then

$$S_2 = S_1 + 0.10 S_1 = 1.1 S_1$$

$$S_2 = 1.1 \begin{bmatrix} 400 & 800 \\ 600 & 500 \end{bmatrix} = \begin{bmatrix} 440 & 880 \\ 660 & 550 \end{bmatrix}$$

c) The entries of $S_1 + S_2$ give the total number of transistors and resistors purchased from suppliers A and B for the first two weeks of January.

$$s_1 + s_2 = \begin{bmatrix} 400 & 800 \\ 600 & 500 \end{bmatrix} + \begin{bmatrix} 440 & 880 \\ 660 & 550 \end{bmatrix} = \begin{bmatrix} 840 & 1680 \\ 1260 & 1050 \end{bmatrix}$$

Example : The following table 1 Shows the number of economy, mid-size and large cars rented by individuals and corporations at a rental agency in a single day Table 1

	Economy	mid-size	large
Individuals	3	2	6
Corporations	5	2	4

Table 2

	Bonus points	Free miles
Economy	20	50
Mid-size	30	100
Large	40	150

Table 2 shows the number of bonus points and free miles given in a promotional program for each of the three car types. Find

- Total bonus points for individuals
- Total free miles for individuals
- Total bonus points for corporations
- Total free miles for corporations

$$\begin{bmatrix} 3 & 2 & 6 \end{bmatrix}$$

The 1 x 3

row matrix from the Table 1 – Represents the number of economy, medium size and large rented by Individuals

- The 3 x 1 column matrix

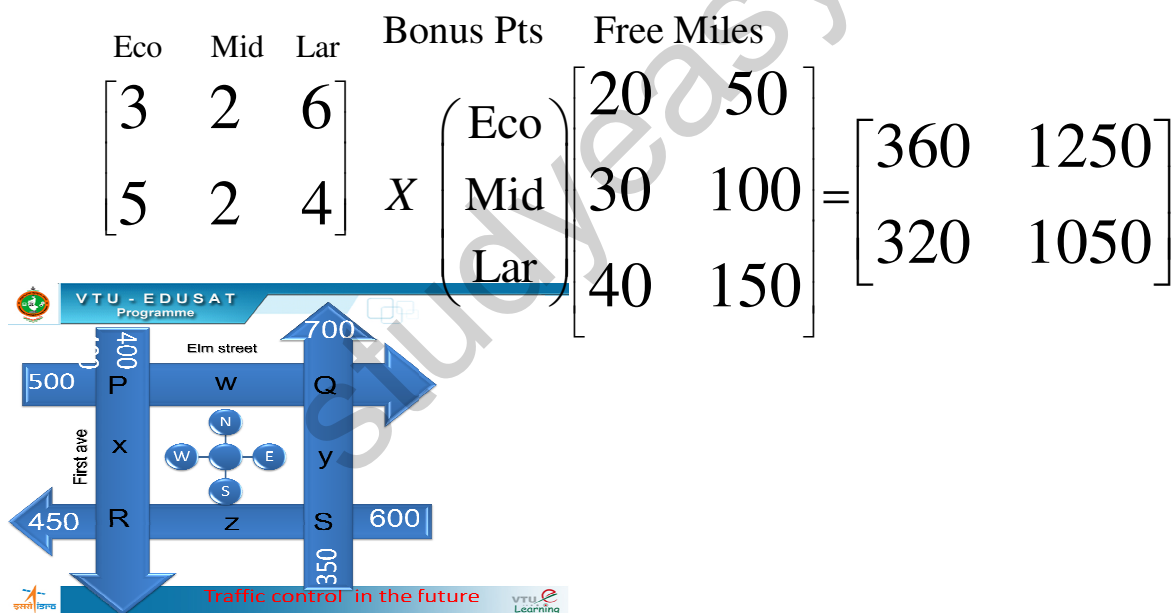
$$\begin{bmatrix} 20 \\ 30 \\ 40 \end{bmatrix}$$

From table 2 represents the bonus points given for each economy, mid-size and large car that is rented.

The product of these two matrices is a 1 x 1 matrix whose entry is the sum of the products of the corresponding entries

$$\begin{bmatrix} 3 & 2 & 6 \end{bmatrix} \begin{bmatrix} 20 \\ 30 \\ 40 \end{bmatrix} = \begin{bmatrix} 360 \end{bmatrix}$$

The product of above two matrices, gives the total number of bonus points given to individuals on the rental of the 11 cars.



Traffic control in the future:

The above picture shows the intersections of four one way streets. 400 cars per hour want to enter intersection P from the north on First Avenue while 300 cars per hour want to head east from intersection Q on Elm Street. The letters w, x, y, and z represent the number of cars per hour passing the four points between these four intersections, as shown in picture.

A) Find values for w, x, y and z that would realize this desired traffic flow.

B) If construction on Oak Street limits z to 300 cars per hour, then how many cars per hour would have to pass w , x , and y .

Solution: The solution to the problem is based on the fact that the number of cars entering an intersection per hour must equal the number leaving that intersection per hour, if the traffic is to keep flowing

Since 900 cars (400 + 500) enter intersection P, 900 must leave, $x + w = 900$. Writing a similar equation for each intersection yields the system of equations.

$$W + x = 900$$

$$W + y = 1000$$

$$X + z = 850$$

$$y + z = 950$$

$$R_2 = R_2 - R_1$$

$$R_2 = R_2 - R_1$$

$$R_4 = R_4 - R_2$$

$$R_{23}$$

$$\begin{bmatrix} 1 & 1 & 0 & 0 & 900 \\ 1 & 0 & 1 & 0 & 1000 \\ 0 & 1 & 0 & 1 & 850 \\ 0 & 0 & 1 & 1 & 950 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 1 & 0 & 0 & 900 \\ 0 & -1 & 1 & 0 & 100 \\ 0 & 1 & 0 & 1 & 850 \\ 0 & -1 & 1 & 0 & 100 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 1 & 0 & 0 & 900 \\ 0 & -1 & 1 & 0 & 100 \\ 0 & 1 & 0 & 1 & 850 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 1 & 0 & 0 & 900 \\ 0 & 1 & 0 & 1 & 850 \\ 0 & -1 & 1 & 0 & 100 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\approx \begin{bmatrix} 1 & 1 & 0 & 0 & 900 \\ 0 & 1 & 0 & 1 & 850 \\ 0 & 0 & 1 & 1 & 950 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$w + x = 900$$

$$w = 900 - x$$

$$x + z = 850$$

$$x = 850 - z$$

$$y + z = 950$$

$$y = 950 - z$$

The system is a dependent system does not have a unique solution. (z is non negative integer)

b) If z is limited to 300 because of construction, then the solution is (350, 550, 650, 300) To keep traffic flowing when $z = 300$, the system must route 350 cars past w , 550 past x and 650 past y .

Definition Matrices and Types of Matrices:

Twenty six female students and twenty-four male students responded to a survey on income in a college algebra class. Among the female students, 5 classified themselves as low-income, 10 as middle-income, and 11 as high-income. Among the male students, 9 were low-income, 2 were middle-income, and 13 were high-income. Each student is classified in two ways, according to gender and income. This information can be written in a matrix:

	L	M	H
Female	5	10	11
Male	9	2	13

Square matrix: In this matrix we can see the class of makeup according to gender and income. A matrix provides a convenient way to organize a two way classification of data.

A matrix is a rectangular array of real numbers. The rows of a matrix run horizontally, and the columns run vertically.

A matrix with m rows and n columns has size $m \times n$ (read “ m by n ”). The number of rows is always given first. For example, the matrix used to classify the students is a 2×3 matrix

When $m = n$ i.e the number of rows is the same as the number of columns the array is called a square matrix. The others are called rectangular matrix.

Row-matrix: In a matrix if there is only one row it is called a row matrix

Column matrix: In a matrix if there is only one column it is called a column matrix

Null matrix : In a matrix, if all the elements of matrix are zero, it is called null matrix and is denoted by $\mathbf{O}$

Equal Matrix: Two matrices are said to be equal if

- They are of the same type i.e they have same number of rows and columns
- The elements in the corresponding positions of the two matrices are equal.

Location of an element: To locate any particular element of a matrix the elements are usually denoted by a letter followed by two suffixes which respectively specify the row and the column in which it appears. Thus the element occurring in p^{th} row and q^{th} column will be written as a_{pq} .

Note: The matrices are generally denoted by capital letters.

Diagonal Elements of a Matrix: An element a_{ij} of a square matrix is said to be a diagonal element if $i = j$.

Scalar Multiple of a matrix: If k is a number and A is a matrix, then kA is defined as the matrix each element of which is k times the corresponding element of A .

Triangular Matrix: if every element above (or below) the diagonal is zero, the matrix is called a triangular matrix

Example:

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 5 \\ 0 & 0 & 6 \end{bmatrix} \text{ upper triangular} \quad \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix} \text{ lower triangular}$$

Transposed Matrix: The matrix of order $n \times m$ obtained by interchanging the rows and columns of a matrix A of order $m \times n$ is called the transpose matrix of A or transpose of the matrix and is denoted by A' or A^t .

Representation of Points: A point is represented in two dimensions by its coordinates. These two values are specified as the elements of a 1×2 matrix i.e.

$$[x, y] \text{ or } \begin{bmatrix} x \\ y \end{bmatrix} 2 \times 1$$

In three-dimensions a 1×3 matrix

$$[x \ y \ z] \text{ is used or } \begin{bmatrix} x \\ y \\ z \end{bmatrix} 3 \times 1$$

Row matrix or column matrix like $[x \ y]$ or $\begin{bmatrix} x \\ y \end{bmatrix}$ are frequently called position vectors.

Elementary Transformations:

An elementary transformation, which is also known as E-transformation, is an operation of any one of the following types.

- i) Interchange of two rows (columns)
- ii) The multiplication of the elements of rows (columns) by a non-zero number.
- iii) The addition to the elements of a row (column), the corresponding elements of a row (column) multiplied by any number

An elementary transformation is said to be row-transformation or column transformation according as it applies to rows or columns.

Symbols to be used for elementary transformation:

- i) R_{ij} for the interchange of i^{th} and j^{th} rows
- ii) $R_i(k)$ for the multiplication of i^{th} row by $k \neq 0$
- iii) $R_{ij}(k)$ for the addition to the i^{th} row, the products of the j^{th} row by k . Similarly we use the symbols C_{ij} , $C_i(k)$, $C_{ij}(k)$ for the corresponding column operations.

Equivalent Matrices:

Two matrices A and B are said to be equivalent if one can be obtained from the other by a sequence of elementary transformations. Two equivalent matrices have the same order. The symbol $\sim$ is used for equivalence.

Minor : Let A be a matrix square or rectangular, from it delete all rows leaving a certain t -rows and all columns leaving a certain t -column. Now if $t > 1$ then the elements that are left, constitute a square matrix of order t and the determinant of this matrix is called a minor of A of order t

Rank of a Matrix:

Definition: A matrix is said to be rank r when

- i) It has at least one non-zero minor of order r and
- ii) Every minor of order higher than r vanishes.

Briefly, the rank of a matrix is the largest order of any non-vanishing minor of the matrix

For example: 1) $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}$, $\rho(A) = 1$

2) $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 9 & 12 \end{bmatrix}$, $\rho(A) = 1$

Note : i) If a matrix has a non-zero minor of order r , its rank is $\geq r$

ii) If all minors of a matrix of order of $r+1$ are zero, its rank is $\leq r$

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}$ $|A| = 2 \neq 0$ $\rho(A) = 3$

$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \neq 0$ $\rho(A) = 3$

Example:. Find the rank of the matrix

$$\begin{aligned}
 & \sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad R_2 = R_2 + R_1; R_3 = R_3 + 2R_1 \quad \sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & -6 & 0 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad \text{operate} \\
 & \sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 1 & -2 & 0 \end{bmatrix} \quad R_3 = R_3 - R_2; R_4 = R_4 - R_2 \quad \sim \begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad R_{12} \\
 & \sim \begin{bmatrix} 1 & 1 & -2 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \rho(A) = 2
 \end{aligned}$$

Example:. Reduce the matrix $A = \begin{bmatrix} 2 & 6 & 5 \\ 2 & 5 & 4 \\ 5 & 16 & 13 \end{bmatrix}$ to a unit matrix by only elementary row transformations

Solution:

$$\begin{aligned}
 A &= \begin{bmatrix} 2 & 6 & 5 \\ 2 & 5 & 4 \\ 5 & 16 & 13 \end{bmatrix} \quad R_2 = R_2 - R_1; R_3 = R_3 - 2R_1 \\
 &\sim \begin{bmatrix} 2 & 6 & 5 \\ 0 & -1 & -1 \\ 1 & 4 & 3 \end{bmatrix} \quad R_{13} \sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & -1 & -1 \\ 2 & 6 & 5 \end{bmatrix} \quad R_3 = R_3 - 2R_1 \\
 &\sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & -1 & -1 \\ 0 & -2 & -1 \end{bmatrix} \quad R_3 = R_3 - 2R_2 \sim \begin{bmatrix} 1 & 4 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \quad R_1 = R_1 - 3R_3; R_2 = R_2 - R_3
 \end{aligned}$$

Example:. Find the rank of the matrix $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\rho(A) = 3$

$$\begin{aligned}
 & \sim \begin{bmatrix} 0 & 2 & 1 \\ 0 & 4 & 2 \\ 0 & 6 & 3 \end{bmatrix} \quad c_2 = c_2 - 2c_3 \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 3 \end{bmatrix} \quad R_2 = R_2 - 2R_1; R_3 = R_3 - 3R_1 \\
 & \sim \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad c_{13} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} I_1 & 0 \\ 0 & 0 \end{bmatrix}
 \end{aligned}$$

Normal form of a Matrix:

By means of a sequence of elementary transformations every matrix A of order m x n can be reduced to one of the following form

$$\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

Example: Reduce the matrix A to its normal form

$$A = \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix}$$

Solution:

$$\begin{aligned} A &= \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix} \xrightarrow{c_{12}} \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 1 & 2 & 3 & 1 \end{bmatrix} \xrightarrow{R_3 = R_3 - R_1} \begin{bmatrix} 1 & 0 & 2 & -2 \\ 0 & 4 & 2 & 6 \\ 0 & 2 & 1 & 3 \end{bmatrix} \xrightarrow{c_3 = c_3 - 2c_1; c_4 = c_4 + 2c_1} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 4 & 2 & 6 \\ 0 & 2 & 1 & 3 \end{bmatrix} \xrightarrow{R_2 = \frac{R_2}{2}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 2 & 1 & 3 \end{bmatrix} \xrightarrow{R_3 = R_3 - R_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{c_2 = c_2 - 2c_3; c_4 = c_4 - 3c_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{c_{23}} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rho(A) = 2 \end{aligned}$$

Consistency of a system of linear equations:

Consider the following system of m linear equations in n-unknowns $x_1, x_2, x_3, \dots, x_n$

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = b_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \dots + a_{mn}x_n = b_n$$

The system of equations can be written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \dots \\ x_n \end{bmatrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \dots \\ b_n \end{bmatrix}$$

$$D = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ a_{31} & a_{32} & \dots & a_{3n} & b_3 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_n \end{bmatrix}$$

Conditions for Consistency:

- ❖ Here the system of equations $AX = B$ is called the non homogeneous set of equations and the system of equations $AX = 0$ is called the homogeneous set of equations.
- ❖ A system of linear equations is said to be consistent if it **possesses a solution** and inconsistent (not consistent) if it does not **possess a solution**
- ❖ A homogeneous system always possesses at least one solution (namely the trivial solution), a homogeneous system is always consistent. But a non-homogeneous system may or may not be consistent.
- ❖ The equation $AX = B$ is consistent i.e. has a solution (unique or infinite) iff the matrices A and D are of the same rank i.e. $\rho(A) = \rho(D)$.
- ❖ i) If $\rho(A) = \rho(D) = n$, the number of unknowns, then only the given system of equations will have a unique solution
- ❖ ii) If $\rho(A) = \rho(D) < n$, then the given system will have infinite solutions.
- ❖ iii) If $\rho(A) \neq \rho(D)$ then the given system of equations will be inconsistent or will not have any solution.

Example: Test the consistency of the following equations

$$4x - 2y + 6z = 8$$

$$x + y - 3z = -1$$

$$15x - 3y + 9z = 21$$

Solution:

$$A = \begin{bmatrix} 4 & -2 & 6 \\ 1 & 1 & -3 \\ 15 & -3 & 9 \end{bmatrix}, \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad B = \begin{bmatrix} 8 \\ -1 \\ 21 \end{bmatrix}$$

$$D = [A : B] = \begin{bmatrix} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{bmatrix} \quad R_{12}, R_3 = R_3 / 3$$

$$\begin{bmatrix} 1 & 1 & -3 & -1 \\ 4 & -2 & 6 & 8 \\ 5 & -1 & 3 & 7 \end{bmatrix} R_2 = R_2 - 4R_1; R_3 = R_3 - 5R_1$$

$$\begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & -6 & 18 & 12 \end{bmatrix} R_3 = R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_2 = -\frac{1}{6}R_2$$

$$\begin{bmatrix} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_1 = R_1 - R_2 \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If $\rho(A) = \rho(D) = 2 < \text{number of unknowns}$, the system is consistent and has an infinite number of solutions

Example: Show that the following equations are inconsistent

$$2x + y + 5z = 4$$

$$3x - 2y + 2z = 2$$

$$5x - 8y - 4z = 1$$

Soln.: $Ax = B$ i.e. $\begin{bmatrix} 2 & 1 & 5 \\ 3 & -2 & 2 \\ 5 & -8 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

$D = [A/B] = \begin{bmatrix} 2 & 1 & 5 & 4 \\ 3 & -2 & 2 & 2 \\ 5 & -8 & -4 & 1 \end{bmatrix} R_1 = R_2 - R_1; R_3 = R_3 - R_2$

$$\begin{bmatrix} 1 & -3 & -3 & -2 \\ 3 & -2 & 2 & 2 \\ 2 & -6 & -6 & -1 \end{bmatrix} R_2 = R_2 - 3R_1; R_3 = R_3 - 2R_1$$

$$\begin{bmatrix} 1 & -3 & -3 & -2 \\ 0 & 7 & 11 & 8 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$\rho(A) = 2, \rho(D)=3 \quad \rho(A) \neq \rho(D)$ Hence the given equations are inconsistent

Example: Show that the system of equations

$$2x - y + z = 4$$

$$3x - y + z = 6$$

$$4x - y + 2z = 7$$

$$-x + y - z = 9 \text{ is consistent.}$$

Solution:

$$A = \begin{bmatrix} 2 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 4 \\ 6 \\ 7 \\ 9 \end{bmatrix}$$

$$D = [A : B] = \begin{bmatrix} 2 & -1 & 1 & 4 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \\ -1 & 1 & -1 & 9 \end{bmatrix} \left\{ \begin{array}{l} R_1 = R_1 + 2R_4 \\ R_2 = R_2 + 3R_4 \\ R_3 = R_3 + 4R_4 \end{array} \right.$$

$$\begin{bmatrix} 0 & 1 & -1 & 22 \\ 0 & 2 & -2 & 33 \\ 0 & 3 & -2 & 43 \\ -1 & 1 & -1 & 9 \end{bmatrix} \left\{ \begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \\ R_4 = R_4 - R_1 \end{array} \right.$$

$$\begin{bmatrix} 0 & 1 & -1 & 22 \\ 0 & 0 & 0 & -11 \\ 0 & 0 & 1 & -23 \\ -1 & 0 & 0 & -13 \end{bmatrix} \left\{ \begin{array}{l} R_1 = R_1 + R_3 \\ R_4 = (-1)R_4 \end{array} \right.$$

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -11 \\ 0 & 0 & 1 & -23 \\ 1 & 0 & 0 & 13 \end{bmatrix} R_{14} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 13 \\ 0 & 0 & 0 & -11 \\ 0 & 0 & 1 & -23 \\ 0 & 1 & 0 & -1 \end{bmatrix} R_{24} \Rightarrow \begin{bmatrix} 1 & 0 & 0 & 13 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & -23 \\ 0 & 1 & 0 & -11 \end{bmatrix} R_{14}$$

$\rho(A) = 3, \rho(D)=4 \quad \rho(A) \neq \rho(D)$ The system is inconsistent

Example : Investigate the values of λ and μ so that the equations

$$2x + 3y + 5z = 9$$

$$7x + 3y - 2z = 8$$

$$2x + 3y + \lambda z = \mu$$

have i) Unique solution ii) no solution iii) an infinite number of solutions

Solution:

$$\begin{bmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 8 \\ \mu \end{bmatrix} \quad (1)$$

- i) If equation (1) has a unique solution then the determinant of the coefficient matrix $A \neq 0$

$$|A| = \begin{vmatrix} 2 & 3 & 5 \\ 7 & 3 & -2 \\ 2 & 3 & \lambda \end{vmatrix} = 15(5 - \lambda) \therefore |A| \neq 0, \lambda \neq 5$$

i.e the given system of equations will have a unique solution if $\lambda \neq 5$ whatever may the value of μ .

$$D = [A / B] = \begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 2 & 3 & \lambda & \mu \end{bmatrix} \quad R_3 = R_3 - R_1$$

$$\begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 0 & 0 & \lambda - 5 & \mu - 9 \end{bmatrix}$$

If $\lambda = 5$ and $\mu \neq 9$, then $\rho(A) = 2$, $\rho(D) = 3$ $\rho(A) \neq \rho(D)$ Hence the given system of equations has no solution

c) If in the augmented D, $\lambda = 5$ and $\mu = 9$ then

$$D = \begin{bmatrix} 2 & 3 & 5 & 9 \\ 7 & 3 & -2 & 8 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

It is clear that $\rho(A) = \rho(D) = 2 (< 3 \text{ the number of unknowns})$ Therefore there exist infinite number of solutions to the given set of equations

Basically, there are TWO types of methods available, viz, Direct methods and Indirect methods or Iterative methods.

Direct Method:

- i) Cramer's rule
- ii) Gaussian Elimination
- iii) Gauss-Jordan method
- iv) Choleskey Method

Indirect Method:

- i) Jacobi Method
- ii) Gauss seidal method
- iii) Successive Over-Relaxation Technique

Gauss Elimination method:

Gauss Elimination method applied to three linear equations. First we explain this method applied to a particular systems of order three given by

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \quad (1)$$

The augmented matrix of the above system

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{pmatrix}$$

To eliminate x_1 from the second equation multiplying the first equation by $-\frac{a_{21}}{a_{11}}$ and add it to the second equation.

Similarly to eliminate x_1 from the third equation multiply the first equation by $-\frac{a_{31}}{a_{11}}$ and add it to the third equation. Thus we get the matrix.

$$\sim \begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & a'_{32} & a'_{33} & b'_3 \end{pmatrix}$$

Where $-\frac{a_{21}}{a_{11}}$, $-\frac{a_{31}}{a_{11}}$ are called the multipliers for the first stage of elimination. The first equation is called the pivot equation and a_{11} ($a_{11} \neq 0$) is called the first pivot a'_{22} is the new pivot and the multiplier is $-\frac{a'_{32}}{a'_{22}}$

i.e. multiply the second row by $-\frac{a'_{32}}{a'_{22}}$ and add it to the third row.

$$\sim \begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & a'_{32} - \frac{a'_{32}}{a'_{22}}a'_{22} & a'_{33} - \frac{a'_{32}}{a'_{22}}a'_{23} & b'_3 - \frac{a'_{32}}{a'_{22}}b'_2 \end{pmatrix}$$

$$\sim \begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a''_{33} & b''_3 \end{pmatrix}$$

The values of x_1 , x_2 and x_3 can be obtained by the substitution

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a'_{22}x_2 + a'_{23}x_3 &= b'_2 \\ a''_{33}x_3 &= b''_3 \end{aligned}$$

$$\text{From the last equation } x_3 = \frac{b''_3}{a''_{33}}$$

Remark: This method fails if one of the pivots say a_{11} , a'_{22} or a''_{33}

Vanishes. We can modify by rearranging the rows so that the pivot is non-zero.

Example: Solve by Gauss elimination Method

$$2x + y + z = 10$$

$$3x + 2y + 3z = 18$$

$$x + 4y + 9z = 16$$

Solution: The augmented matrix is

$$\begin{pmatrix} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{pmatrix} \quad R_2 = R_2 - \frac{3}{2}R_1; R_3 = R_3 - \frac{1}{2}R_1$$

Eliminating x from second and third equation

The augmented matrix is

$$\sim \begin{pmatrix} 2 & 1 & 1 & 10 \\ 3 - \frac{3}{2} & 2 - \frac{3}{2} & 3 - \frac{3}{2} & 18 - \frac{3}{2} \cdot 10 \\ 1 - \frac{1}{2} & 4 - \frac{1}{2} & 9 - \frac{1}{2} & 16 - \frac{1}{2} \cdot 10 \end{pmatrix}$$

$$\sim \begin{pmatrix} 2 & 1 & 1 & 10 \\ 0 & \frac{1}{2} & \frac{3}{2} & 3 \\ 0 & \frac{7}{2} & \frac{17}{2} & 11 \end{pmatrix}$$

Eliminating y from the third equation $R_3 = R_3 - 7R_2$

$$\begin{pmatrix} 2 & 1 & 1 & 10 \\ 0 & \frac{1}{2} & \frac{3}{2} & 3 \\ 0 & 0 & -2 & -10 \end{pmatrix}$$



$$\begin{aligned} 2x + y + z &= 10 \\ \frac{1}{2}y + \frac{3}{2}z &= 3 \\ -2z &= 10 \end{aligned}$$

$$z = 5, y = -9, x = 7$$

Gauss-Jordan Method:

Instead of eliminating x_2 only in the third equation, we could have eliminated it from the first equation also, so that at the end of the second stage, the augmented matrix becomes.

Gauss-Jordan Method: consider

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \quad (1)$$

The augmented matrix of the above system

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{pmatrix} \quad (2)$$

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & b_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & a'_{32} - \frac{a_{32}}{a'_{22}}a'_{22} & a'_{33} - \frac{a_{32}}{a'_{22}}a'_{23} & b'_3 - \frac{a_{32}}{a'_{22}}b'_2 \end{pmatrix}$$

$$\begin{pmatrix} a'_{11} & 0 & a'_{13} & b'_1 \\ 0 & a'_{22} & a'_{23} & b'_2 \\ 0 & 0 & a'_{33} & b'_3 \end{pmatrix}$$

This modification of Gaussian elimination is called the Gauss-Jordan method

Example: Solve by Gauss-Jordan method

$$\begin{aligned} 2x + y + z &= 10 \\ 3x + 2y + 3z &= 18 \\ x + 4y + 9z &= 16 \end{aligned}$$

Solution: The augmented matrix is

$$\begin{pmatrix} 2 & 1 & 1 & 10 \\ 3 & 2 & 3 & 18 \\ 1 & 4 & 9 & 16 \end{pmatrix} \quad R_2 = R_2 - \frac{3}{2}R_1; R_3 = R_3 - \frac{1}{2}R_1$$

Eliminating x from second and third equation

$$\begin{pmatrix} 2 & 1 & 1 & 10 \\ 3-\frac{3}{2} & 2-\frac{3}{2} & 3-\frac{3}{2} & 18-\frac{3}{2} \\ 1-\frac{1}{2} & 4-\frac{1}{2} & 9-\frac{1}{2} & 16-\frac{1}{2} \end{pmatrix} \begin{matrix} 2 \\ 10 \\ 10 \end{matrix} \text{ i.e } R_2 = R_2 - \frac{3}{2}R_1; R_3 = R_3 - \frac{1}{2}R_1$$

$$\begin{pmatrix} 2 & 1 & 1 & 10 \\ 0 & \frac{1}{2} & \frac{3}{2} & 3 \\ 0 & \frac{7}{2} & \frac{17}{2} & 11 \end{pmatrix} \begin{matrix} R_1 = R_1 - 2R_2 \\ R_3 = R_3 - 7R_2 \end{matrix} \begin{pmatrix} 2 & 0 & -2 & 4 \\ 0 & \frac{1}{2} & \frac{3}{2} & 3 \\ 0 & 0 & -2 & -10 \end{pmatrix}$$

$$2x - 2z = 4$$

$$(\frac{1}{2})y + (\frac{3}{2})z = 3$$

$$-2z = -10 \Rightarrow z = 5, y = -9, x = 7$$

Example :Solve the following system of linear equations

$$x + 2y + z = 8$$

$$2x + 3y + 4z = 20$$

$$4x + 3y + 2z = 16 \text{ Using Gauss-Jordan method.}$$

Solution: Let the given system of equations be written in the matrix form $AX = B$ where

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 4 \\ 4 & 3 & 2 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 8 \\ 20 \\ 16 \end{bmatrix}$$

The augmented matrix is

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & 3 & 4 & 20 \\ 4 & 3 & 2 & 16 \end{bmatrix}$$

$$[A:B] = \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & 3 & 4 & 20 \\ 4 & 3 & 2 & 16 \end{bmatrix} \begin{matrix} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 4R_1 \end{matrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & -5 & -2 & -16 \end{bmatrix} \quad R_3 = R_3 - 5R_2$$

$$\begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & -1 & 2 & 4 \\ 0 & 0 & -12 & -36 \end{bmatrix} \quad R_1 + 1/12R_3, R_2 + 1/6R_3$$

$$\begin{bmatrix} 1 & 2 & 0 & 5 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -12 & -36 \end{bmatrix} \quad R_1 = R_1 + 2R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & -1 & 0 & -2 \\ 0 & 0 & -12 & -36 \end{bmatrix}$$

We get the following system of equations $x = 1$, $y = 2$ and $12z = 36$

Then required solution is $x = 1$, $y = 2$, $z = 3$

Unit-VIII: Linear Algebra-II

Linear Transformations: Definition:

A transformation (or function or mapping):

A transformation T from R^n to R^m is a rule that assigns to each vector x in R^n a vector $T(x)$ in R^m .

The set R^n is called the domain of T , and R^m is called the co-domain of T . The notation $T:R^n \rightarrow R^m$ indicates that the domain of T is R^n and the co-domain is R^m .

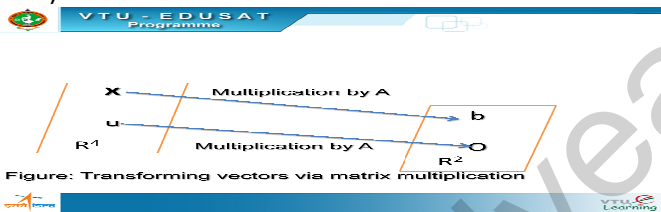
For x in R^n , the vector $T(x)$ in R^m is called the image of x (under the action of T).

The set of all images $T(x)$ is called the range of T .

For example: $AX = b$ and $Au = 0$

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Say that multiplication by A transforms X into b and similarly multiplication by A transforms u into the 0 (zero) vector.



Matrix Transformation : $T:R^n \rightarrow R^m$ for each x in R^n , $T(x)$ is computed as $A(x)$, where A is an $m \times n$ matrix, For simplicity, we sometimes denote such a matrix transformation by $x \rightarrow Ax$.

Example: Let $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $u = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $b = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$

And define a transformation $T:R^2 \rightarrow R^3$ by $T(x) = Ax$ so that $T(x) = A(x) = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$

Find $T(u)$, the image of u under the transformation T .

$$T(u) = Au = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ -9 \end{bmatrix}$$

Definition: A transformation (or mapping) T is linear if

- i) $T(u + v) = T(u) + T(v)$ for all u, in the domain of T
- ii) $T(cu) = c T(u)$ for all u and all scalars c.

Characteristic Equation of a Matrix :

If $A = (a_{ij})$ is any square matrix of order n and λ be any variable, then the matrix $[A - \lambda I]$ is called the characteristic matrix of A. The determinant of this matrix i.e. $|A - \lambda I|$ is called the characteristic function. The equation $|A - \lambda I| = 0$ is the characteristic equation of A and of degree n. The roots of the equation $|A - \lambda I| = 0$ are called the characteristics or Eigen values of A.

Example: Determine the characteristic roots of

$$A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$$

Solution: The characteristic equation is $|A - \lambda I| = 0$

i.e.

$$\begin{vmatrix} 2-\lambda & 2 & 1 \\ 1 & 3-\lambda & 1 \\ 1 & 2 & 2-\lambda \end{vmatrix} = 0 \quad \Rightarrow \quad -\lambda^3 + 7\lambda^2 - 11\lambda + 5 = 0$$

By synthetic division method

$$\begin{array}{r|rrrr} 1 & -1 & 7 & -11 & 5 \\ & & -1 & 6 & -5 \\ \hline & -1 & 6 & 5 & 0 \end{array}$$

$$-\lambda^2 + 6\lambda - 5 = 0 \Rightarrow (-\lambda + 1)(\lambda - 5) = 0$$

$\lambda_1 = 1, \lambda_2 = 1, \lambda_3 = 5$ are the require roots or Eigen values

Example: Find the eigen values and the eigenvector of the matrix $A = \begin{pmatrix} 5 & 4 \\ 1 & 2 \end{pmatrix}$

Solution: The characteristic equation of A is $|A - \lambda I| = 0$

$$\text{i.e. } \begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0 \quad \text{expanding we get} \quad \lambda^2 - 7\lambda + 6 = 0$$

The roots of this equation are, $\lambda = 6, \lambda = 1$ which are the eigen values of A. To find the eigen vectors, consider the matrix equation is

$$[A - \lambda I]X = 0 \quad \text{where} \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\text{i.e.} \begin{bmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This gives 2 homogeneous linear equations

$$(5-\lambda)x_1 + 4x_2 = 0$$

$$x_1 + (2-\lambda)x_2 = 0$$

i) For $\lambda = 6$ above equations becomes

$$-x_1 + 4x_2 = 0$$

$$x_1 - 4x_2 = 0 \quad \text{Both represent the same equation } x_1 - 4x_2 = 0$$

$$\therefore \frac{x_1}{4} = \frac{x_2}{1}$$

giving the eigenvector

$$X_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$$

ii) Again for $\lambda = 1$,

$$4x_1 + 4x_2 = 0$$

$$x_1 + x_2 = 0 \quad \text{Both represent the same } x_1 + x_2 = 0 \quad \therefore \frac{x_1}{1} = \frac{x_2}{-1}$$

giving the eigenvector

$$X_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Example: Find the eigen value and the eigen vectors of the matrix

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Solution:

$$\text{Let } A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

The characteristic equation of the matrix A is $|A - \lambda I| = 0$

$$\text{i.e } \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} = 0$$

Expanding, we get $\lambda^3 - 18\lambda^2 + 45\lambda = 0$: $\lambda^3 - 18\lambda^2 + 45\lambda = 0$ $\lambda(\lambda^2 - 18\lambda + 45) = 0$

i.e $\lambda = 0$ or $(\lambda^2 - 18\lambda + 45) = 0$

i.e $\lambda = 0$ or $(\lambda - 3)(\lambda - 15) = 0$

i.e $\lambda_1 = 0, \lambda_2 = 3, \lambda_3 = 15$ These are the eigen values of A

To find the eigen vector consider the matrix equation is

$$[A - \lambda I]X = 0. \text{ where } X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

i.e

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

This gives the three homogeneous equations

$$\begin{aligned} (8-\lambda)x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + (7-\lambda)x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + (3-\lambda)x_3 &= 0 \end{aligned} \quad (1)$$

i) $\lambda_1 = 0$ equation (1) become

$$\begin{aligned} 8x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + 7x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + 3x_3 &= 0 \end{aligned} \quad (2)$$

Solving any two non-identical equations of (2) for x_1, x_2, x_3 by the rule of cross multiplication. We get

$$\frac{x_1}{24-14} = \frac{-x_2}{-32+12} = \frac{x_3}{56-36} \Rightarrow \frac{x_1}{10} = \frac{-x_2}{20} = \frac{x_3}{20} \Rightarrow \frac{x_1}{1} = \frac{-x_2}{2} = \frac{x_3}{2}$$

$$x_1 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Therefore the eigenvector corresponding to $\lambda_1 = 0$ is

ii) $\lambda_2 = 3$ equation (1) become

$$\begin{aligned} 5x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 + 4x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 + 0x_3 &= 0 \end{aligned} \quad (3)$$

Solving by the rule of cross multiplication We get

$$\frac{x_1}{2} = \frac{x_2}{1} = \frac{x_3}{-2}$$

Therefore the eigenvector corresponding to $\lambda_2 = 3$ is

$$x_2 = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$$

iii) $\lambda_3 = 15$ equation (1) become

$$\begin{aligned} -7x_1 - 6x_2 + 2x_3 &= 0 \\ -6x_1 - 8x_2 - 4x_3 &= 0 \\ 2x_1 - 4x_2 - 12x_3 &= 0 \end{aligned} \quad (3)$$

Solving by the rule of cross multiplication We get

$$\frac{x_1}{2} = \frac{x_2}{-2} = \frac{x_3}{1}$$

Therefore the eigenvector corresponding to $\lambda_3 = 15$ is

$$x_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$$

❖ Orthogonal Matrix:

A square matrix A is said to be orthogonal if $AA' = I = A'A$

❖ Singular matrix and Nonsingular matrix:

A square matrix A is said to be singular if its determinant is zero and A is said to be nonsingular if its determinant is not equal to zero.

❖ Inverse of a square matrix:

If A and B are two square matrices of the same order such that $AB = BA = I$ then B is called the inverse of A denoted by A^{-1} .

❖ Similarity of Matrices:

Two square matrices A and B of the same order are said to be similar if there exists a non singular matrix P such that $B = P^{-1}AP$

Here B is said to be similar to A.

Example:

Reduce the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ to the diagonal form .

Solution: The characteristic equation of A is $|A - \lambda I| = 0$

$$i.e \begin{bmatrix} -1-\lambda & 3 \\ -2 & 4-\lambda \end{bmatrix} = 0 \Rightarrow \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1 \text{ and } 2$$

Are the eigen values of A.

Consider $[A - \lambda I][X] = [0]$

$$i.e \begin{bmatrix} -1-\lambda & 3 \\ -2 & 4-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$i.e -2x + (4-\lambda)y = 0$$

Casei) let $\lambda=1$, we get $-2x+3y=0$ or $2x=3y$ or $x/3 = y/2$

$X_1 = (3,2)'$ is the eigen vector corresponding to $\lambda=1$

Caseii) let $\lambda=2$,

we get $-3x+3y=0$ or $x=y$ or $x/1 = y/1$

$X_2 = (1,1)'$ is the eigen vector corresponding to $\lambda=2$.

$$P = [X_1 \quad X_2] = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$$

we have $|P| = 1$ and $P^{-1} = \frac{1}{|P|} (\text{Adj} P)$

$$P^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}$$

$$P^{-1}AP = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Thus $P^{-1}AP = D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ is the diagonal matrix

Reduce the following matrices into diagonal matrices

$$A = \begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix} \quad \text{Ans.} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \quad \text{Ans.} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}$$